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Locally ringed spaces are fundamental objects in algebraic geometry, forming a categorical framework for schemes. While the category of ringed spaces admits colimits, constructing colimits in the subcategory of locally ringed spaces (LRS) requires special attention to the preservation of the local structure. In this thesis, we construct colimits in LRS by describing coproducts and coequalisers in LRS in detail and verifying their universal properties. We show that coproducts in LRS arise from coproducts of underlying topological spaces, equipped with products of structure sheaves. Coequalisers in LRS are obtained from coequalisers of underlying topological spaces, together with equalisers of corresponding structure sheaves. These constructions provide a concrete realisation of colimits in LRS, highlighting the role of categorical methods in geometric settings.

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# 1 Introduction

In this thesis, we explicitly construct colimits in the category of locally ringed spaces (LRS), a construction that plays a key role in algebraic geometry. Category theory offers a unifying language across diverse mathematical areas, including algebra, topology, and measure theory. It abstracts structures—such as groups, rings, and topological spaces—and focuses on structure-preserving maps, clarifying hidden analogies. Moreover, in fields like algebraic geometry, algebraic topology, and homological algebra, category theory allows us to describe complex constructions—such as sheaves—systematically. A formal introduction to categories is provided in Chapter 3.

Limits and colimits provide a general toolkit for building new objects from diagrams, unifying constructions such as products, coproducts, equalisers, and coequalisers, as discussed in Chapter 6. Furthermore, by appealing to universal properties, limits and colimits allow us to use structural reasoning instead of complicated element-wise arguments.

The Yoneda Construction and the Yoneda Lemma in Chapter 5 illustrate the power of universal properties. Universal properties allow us to characterise limits and colimits in terms of representable functors, which makes the definition intrinsic rather than arbitrary.

Functors, which are introduced in Chapter 4, are maps between categories preserving certain intrinsic structures. Functors like the contravariant  $\text{Hom}$ -functor illustrate dualities and reveal structural correspondences.

Locally ringed spaces generalise both manifolds and varieties, providing a unified framework in which each point of a locally ringed space captures local algebraic or geometric information. They allow us to study algebraic objects locally and glue them together globally. The construction of colimits in LRS is non-trivial, yet foundational, since LRS forms a natural category for schemes, which are central objects in

algebraic geometry. Building schemes involves gluing affine schemes along overlaps—which is precisely the colimit construction. The existence of colimits allows categorical flexibility for carrying out global geometric constructions. As we see in Chapter 7, locally ringed spaces are topological spaces equipped with a sheaf of local rings, both of which must be glued consistently. While the colimit in the category of topological spaces is rather straightforward, we also have to consider sheaves, which makes the construction far more subtle, since the colimit has to respect locality and gluing conditions. Ensuring that the resulting structure is a locally ringed space requires additional care.

To ensure accessibility, the foundational definitions are presented in Chapter 2. Elaborations can be found in the Appendix.

## 2 Preliminaries

In this chapter, we introduce and revisit several fundamental definitions. We begin by revisiting set-theoretic foundations—as they underpin the arguments developed throughout the thesis—and touch some graph-theoretic concepts. We continue with foundational topological concepts, since they are elementary in the definition of locally ringed spaces and provide illustrative examples of categories, limits, and colimits. This chapter concludes with the foundations of ring theory, since commutative rings—and especially local rings—are essential in the definition and handling of locally ringed spaces.

### 2.1 Sets and Functions

In this thesis, several set-theoretic concepts are used. Set-theory itself is not introduced, and we use the terms **set** and **class** informally, without engaging in foundational set theory—such as the Zermelo-Fraenkel axioms. Classes can be thought of **collections** that are too large to be sets—such as the power set of all sets. Moreover, the term "collection" is a placeholder for sets or classes. More on set-theory and classes can be found in [6] or [5].

In Chapter 6 we discuss the functoriality of limits, which relies on the axiom of choice.

**Axiom 2.1** (Axiom of choice). Let  $\mathcal{M}$  be a collection of disjoint non-empty sets. There exists a set  $M$  consisting of exactly one element from each element of  $\mathcal{M}$ . In particular, for each  $m \in \mathcal{M}$  the intersection  $m \cap M$  contains one single element (see [13, Ch. 1, § 9, p. 59]).

When it comes to stalks—which are special directed colimits and essential to locally ringed spaces—directed posets play an essential role. We use directed posets in the definition of stalks—which are essential in the definition of locally ringed spaces—as well as in the proof of Theorem C.1, which allows us to commute directed colimits and finite limits, as well as indexing categories—for example in directed colimits in Chapter 6.

**Definition 2.2** (Directed poset). Let  $(I, \leq)$  be a poset, that is, a set with a partial order relation. It is called **directed** if for every finite subset  $\{i_1, \dots, i_n\} \subseteq I$ , there exists an element  $k \in I$  such that  $i_j \leq k$  for all  $j \in \{1, \dots, n\}$  (see [3, Ch. 6, § 5, Def. 6.5.15]).

Cofinal subsets allow us to simplify complex indexing sets to manageable subcollections, a critical step when we compute stalks, or any directed colimit in both, algebraic and topological contexts.

**Definition 2.3** (Cofinal). Let  $(M, \leq)$  be a preordered set, that is, a set  $M$  equipped with a binary relation  $\leq$  that is reflexive and transitive but not necessarily antisymmetric. A subset  $S \subseteq M$  is **cofinal** in  $M$  if for all  $m \in M$  there exists an  $s \in S$  such that  $m \leq s$  (see [14]).

When we discuss limits in Chapter 6, we use the notion of arbitrary Cartesian products.

**Definition 2.4** (Cartesian product). Let  $J$  be an index set and let  $A$  be a set. A  $J$ -tuple of elements of  $A$  is formally a function

$$a : J \rightarrow A.$$

For  $j \in J$ , we denote the value of  $a$  at  $j$  as  $a_j$  (instead of  $a(j)$ ) and call it the **j-th coordinate** of  $a$ . The function itself is often represented by the tuple  $(a_j)_{j \in J}$ .

For an indexed family of sets  $(A_j)_{j \in J}$ , the (infinite) **Cartesian product** is the set of all  $J$ -tuples  $(a_j)_{j \in J}$  such that  $a_j \in A_j$  for each  $j \in J$  (see [13, Ch. 2, § 19, p. 113]).

In Chapter 3, we see that categories and diagrams are closely related. To understand diagrams, we use directed multigraphs.



**Definition 2.5** (Directed multigraph). A **directed multigraph** is a pair  $(V, E)$ , where  $V$  is a set of vertices, and  $E$  is a set of directed edges, such that every edge  $e$  has a source vertex  $s$  and a target vertex  $t$  for  $s \neq t$ . We may write this as  $e : s \rightarrow t$  or  $e = (s, t)$  (see [3, Ch. 2, § 2, Ex. 2.2.22]).

*Remark 2.6.* 1. If between any *ordered* pair of vertices  $(s, t)$  there exists at most one edge  $e : s \rightarrow t$ , a graph is called **directed**.

2. A **loop** is an edge  $e : v \rightarrow v$ .

3. If we allow a graph to have multiple edges between two vertices, it is called a **multigraph**. Following [4], loops are not allowed in multigraphs. Other authors allow loops in multigraphs.

4. A graph that permits multiple edges between two vertices, as well as loops, is called a **pseudograph**.

For reference see [4, Ch. 2, p. 10].

In Chapter 7, we want to show that limits commute with limits (Theorem 6.28). The theorem relies on the Cartesian product of directed graphs.

**Definition 2.7** (Cartesian product of graphs). Let  $\Gamma := (V, E)$  and  $\Lambda := (W, F)$  be directed graphs. The **Cartesian product**  $\Gamma \otimes \Lambda$  is the directed graph

$$\Gamma \otimes \Lambda := (V \times W, V \times F \sqcup E \times W),$$

that is, there exists an edge  $(v, w) \rightarrow (v', w')$  in  $\Gamma \otimes \Lambda$  if and only if one of the following cases is true:

1.  $v = v'$  and there exists an edge  $w \rightarrow w'$  in  $\Lambda$ .

2. There exists an edge  $v \rightarrow v'$  in  $\Gamma$  and  $w = w'$  in  $\Lambda$ .

For reference, see [3, Ch. 6, § 6, Def. 6.6.1].

The following example illustrates the concept of the Cartesian product of graphs concretely:

**Example 2.8.** The Cartesian product of the two directed graphs

$$\Gamma := v_1 \longrightarrow v_2 \longrightarrow v_3 \quad \text{and} \quad \Lambda := \begin{array}{c} w_1 \\ \downarrow \\ w_2 \end{array}$$

is

$$\Gamma \otimes \Lambda = \begin{array}{ccccc} (v_1, w_1) & \longrightarrow & (v_2, w_1) & \longrightarrow & (v_3, w_1) \\ \downarrow & & \downarrow & & \downarrow \\ (v_1, w_2) & \longrightarrow & (v_2, w_2) & \longrightarrow & (v_3, w_2) \end{array}$$

(see [3, Ch. 6, § 6, Def. 6.6.1]).

*Remark 2.9.* The Cartesian product of graphs is not the product in the categorical sense. The categorical product of graphs is the tensor product, which is defined differently (see [3, Ch. 6, § 6, Rem. 6.6.2]).

## 2.2 Topological Concepts

Topological concepts are not only indispensable in the definition of locally ringed spaces, they also provide exciting results, such as that a doughnut is homeomorphic to coffee cups. The word "topology" originates from the Greek " $\tau\acute{o}\pi\omicron\sigma$ " (topos), which translates to "place", and " $\lambda\acute{o}\gamma\omicron\varsigma$ " (logos), which means "study" or "discourse". In Analysis, they allow us to define openness and compactness of sets, as well as continuity of functions, among other things.

**Definition 2.10** (Topological space). A **topological space** is a pair  $(X, \tau)$ , where  $X$  is a set and  $\tau \subseteq \mathcal{P}(X)$  is a set of subsets of  $X$  satisfying the following conditions:

1.  $\emptyset$  and  $X$  are elements of  $\tau$ .
2. If  $T_i \in \tau$  for all  $i \in I$ , then the union  $\bigcup_{i \in I} T_i$  is in  $\tau$  for any index set  $I$ .
3. If  $T_i \in \tau$  for all  $i \in I$ , then the finite intersection  $\bigcap_{i \in I} T_i$  is in  $\tau$  for every finite index set  $I$ .

The elements of  $\tau$  are called **open sets** (see [13, Ch. 2, § 12, p. 76]).

*Remark 2.11.* In order to improve the reading flow, we abuse notation slightly by denoting a topological space  $(X, \tau)$  by  $X$ .

Sometimes we are interested in local descriptions of topologies and topological concepts. For example, we want to know if a property is satisfied in a certain point of a topological space. Since topological spaces are defined via open sets, the point of interest itself might not be in the topology, but in an open set, which is in the topology. This motivates the use of neighbourhoods.

**Definition 2.12** (Neighbourhood). Let  $X$  be a topological space and  $x \in X$ . A **neighbourhood** of  $x$  is a set  $U$  that contains an open set  $V$  containing  $x$ . The collection of all neighbourhoods of  $x$ , denoted by  $\mathcal{U}_x$ , is called the **neighbourhood system** at  $x$  (see [20, Ch. 2, § 4, Def. 4.1]).

A more structured framework to neighbourhood systems is the neighbourhood base.

**Definition 2.13** (Neighbourhood base). A **neighbourhood base** at  $x$  in a topological space  $X$  is a subcollection  $\mathcal{B}_x$  of  $\mathcal{U}_x$  such that for any  $U \in \mathcal{U}_x$  there exists a  $B \in \mathcal{B}_x$  such that  $B \subseteq U$ . That is

$$\mathcal{U}_x = \{U \subseteq X \mid B \subseteq U \text{ for some } B \in \mathcal{B}_x\}$$

(see [20, Ch. 2, § 4, Def. 4.3]).

*Remark 2.14.* This allows us to understand neighbourhood systems as cofinal systems. Let  $X$  be an arbitrary topological space and let  $\mathcal{U}_x$  be a system of all neighbourhoods of  $x \in X$ . We can define an order relation  $\leq$  via

$$U \leq V \iff V \subseteq U.$$

This gives rise to a neighbourhood basis  $\mathcal{B} \subseteq \mathcal{U}_x$  such that for any  $V \in \mathcal{U}_x$  there exists  $B \in \mathcal{B}$  such that  $B \subseteq V$ . Thus,  $\mathcal{B}$  is a cofinal subset of  $(\mathcal{U}_x, \supseteq)$ .

One can define different topologies on the same underlying set. The notion of coarseness allows to compare these topologies.

**Definition 2.15** (Coarseness). Let  $\tau$  and  $\tau'$  be two topologies on a set  $X$ .

1. If  $\tau \subset \tau'$ , then  $\tau'$  is **finer** than  $\tau$ , and  $\tau$  is **coarser** than  $\tau'$ .
2. If  $\tau \subsetneq \tau'$ , then  $\tau'$  is **strictly finer** than  $\tau$ , and  $\tau$  is **strictly coarser** than  $\tau'$ .

Two topologies are **comparable** if  $\tau \subseteq \tau'$  or  $\tau' \subseteq \tau$  (see [13, Ch. 2, § 12, p. 77]).

**Definition 2.16** (Continuity). Let  $X$  and  $Y$  be topological spaces. A function  $f : X \rightarrow Y$  is called **continuous** if and only if the preimage of any open subset  $V \subseteq Y$  is open in  $X$  (see [8, Ch. 1, § 5]).

*Remark 2.17.* The composite of continuous functions, as well as the identity map, are continuous. The composition of continuous functions is associative (see [13, Ch. 2, §18, pp. 102, 107], [8, Ch. 1, § 5]).

**Example 2.18** (Product topology). Let  $I$  be a non-empty index set and let  $((X_i, \tau_i))_{i \in I}$  be a family of topological spaces, and let

$$X := \prod_{i \in I} X_i$$

be the Cartesian product of the underlying sets  $X_i$ . For every  $i \in I$  we define the  **$i$ -th canonical projection** as

$$\begin{aligned} \text{pr}_i : \prod_{j \in I} X_j &\rightarrow X_i \\ (x_j)_{j \in I} &\mapsto x_i. \end{aligned} \tag{2.19}$$

The **product topology**  $\tau$ , also called **Tychonoff topology**, is the coarsest topology, in essence, the topology with the fewest open sets, such that all projections (2.19) are continuous. The Cartesian product  $X$  endowed with the product topology  $\tau$  is called the **product space** (see [8, Ch. 6, § 2], [13, Ch. 2, § 19, pp. 113]).

An alternative definition of the product topology using the basis (which is introduced in Definition A.1) can be found in the appendix (see Definition A.4), alongside with a concrete example. Furthermore, it is not immediately clear that a function into the product topology

$$f : Y \rightarrow \prod_{i \in I} X_i$$

given by

$$f(y) = (f_i(y))_{i \in I}$$

with  $f_i : Y \rightarrow X_i$  for all  $i \in I$ , is continuous for an infinite set  $I$ .

**Theorem 2.20.** Let  $\prod_{i \in I} X_i$  be the product topology and let

$$f : Y \rightarrow \prod_{i \in I} X_i$$

be given by

$$f(y) = (f_i(y))_{i \in I}$$

with  $f_i : Y \rightarrow X_i$  for all  $i \in I$ . The function  $f$  is continuous if and only if every function  $f_i$  is continuous (see [13, Ch. 2, § 19, Thm. 19.6]).

The proof can be consulted in the appendix, [A](#).

*Remark 2.21.* A topology that is similar to the product topology is the **box topology**. While the box topology and the product topology agree for a finite set of topological spaces, they do not agree for infinitely many topological spaces. For infinite topological spaces, the box topology is finer than the product topology. The differences are discussed in detail in [13, Ch. 2, § 19, pp. 114]. Theorem 2.20 is part of the reason why we do not use the box topology, since the theorem holds explicitly for the product topology, as the use of the box topology would lead to issues with the continuity of the function  $f$  (compare [13, Ch. 2, § 19, p. 115]).

**Example 2.22** (Quotient topology). Let  $X$  be a topological space and let  $\sim$  be an equivalence relation on  $X$ . The **quotient set**

$$Y = X / \sim = \{[x] \mid x \in X\}$$

is the set of equivalence classes, where

$$[x] = \{y \in X \mid y \sim x\}$$

is the equivalence class of  $x$ . We construct the set  $Y$  via the surjection

$$\begin{aligned}\pi : X &\rightarrow Y \\ x &\mapsto [x].\end{aligned}$$

The **quotient space** under  $\sim$  is the set  $Y$ , equipped with the **quotient topology**, which is the finest topology on the quotient space, where the open sets  $U$  are subsets  $U \subseteq Y$  such that the preimage  $\pi^{-1}(U)$  is open in  $X$  (see [13, Ch. 2, §22, pp. 136]). A concrete example is given in A.8.

**Example 2.23** (Disjoint union topology). Let  $((X_i, \tau_i))_{i \in I}$  be a family of topological spaces and let  $X := \coprod_{i \in I} X_i$  be the disjoint union of the underlying sets. We define the **canonical injection** for each  $i \in I$  as

$$\begin{aligned}\iota_i : X_i &\rightarrow X \\ \iota_i(x) &= (x, i).\end{aligned}\tag{2.24}$$

The **disjoint union topology** on  $X$  is the finest topology on  $X$ , such that all canonical injections (2.24) are continuous (see [8, Ch. 1, § 3], [3, Ch. 6, § 5, Thm. 6.5.6]). For a concrete example, see A.9.

**Example 2.25** (Subspace topology). Let  $(X, \tau)$  be a topological space and  $Y \subset X$  a subset, then the topology

$$\tau|_Y := \{U \cap Y \mid U \in \tau\}$$

is the **subspace topology** of  $(X, \tau)$ . A subset of  $Y$  is open if and only if it is the intersection of  $Y$  with an open set of  $X$  (see [8, Ch. 1, § 3]). In A.10, a concrete example is given.

Another central concept is the cover. We use coverings to define the sheaves in locally ringed spaces in Chapter 7, Definition 7.1.

**Definition 2.26** (Covering). Let  $X$  be a topological space, and let  $I$  be an index set. A collection  $\mathcal{U}$  of sets  $\mathcal{U} := \{U_i \subseteq X, i \in I\}$  is called a **covering** (or **cover**) of  $X$  if

$$\bigcup_{i \in I} U_i = X$$

holds (see [13, Ch. 3, § 26, p. 164]).

## 2.3 Ring Theoretic Basics

Rings, particularly commutative rings, play a foundational role in algebraic geometry and topology. They carry local information, for example as commutative rings of continuous functions, defined on topological spaces. This allows us to bridge the gap between algebraic and geometric data. In the study of locally ringed spaces, each point in a topological space is equipped with a commutative ring that encodes properties arbitrarily "near" that point. Thus, the notion of commutative rings is indispensable for the study of locally ringed spaces. We recall a few essential definitions.

**Definition 2.27** (Ring). A **ring** is a set  $R$ , equipped with two binary operations: Addition  $+$  and multiplication  $\cdot$ , such that the following axioms are satisfied:

1.  $(R, +)$  is an Abelian group (see Definition A.11) with an additive identity element that is usually denoted by 0.
2. The multiplication  $\cdot$  is associative with a multiplicative identity element that is usually denoted as 1.
3. The distributive laws hold, in essence, for all  $a, b, c \in R$ ,

$$(a + b) \cdot c = a \cdot c + b \cdot c$$

and

$$a \cdot (b + c) = ab + ac.$$

A ring is **commutative** if the multiplication is commutative (see [1, Ch. 10, § 1, Def. 1.3]).

*Remark 2.28.* Sometimes, the definition of rings is even more general, where the existence of a multiplicative identity is not demanded (see [1, Ch. 10, § 1]). **In this thesis, only rings with a multiplicative identity shall be considered.** This is essential in the proof of Lemma 7.18, which are used to prove that a stalk is a local ring.

Units generalise the notion of multiplicative inverses in rings, which form a group.

**Definition 2.29** (Unit). Let  $R$  be a ring. A **unit** is an element  $r$  that has a multiplicative inverse, that is, there exists an element  $s$  such that

$$r \cdot s = 1.$$

The set of all units is denoted by  $R^\times$  (see [1, Ch. 10, § 1]).

Ring homomorphisms are structure-preserving maps between rings.

**Definition 2.30** (Ring homomorphism). Let  $R_1$  and  $R_2$  be two rings. A **ring homomorphism**  $\phi : R_1 \rightarrow R_2$  is a mapping that preserves the binary operations: for two arbitrary elements  $a, b \in R_1$ , the following properties hold:

$$\phi(a +_{R_1} b) = \phi(a) +_{R_2} \phi(b)$$

$$\phi(a \cdot_{R_1} b) = \phi(a) \cdot_{R_2} \phi(b)$$

(see [1, Ch. 10, § 3]).

- Remark 2.31.*
1. If we consider two rings  $R_1$  and  $R_2$  with a multiplicative identity and a ring homomorphism  $\phi : R_1 \rightarrow R_2$ , then ring homomorphisms must preserve the multiplicative identity, that is  $\phi(1_{R_1}) = 1_{R_2}$  (see [1, Ch. 10, § 3]).
  2. The (infinite) Cartesian product of a family of (commutative) rings  $(R_i)_{i \in I}$  is defined component-wise (see [1, Ch. 10, Exercise 14]).
  3. The composite of ring homomorphisms is a ring homomorphism and the composition is associative. The identity  $\text{id} : R \rightarrow R$  is a ring homomorphism (see [7, Ch. 3, § 1, Def. 1.2]).
  4. The **zero ring**  $\{0\}$  is a ring with one object. The multiplicative identity element is the same as the additive identity element, that is  $1 = 0$ . The zero ring is the only ring with  $1 = 0$  (see [1, Ch. 10, § 1]).

Ideals are central tools in ring theory. Maximal ideals, a special collection of ideals, are of extreme importance in the definition of local commutative rings.



**Definition 2.32** (Ideal). Let  $R$  be a ring. A subset  $\mathfrak{r}$  of  $R$  is called an **ideal** if the following properties hold:

1.  $\mathfrak{r}$  is an additive subgroup of  $(R, +)$
2. For every  $r \in \mathfrak{r}$  and for every  $x \in R$ , it holds that  $x \cdot r \in \mathfrak{r}$  and  $r \cdot x \in \mathfrak{r}$

(see [7, Ch. 3, § 1, Def. 1.3]).

**Definition 2.33** (Maximal ideal). Let  $R$  be a ring and let  $\mathfrak{r}$  be an ideal. We call  $\mathfrak{r}$  **maximal** if the following properties hold:

1.  $\mathfrak{r} \neq R$
2. There exists no other ideal  $\mathfrak{s}$  of  $R$  such that  $\mathfrak{r} \subsetneq \mathfrak{s} \subsetneq R$

(see [7, Ch. 3, § 3, Def. 3.2]).

**Definition 2.34** (Local ring). A commutative ring with a unique maximal ideal is called **local** (see [7, Ch. 3, Exercise 14]).

The following lemma states several equivalent conditions that characterise local rings.

**Lemma 2.35.** A ring  $R$  is a local ring if it has any of the following equivalent properties:

1.  $R$  has a unique maximal (left/right) ideal.
2.  $1 \neq 0$  and the sum of two non-units is a non-unit.
3.  $1 \neq 0$  and for an element  $x \in R$ , it holds that either  $x$  or  $1 - x$  is a unit.
4. If a finite sum is a unit, then it has a term that is a unit

(see [11, Ch. 1, Prop. 1.6] and [15, Lemma 10.18.3])

This provides a solid base of fundamental definitions and concepts we use throughout this thesis.

### 3 Introduction to Categories

This chapter introduces category theory. We begin by defining categories, diagrams, subcategories, and present illustrative examples. The chapter concludes with the essential concept of duality.

**Definition 3.1** (Category). A category  $\mathcal{C}$  consists of the following data:

(C1) A *class* of objects  $\text{Ob}(\mathcal{C})$ , whose elements are called **objects**.

(C2) Between every two objects  $A$  and  $B$  of  $\mathcal{C}$ , there exists a class (which may be empty) of **morphisms** from  $A$  to  $B$ , denoted by  $\text{Hom}_{\mathcal{C}}(A, B)$ .

For arbitrary  $A, B, C, D \in \text{Ob}(\mathcal{C})$  and arbitrary morphisms  $f : A \rightarrow B$ ,  $g : B \rightarrow C$  and  $h : C \rightarrow D$  in  $\mathcal{C}$ , the following axioms hold:

(C3) The composite  $g \circ f : A \rightarrow C$  is a morphism in  $\mathcal{C}$ .

(C4) The composition is associative, that is

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

(C5) For all  $A \in \text{Ob}(\mathcal{C})$  there exists a unique morphism  $\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$ , such that for every morphism  $f \in \text{Hom}_{\mathcal{C}}(A, B)$

$$f \circ \text{id}_A = f = \text{id}_B \circ f.$$

We call the morphism  $\text{id}_A$  the **identity** of the object  $A$

(see [3, Ch. 2, § 2, Def. 2.2.2]).

*Remark 3.2.* 1. In this thesis, a **class** is a concrete mathematical object. Informally, it is a collection of mathematical objects, which is too large to be a set

(see [3, Ch. 2, § 2, Rem. 2.2.1], [6]). The term "collection" is not a formal mathematical object, but used as a placeholder.

2. The morphisms are also called **arrows** or **maps** (see [10, Ch. 2, § 2.1, Rem. 2.1.4]).
3. Since the composition is associative, we may write  $h \circ g \circ f$  instead of  $h \circ (g \circ f)$  or  $(h \circ g) \circ f$  (see [3, Ch. 2, § 2, Rem. 2.2.4]).
4. Assume that there were two identities  $\text{id}_A$  and  $\tilde{\text{id}}_A$  of an arbitrary object  $A \in \mathcal{C}$ . Since  $\text{id}_A = \text{id}_A \circ \tilde{\text{id}}_A = \tilde{\text{id}}_A$ , the identity is unique (see [3, Ch. 2, § 2, Rem. 2.2.3]).
5. A category is called **locally small** if the class of morphisms between every two objects is a set. If the class of objects and the morphisms between every two objects are sets, we call the category **small** (see [3, Ch. 2, § 2, Rem. 2.2.6, Def. 2.2.29]). If at least one of the collections between two objects is a proper class, the category is called **large** (see [10, Ch. 2, § 1, Def. 2.1.5]).
6. One could also define categories via directed graphs or collections of objects and morphisms, like it was done in [9] or in [10]. In order to define categories via directed graphs, we may define directed graphs on a class of vertices and a class of edges (see [10]).
7. The notation  $A \in \mathcal{C}$  is often used instead of  $A \in \text{Ob}(\mathcal{C})$ . In case that it is clear which category is considered,  $\text{Hom}(A, B)$  is used instead of  $\text{Hom}_{\mathcal{C}}(A, B)$  (see [3, Ch. 2, § 2, Rem. 2.2.4]).
8. A category is called **discrete** if the only morphisms are the identities (see [3, Ch. 2, § 2, Ex. 2.2.31]).
9. A category is called **indiscrete** if between every two objects, there exists a unique morphism (see [3, Ch. 2, § 2, Ex. 2.2.31]).

A common strategy in category theory is *chasing diagrams*, that is, testing them for commutativity. We use the diagram chase for example in Chapter 5 in the proof of the Yoneda correspondence 5.1. Thus, we need to understand what a diagram is and what it means when a diagram is commutative, and what a directed diagram is.

**Definition 3.3** (Diagram). Let  $\mathcal{C}$  be a category and let  $\Gamma = (V, E)$  be a directed multigraph. A **diagram**  $X$  of shape  $\Gamma$  in  $\mathcal{C}$  consists of the following data:

1. For every vertex  $v \in V$ ,  $X(v)$  is an object in  $\mathcal{C}$ .
2. For every edge  $e := uv \in E$ ,  $X(e) : X(u) \rightarrow X(v)$  is a morphism in  $\text{Hom}_{\mathcal{C}}(X(u), X(v))$

(see [3, Ch. 2, § 4, Def. 2.4.3]).

**Definition 3.4** (Commutative diagram). Let  $\Gamma = (V, E)$  be a graph and let  $X$  be a diagram of shape  $\Gamma$  in  $\mathcal{C}$ . Let  $(e_1, \dots, e_n)$  and  $(f_1, \dots, f_m)$  be two  $u - v$ -paths in  $\Gamma$ . A diagram is called **commutative** if for every two  $u - v$ -paths

$$X(e_n) \circ \dots \circ X(e_1) = X(f_m) \circ \dots \circ X(f_1)$$

(see [3, Ch. 2, § 4, Def. 2.4.3]).

*Remark 3.5.* We can assign a diagram  $X(\mathcal{C})$  to every small category  $\mathcal{C}$ . The only difference is that diagrams have neither compositions, nor identities (see [3, Ch. 2, § 2, Def. 2.2.29]).

**Example 3.6.** We define maps  $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$  with  $f(x) := x^3$ ,  $g(x) := x - 1$  and  $h(x) := x$ . These maps can be visualised as

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{h} & \mathbb{R} \\ & \searrow f & \nearrow g \\ & \mathbb{R} & \end{array},$$

which is not commutative, since  $g \circ f \neq h$ .

In contrast if we consider  $(g \circ f)$  instead of  $h$ , then the resulting diagram would commute.

In Chapter 2, we saw the Cartesian product of directed graphs (recall Definition 2.7). Analogous, we can define the Cartesian product of diagrams. While we cannot guarantee that the Cartesian product of commutative diagrams commutes, we can name the property of those Cartesian products that do:

**Definition 3.7** (Legal diagram). A diagram  $X$  of shape  $\Gamma \otimes \Lambda$  is called **legal** if for every edge  $v \rightarrow v'$  in  $\Gamma$  and for every edge  $w \rightarrow w'$  in  $\Lambda$ , the induced diagram

$$\begin{array}{ccc} X(v, w) & \longrightarrow & X(v, w') \\ \downarrow & & \downarrow \\ X(v', w) & \longrightarrow & X(v', w') \end{array}$$

commutes. Thus, every edge  $e : v \rightarrow v'$  in  $\Gamma$  induces a morphism between diagrams  $X(v, -) \rightarrow X(v', -)$  of shape  $\Lambda$  (see [3, Ch. 6, § 6, Def. 6.6.3]).

Although diagrams look like directed graphs, they are not to be confused with directed diagrams:

**Definition 3.8** (Directed diagram). Let  $(I, \leq)$  be a directed poset (see (2.2)). It gives rise to a diagram with vertex set  $I$  and for all  $i, j \in I$  whenever  $i \leq j$ , there exists a unique morphisms  $e : i \rightarrow j$ . We call such a diagram **directed diagram**. Note that this diagram is commutative (see [3, Ch. 6, § 5, Def. 6.5.15]).

*Remark 3.9.* We see in Example 4.7 that we can regard a diagram as a functor

$$X : (I, \leq) \rightarrow \mathcal{C}.$$

We present some initial examples of categories, several of which play an important role throughout this thesis.

**Example 3.10** (Trivial categories). Two trivial categories are the empty category  $\emptyset$ , where the class of objects and therefore the class of morphisms are both empty, and the category  $\{\star\}$ , that contains just the object  $\star$  and its identity (see [3, Ch. 2, § 2, Ex. 2.2.21]).

**Example 3.11** (The category of sets). One of the most important categories is the category of sets. The objects are sets and the morphisms are maps between sets. The category is denoted by  $\mathbf{Set}$ . Note that this is a large category, since the collection of all sets is a real class (see [3, Ch. 2, § 2, Def. 2.2.7], [10, p. 15]).

**Example 3.12** (The category of posets). In the category of posets, denoted by  $\mathbf{Pos}$ , the objects are posets and the morphisms are increasing functions. In essence, consider two posets  $(M, \leq)$  and  $(N, \leq')$ . We can define a morphism

$$f : (M, \leq) \rightarrow (N, \leq')$$

such that  $f : M \rightarrow N$  is an increasing function, that is, a function such that for every  $m_1, m_2 \in M$  with  $m_1 \leq m_2$ , it follows that  $f(m_1) \leq' f(m_2)$ .

The transitivity of posets allows us to define the composite of morphisms, and the existence of the identity follows from the reflexivity of posets (see [3, Ch. 2, § 2, Ex. 2.2.15]).

Furthermore, we may consider a poset  $(I, \leq)$  itself a category, where the objects are the elements of  $I$  and there exists a morphism  $f : i \rightarrow j$  if and only if  $i \leq j$  (see [10, p. 17]).

**Example 3.13** (The category of topological spaces). In the category of topological spaces, denoted by  $\mathbf{Top}$ , we view topological spaces as objects and continuous functions as morphisms. As we saw in remark 2.17, the composite of continuous functions is continuous as well as associative. The identity is continuous. Thus, (3.1), (3.1) and (3.1) are satisfied and the topological spaces form a category (see [3, Ch. 2, § 2, Ex. 2.2.13]).

**Example 3.14** (The category of open sets). Let  $X$  be a topological space and let  $U, V \subseteq X$  be two open subsets. Whenever  $U$  is a subset of  $V$ , we define a morphism  $f : U \rightarrow V$ . Thus, the open sets in a topological space, together with  $\subseteq$  form a category, denoted by  $\mathbf{Open}(X)$  (see [3, Ch. 3, § 5, Ex. 3.5.9]).

**Example 3.15** (The category of commutative rings). We consider commutative rings (see (2.27)) with a multiplicative identity as objects and ring homomorphisms (see (2.30)) as morphisms. They form a category, since the identity is a ring homomorphism and the composite of ring homomorphisms is associative and a ring homomorphism, as we saw in remark 2.31, 3. This category is denoted by  $\mathbf{CRing}$  (see [3, Ch. 2, § 2, EX. 2.2.11]).

**Example 3.16** (The category of diagrams). Let  $\Gamma = (V, E)$  be a directed multigraph (see 2.5) and let  $\mathcal{C}$  be an arbitrary (small) category. Let  $X$  and  $Y$  be two diagrams in  $\mathcal{C}$  of shape  $\Gamma$ . We define a morphism  $f : X \rightarrow Y$ , such that the diagram

$$\begin{array}{ccc} X(v) & \xrightarrow{X(e)} & X(w) \\ \downarrow f(v) & & \downarrow f(w) \\ Y(v) & \xrightarrow{Y(e)} & Y(w) \end{array}$$

commutes. The identity is defined as  $(\text{id}_X)(v) := \text{id}_{X(v)}$ . For the composite, we take another diagram  $Z$  of shape  $\Gamma$  in  $\mathcal{C}$  and a morphism  $g : Y \rightarrow Z$  into consideration. We define the composite as

$$(g \circ f)(v) := g(v) \circ f(v).$$

In terms of a diagram, the visualisation of the composite is

$$\begin{array}{ccc} X(v) & \xrightarrow{X(e)} & X(w) \\ (g \circ f)(v) \downarrow & & \downarrow (g \circ f)(w) \\ Z(v) & \xrightarrow{Z(e)} & Z(w) \end{array}$$

and the diagram

$$\begin{array}{ccc} X(v) & \xrightarrow{X(e)} & X(w) \\ \text{id}(v) \downarrow & & \downarrow \text{id}(w) \\ X(v) & \xrightarrow{X(e)} & X(w) \end{array}$$

is the identity. Since the composite is associative by definition, the diagrams in  $\mathcal{C}$  of shape  $\Gamma$  form a category, denoted by  $\text{Diag}_\Gamma(\mathcal{C})$  (see [3, Ch. 2, § 6, Def. 2.6.18]).

Sometimes, we are not interested in the whole category, but only in certain parts of it, for example, in objects with a special property. This motivates the definition of subcategories. We make use of this property when working with sheaves and presheaves in the context of locally ringed spaces in Chapter 7.

**Definition 3.17** (Subcategory). Let  $\mathcal{C}$  be a category. The objects of a **subcategory**  $\mathcal{D}$  of  $\mathcal{C}$  consists of a subclass of objects  $\text{Ob}(\mathcal{D}) \subseteq \text{Ob}(\mathcal{C})$ , and for every two objects

$A, B \in \text{Ob}(\mathcal{D})$ , we consider a subclass of morphisms  $\text{Hom}_{\mathcal{D}}(A, B) \subseteq \text{Hom}_{\mathcal{C}}(A, B)$  such that the following properties are satisfied:

1.  $\text{id}_A \in \text{Hom}_{\mathcal{D}}(A, A)$  for all  $A \in \text{Ob}(\mathcal{D})$ .
2. For every two morphisms  $f \in \text{Hom}_{\mathcal{D}}(A, B)$ ,  $g \in \text{Hom}_{\mathcal{D}}(B, C)$ , it holds that  $g \circ f \in \text{Hom}_{\mathcal{D}}(A, C)$ .

It inherits the identities and composites of the original category  $\mathcal{C}$ . Furthermore if

$$\text{Hom}_{\mathcal{D}}(A, B) = \text{Hom}_{\mathcal{C}}(A, B)$$

holds for all  $A, B \in \text{Ob}(\mathcal{D})$ , then  $\mathcal{D}$  is called **full** (see [3, Ch. 2, § 6, Def. 2.6.1]).

**Example 3.18.** 1. As we see in Remark 7.3, 2, the category of sheaves  $\text{Sh}$  is a full subcategory of presheaves  $\text{PSh}$  (see [3, Ch. 7, § 6, Rem. 7.6.11]).

2. The discrete category is always a non-full subcategory of non-discrete categories.

A category that generalises the idea of directed sets is the filtered category. Filtered categories arise naturally when we study stalks in locally ringed spaces, since the definition of filtered limits is based on filtered categories.

**Definition 3.19** (Filtered category). A **filtered category**  $\mathcal{I}$  is a non-empty category such that the following properties are satisfied:

1. For any pair  $i, j \in \text{Ob}(\mathcal{I})$ , there exists  $k \in \text{Ob}(\mathcal{I})$  and morphisms  $i \rightarrow k$  and  $j \rightarrow k$ .
2. For any pair of parallel morphisms  $f, g : i \rightrightarrows j$  there exists a morphism  $h : j \rightarrow k$  such that  $h \circ f = h \circ g$

(see [9, p. 211]).

In some categories, there exist objects with certain properties. We use these properties in Chapter 7, where we construct the coproduct of locally ringed spaces.

**Definition 3.20** (Initial-, terminal- and zero objects). Let  $\mathcal{C}$  be a category.



1. An object  $A \in \text{Ob}(\mathcal{C})$  is **initial** if for every object  $B \in \text{Ob}(\mathcal{C})$  there exists a unique morphism  $A \rightarrow B$  in  $\mathcal{C}$ .
2. An object  $A \in \text{Ob}(\mathcal{C})$  is **terminal** if for every object  $B \in \text{Ob}(\mathcal{C})$  there exists a unique morphism  $B \rightarrow A$  in  $\mathcal{C}$ . Terminal objects are sometimes called **final**.
3. An object which is initial and terminal is a **zero object** (see [3, Ch. 2, § 5, Def. 2.5.1]).

**Example 3.21.** The zero ring  $\{0\}$  is a terminal object in the category  $\mathbf{CRing}$ . For every commutative ring  $R$ , there exists a unique ring homomorphism

$$\begin{aligned}\phi : R &\rightarrow \{0\} \\ \phi(r) &\mapsto 0 \text{ for all } r \in R\end{aligned}$$

(see [3, Ch. 2, § 5, Ex. 2.5.4, 6]).

Essential in category theory is the factorisation of a morphism. We use it for example in the formulation of the universal property of equalisers in Chapter 6.

**Definition 3.22** (Factorisation of a morphism). The **factorisation of a morphism**  $f : A \rightarrow B$  in a category  $\mathcal{C}$  is a commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ & \searrow g & \nearrow h \\ & C & \end{array} \quad (3.23)$$

Given two morphisms  $f$  and  $g$  and assuming a (unique) morphism  $h$  exists, such that  $f = h \circ g$ , we say that  $f$  **factors (uniquely) through**  $g$ , or equivalently, that  $f$  **continues along**  $g$  **to**  $h$  (see [3, Ch. 2, § 4, Def. 2.4.8]).

We are already familiar with isomorphisms in other contexts. Here, we define them categorically:

**Definition 3.24** (Isomorphism). Let  $\mathcal{C}$  be a category and let  $f \in \text{Hom}_{\mathcal{C}}(A, B)$  be a morphism. If there exists a morphism  $g \in \text{Hom}_{\mathcal{C}}(B, A)$ , such that

1.  $f \circ g = \text{id}_B$

$$2. \ g \circ f = \text{id}_A,$$

then  $f$  is called an **isomorphism**. If there exists an isomorphism between two objects  $A$  and  $B$ , the objects are isomorphic, denoted as  $A \cong B$ . The morphism  $g$  is unique:

If we assume that there exists another morphism  $h \in \text{Hom}_{\mathcal{C}}(B, A)$  such that (1) and (2) hold, we obtain

$$g = \text{id}_A \circ g = (h \circ f) \circ g = h \circ (f \circ g) = h \circ \text{id}_B = h.$$

Instead of  $g$ , we may write  $f^{-1}$  and call this morphism the **inverse** morphism of  $f$  (see [3, Ch. 2, § 3, Def. 2.3.1]).

We can ask ourselves if we have properties like surjectivity and injectivity in categories. The answer is no-not in general. But we do have a similar concept, monomorphisms and epimorphisms. We use monomorphisms in Chapter 6 in the definition of equalisers.

**Definition 3.25** (Epimorphisms and monomorphisms). Let  $f \in \text{Hom}_{\mathcal{C}}(A, B)$  and  $C \in \text{Ob}(\mathcal{C})$ .

1. A morphism  $f$  is a **monomorphism** if for all morphisms  $h_1, h_2 \in \text{Hom}_{\mathcal{C}}(C, A)$

$$f \circ h_1 = f \circ h_2 \Rightarrow h_1 = h_2.$$

We write  $f \hookrightarrow B$ .

2. A morphism  $f$  is an **epimorphism** if for all morphisms  $h_1, h_2 \in \text{Hom}_{\mathcal{C}}(B, C)$

$$h_1 \circ f = h_2 \circ f \Rightarrow h_1 = h_2.$$

We write  $f : A \twoheadrightarrow B$

(see [3, Ch. 6, § 7, Def. 6.7.1]).

*Remark 3.26.* In  $\text{Set}$ , monomorphisms are the same as injections, and epimorphisms are the same as surjections (see [3, Ch. 6, § 7, Thm. 6.7.3]).

One of the most central concepts on category theory is the duality. We see this for example in the definition of locally ringed spaces in Chapter 7, as well as in certain definitions in Chapter 6, where we see that cocones are cones in the dual category.

**Definition 3.27** (Dual category). Let  $\mathcal{C}$  be a category. The **dual category**  $\mathcal{C}^{\text{op}}$  consists of the following data:

1. The category  $\mathcal{C}^{\text{op}}$  has same objects as  $\mathcal{C}$ , that is  $\text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{C}^{\text{op}})$ .
2.  $\text{Hom}_{\mathcal{C}^{\text{op}}}(A, B) := \text{Hom}_{\mathcal{C}}(B, A)$
3. For arbitrary morphisms  $f \in \text{Hom}_{\mathcal{C}^{\text{op}}}(A, B)$ ,  $g \in \text{Hom}_{\mathcal{C}^{\text{op}}}(B, C)$ , we define the composite

$$\circ^{\text{op}} : \text{Hom}_{\mathcal{C}^{\text{op}}}(A, B) \times \text{Hom}_{\mathcal{C}^{\text{op}}}(B, C) \rightarrow \text{Hom}_{\mathcal{C}^{\text{op}}}(A, C) \quad (3.28)$$

$$\text{via } g \circ^{\text{op}} f := f \circ g.$$

Thus,  $\mathcal{C}^{\text{op}}$  is indeed a category (see [3, Ch. 2, § 6, Def. 2.6.3]).

**Example 3.29.** Recall the category  $\text{Open}(X)$  in Example 3.14. Its dual category has the same objects as the category of open sets in  $X$ , but the morphisms are reversed in  $\text{Open}(X)^{\text{op}}$ . Thus, for two open sets  $U, V \subseteq X$ , there exists a morphism  $f : U \rightarrow V$  if and only if  $U \supseteq V$  (see [3, Ch. 3, § 5, Ex. 3.5.9]).

Now that we were introduced to the foundational notations of category theory, we can go one step further and talk about categories of categories, functors, and natural transformations.

## 4 Introduction to Functors

Some categories are naturally related. Functors formalise relationships between categories and allow us to study and explore structures across different categories. As hinted in Definition 3.8, functors enable us to connect directed diagrams and small categories. Furthermore, the structure of locally ringed spaces can be naturally described by functors between the category of open sets of a topological space and the category of commutative rings. Thus, in preparation for understanding locally ringed spaces and structure sheaves, we have to understand functors and natural transformations.

### 4.1 Functors

In categorical terms, functors can be thought of as maps between categories. They allow us to "translate" data, such as objects and morphisms, from one category to another category. While some functors—like the forgetful functor—discard parts of the structure, other functors preserve it.

**Definition 4.1 (Functor).** Let  $\mathcal{C}$  and  $\mathcal{D}$  be two categories. A **functor**  $F$  is a morphism  $F : \mathcal{C} \rightarrow \mathcal{D}$  such that the following properties are satisfied:

- (F1) For every  $A \in \text{Ob}(\mathcal{C})$  there exists an object  $F(A) \in \text{Ob}(\mathcal{D})$ .
- (F2) For any  $f \in \text{Hom}_{\mathcal{C}}(A, B)$  there exists a morphism  $F(f) \in \text{Hom}_{\mathcal{D}}(F(A), F(B))$ .
- (F3) For every object  $A \in \text{Ob}(\mathcal{C})$ , it holds that  $F(\text{id}_A) = \text{id}_{F(A)}$ .
- (F4) For any two morphisms  $f \in \text{Hom}_{\mathcal{C}}(A, B)$ ,  $g \in \text{Hom}_{\mathcal{C}}(B, C)$ , the equality  $F(g \circ f) = F(g) \circ F(f)$  holds

(see [3, Ch. 3, § 2, Def. 3.2.1]).

- Remark 4.2.** 1. A functor defined like in Definition 4.1 is called **covariant** (see [3, Ch. 3, § 2, Rem. 3.2.3]).
2. A **contravariant** functor is the dual notion to a covariant functor, that is,  $F : \mathcal{C} \rightarrow \mathcal{D}$  is the same as a covariant functor  $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ , such that the composite and the identities are preserved, but the morphisms are reversed in direction (see [3, Ch. 3, § 2, Rem. 3.2.3]).
3. An essential property of functors is that they preserve the commutativity of diagrams (see [3, Ch. 3, § 2, Rem. 3.2.2]).

In the following, we see illustrating examples of functors.

**Example 4.3** (Identity functor). The **identity functor**  $\text{id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$  with the following data:

- $\text{id}_{\mathcal{C}}(A) := A$  for all  $A \in \text{Ob}(\mathcal{C})$ .
- $\text{id}_{\mathcal{C}}(f) := f$  for all morphisms  $f$  in  $\mathcal{C}$

(see [3, Ch. 3, § 2, Ex. 3.2.21, 1]).

**Example 4.4** (Constant functor). Let  $T$  be an arbitrary object in  $\text{Ob}(\mathcal{D})$ . For every category  $\mathcal{C}$  the **constant functor**  $\Delta(T) : \mathcal{C} \rightarrow \mathcal{D}$  exists with the following morphisms:

- $\Delta(T)(A) := T$  for all  $A \in \text{Ob}(\mathcal{C})$ .
- $\Delta(T)(f) := \text{id}_T$  for all morphisms in  $\mathcal{C}$

(see [3, Ch. 3, § 2, Ex. 3.2.21, 2]).

**Example 4.5.** For every category  $\mathcal{C}$ , there exist unique functors  $\emptyset \rightarrow \mathcal{C}$  and  $\mathcal{C} \rightarrow \{\star\}$  (recall Example 3.10) (see [3, Ch. 3, § 2, Ex. 3.2.21, 3]).

**Example 4.6** (Inclusion functor). Let  $\mathcal{D}$  be a full subcategory of  $\mathcal{C}$ . The **inclusion functor**  $\iota : \mathcal{D} \rightarrow \mathcal{C}$  is defined via

$$\begin{aligned}\iota(A) &:= A \\ \iota(f) &:= f\end{aligned}$$

for every  $A \in \text{Ob}(\mathcal{C})$  and every morphism  $f \in \mathcal{C}$  (see [10, Ch. 5, § 5.4]).

**Example 4.7** (Diagrams as functors). As mentioned, functors allow us to understand a directed diagram  $X$  of shape  $\Gamma$  in a category  $\mathcal{C}$  as a functor

$$X : (I, \leq) \rightarrow \mathcal{C},$$

where  $(I, \leq)$  is a directed poset. For every  $i \in I$ ,  $X(i)$  is an object in  $\mathcal{C}$ , and  $X(i \leq j)$  gives rise to a morphism  $X(i) \rightarrow X(j)$  in  $\mathcal{C}$ .

The functoriality of  $X$  ensures that the identity is preserved, that is,

$$X(i \leq i) = \text{id}_{X(i)},$$

as well as the composite,

$$X(i \leq k) = X(j \leq k) \circ X(i \leq j)$$

whenever  $i \leq j \leq k$  in  $(I, \leq)$  (see [10, Ch. 4, § 4.1, Def. 4.1.1]).

Functors have a composition operation, like morphisms do. Although locally ringed spaces form a category, constructions involving them often rely on closely related categories and therefore on functors. When we construct colimits on locally ringed spaces in Chapter 7, we are interested in the composite of functors. Furthermore, we are interested in adjoint functors, that rely on the horizontal composite.

**Definition 4.8** (Composition of functors). For arbitrary functors  $F : \mathcal{C} \rightarrow \mathcal{D}$  and  $G : \mathcal{D} \rightarrow \mathcal{E}$ , the **composition of functors**  $G \circ F : \mathcal{C} \rightarrow \mathcal{E}$  is defined by

1.  $(G \circ F)(A) := G(F(A))$  for all  $A \in \text{Ob}(\mathcal{C})$ .
2.  $(G \circ F)(f) := G(F(f))$  for all morphisms  $f$  in  $\mathcal{C}$

(see [3, Ch. 3, § 3, Def. 3.3.1]).

*Remark 4.9.* We can show that the composite is a functor:

For every object  $A \in \text{Ob}(\mathcal{C})$ , it holds that  $G(F(A)) \in \text{Ob}(\mathcal{E})$  (by Definition 4.1).

Let  $f \in \text{Hom}_{\mathcal{C}}(A, B)$ . Then  $(G \circ F)(f) \in \text{Hom}_{\mathcal{E}}((G \circ F)(A), (G \circ F)(B))$ . Thus, composites of functors preserve identities as  $(G \circ F)(\text{id}_A) = \text{id}_{(G \circ F)(A)}$ .

For  $g \in \text{Hom}_{\mathcal{C}}(B, C)$ , the composition gives rise to

$$\begin{aligned}
 (G \circ F)(g \circ f) &\stackrel{4.8}{=} (G(F(g \circ f))) \\
 &\stackrel{4.1}{=} G(F(g) \circ F(f)) \\
 &\stackrel{4.1}{=} G(F(g)) \circ G(F(f)) \\
 &= (G \circ F)(g) \circ (G \circ F)(f).
 \end{aligned}$$

Thus, the composite of functors is a functor. Furthermore, it allows us to define the category of small categories  $\text{Cat}$ . The objects are all small categories and the morphisms are the functors.

The definition of isomorphisms in categories (see Definition 3.24) allows us to compare small categories (see [3, Ch. 3, § 3, Def. 3.3.1]).

## 4.2 Natural Transformations

In order to understand limits and colimits, we have to understand universal properties and the Yoneda correspondence in Chapter 5. It relies on natural transformation, that is a transformation of one functor into another functor, such that it is intrinsic to the relationship of the functors, reflecting the structure of the categories involved.

**Definition 4.10** (Natural transformation). Let  $F, G : \mathcal{C} \rightarrow \mathcal{D}$  be functors. We define a morphism  $\alpha : F \rightarrow G$ , such that the following properties are satisfied:

1. For every  $A \in \text{Ob}(\mathcal{C})$ , it holds that  $\alpha(A) : F(A) \rightarrow G(A)$ .
2. For every morphism  $f \in \text{Hom}_{\mathcal{C}}(A, B)$ , the diagram

$$\begin{array}{ccc}
 F(A) & \xrightarrow{\alpha(A)} & G(A) \\
 F(f) \downarrow & & \downarrow G(f) \\
 F(B) & \xrightarrow{\alpha(B)} & G(B)
 \end{array} \tag{4.11}$$

commutes.

A morphism with these properties is called **natural transformation**. The commutativity of this diagram is called **naturality**. For  $A \in \text{Ob}(\mathcal{C})$ ,

$$\alpha(A) : F(A) \rightarrow G(A)$$

are called **components**.

We can define the **identity natural transformation**  $\text{id}_F : F \rightarrow F$  via

$$\text{id}_F(A) = \text{id}_{F(A)}$$

for all  $A \in \text{Ob}(\mathcal{C})$ . This identity is a natural transformation (see [3, Ch. 3, § 4, Def. 3.4.1]).

We have thus introduced morphisms between objects, morphisms between categories (functors) and morphisms between functors (natural transformations). We know of the composition of morphisms inside one category (see Definition 3.1) and the composite of functors. Not only the composition of natural transformations can be defined, but also the composition of natural transformations with functors. We use the composition of natural transformations for example in the construction of locally ringed spaces in Chapter 7.

**Definition 4.12** (Vertical composition). Let  $F, G, H : \mathcal{C} \rightarrow \mathcal{D}$  be functors. Let  $\alpha : F \rightarrow G$  and  $\beta : G \rightarrow H$  be natural transformations. The composition of natural transformations  $\beta \circ \alpha : F \rightarrow H$  is defined as

$$(\beta \circ \alpha)(A) = \beta(A) \circ \alpha(A) : F(A) \rightarrow H(A)$$

for all  $A \in \text{Ob}(\mathcal{C})$ . This composition is called **vertical composition**. The diagrams the vertical composition in the following diagram:

$$\begin{array}{ccccc} & & F(A) & \xrightarrow{F(f)} & F(B) \\ & & \alpha(A) \downarrow & & \downarrow \alpha(B) \\ A & \xrightarrow{f} & G(A) & \xrightarrow{G(f)} & G(B) \\ \downarrow f & \sim & \downarrow \beta(A) & & \downarrow \beta(B) \\ B & & H(A) & \xrightarrow{H(f)} & H(B) \end{array}$$



(see [3, Ch. 3, § 5, Def. 3.5.2]).

**Definition 4.13** (Horizontal composition). Let  $F, G : \mathcal{C} \rightarrow \mathcal{D}$  be functors and let  $\alpha : F \rightarrow G$  be a natural transformation.

1. Let  $H : \mathcal{D} \rightarrow \mathcal{E}$  be another functor. Then the **horizontal composition from the left** is defined as

$$(H \bullet \alpha) : (H \circ F) \rightarrow (H \circ G)$$

via

$$(H \bullet \alpha)(A) := H(\alpha(A))$$

for all  $A \in \text{Ob}(\mathcal{C})$ . The diagrams

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\quad F \quad} & \mathcal{D} \\ \downarrow \alpha & & \downarrow \alpha \\ \mathcal{C} & \xrightarrow{\quad G \quad} & \mathcal{D} \end{array} \xrightarrow{H} \mathcal{E} \sim \begin{array}{ccc} \mathcal{C} & \xrightarrow{\quad H \circ F \quad} & \mathcal{E} \\ \downarrow H \circ \alpha & & \downarrow H \circ \alpha \\ \mathcal{C} & \xrightarrow{\quad H \circ G \quad} & \mathcal{E} \end{array}$$

visualises the horizontal composition from the left.

2. Let  $K : \mathcal{E} \rightarrow \mathcal{C}$  be another functor. Then the **horizontal composition from the right** is defined as

$$(\alpha \bullet K) : (F \circ K) \rightarrow (G \circ K)$$

via

$$(\alpha \bullet K)(A) := \alpha(K(A))$$

for all  $A \in \text{Ob}(\mathcal{E})$ . The diagrams

$$\mathcal{E} \xrightarrow{K} \begin{array}{ccc} \mathcal{C} & \xrightarrow{\quad F \quad} & \mathcal{D} \\ \downarrow \alpha & & \downarrow \alpha \\ \mathcal{C} & \xrightarrow{\quad G \quad} & \mathcal{D} \end{array} \sim \begin{array}{ccc} \mathcal{E} & \xrightarrow{\quad F \circ K \quad} & \mathcal{D} \\ \downarrow \alpha \circ K & & \downarrow \alpha \circ K \\ \mathcal{E} & \xrightarrow{\quad G \circ K \quad} & \mathcal{D} \end{array}$$

visualise the horizontal composition from the right.

We may combine the horizontal composition from the left and the horizontal com-

position from the right:

$$(H \bullet (\alpha \bullet K)) = ((H \bullet \alpha) \bullet K) : (H \circ F \circ K) \rightarrow (H \circ G \circ K).$$

Furthermore, for functors  $\tilde{F}, \tilde{G} : \mathcal{D} \rightarrow \mathcal{E}$  and a natural transformation  $\beta : \tilde{F} \rightarrow \tilde{G}$  the diagram

$$\begin{array}{ccc} \tilde{F} \circ F & \xrightarrow{\tilde{F} \bullet \alpha} & \tilde{F} \circ G \\ \beta \bullet F \downarrow & & \downarrow \beta \bullet G \\ \tilde{G} \circ F & \xrightarrow{\tilde{G} \bullet \alpha} & \tilde{G} \circ G \end{array} \quad (4.14)$$

commutes. Thus, the horizontal composition of natural transformations is defined as

$$\alpha \bullet \beta : (\tilde{F} \circ F) \rightarrow (\tilde{G} \circ G).$$

This is exactly the **diagonal** of the diagram 4.14, that is

$$\alpha \bullet \beta := (\beta \bullet G) \circ (\tilde{F} \bullet \alpha) = (\tilde{G} \bullet \alpha) \circ (\beta \bullet F).$$

This leads to the following visualisation:

$$\begin{array}{ccccc} \mathcal{C} & \xrightarrow{F} & \mathcal{D} & \xrightarrow{\tilde{F}} & \mathcal{E} \\ & \searrow \alpha & & \searrow \beta & \\ & G & & \tilde{G} & \\ & \nearrow & & \nearrow & \\ \mathcal{C} & \xrightarrow{G} & \mathcal{D} & \xrightarrow{\tilde{G}} & \mathcal{E} \end{array} \rightsquigarrow \begin{array}{ccc} \mathcal{C} & \xrightarrow{\tilde{F} \circ F} & \mathcal{E} \\ & \searrow \beta \bullet \alpha & \\ & \tilde{G} \circ G & \\ & \nearrow & \\ \mathcal{C} & \xrightarrow{\tilde{G} \circ G} & \mathcal{E} \end{array}$$

(see [3, Ch. 3, § 5, Def. 3.5.10 - Def. 3.5.12]).

*Remark 4.15.* The composition of natural transformations allows us to define the following category: Let  $\mathcal{C}$  be a small category, and let  $\mathcal{D}$  be an arbitrary category. We may consider the functors  $\mathcal{C} \rightarrow \mathcal{D}$  as objects and the natural transformations as morphisms.

The identity was already defined in Definition 4.10 and the composite is defined as in Definition 4.12. We denote the category of functors by  $\underline{\text{Hom}}_{\text{Cat}}(\mathcal{C}, \mathcal{D})$  (see [3, Ch. 3, § 5, Def. 3.5.5], [10, Ch. 11, § 3, Def. 11.3.1]).

The composition of natural transformations motivates isomorphisms between functors. We use the isomorphism of functors in Chapter 5, which we use in the defini-

tion of representable functors (5.5).

**Definition 4.16** (Natural isomorphism). A natural transformation  $\alpha : F \rightarrow G$  is a **natural isomorphism** between functors if all components  $\alpha(A) : F(A) \rightarrow G(A)$  are isomorphisms (see [3, Ch. 3, § 5, Def. 3.5.3]).

Objects in categories can be highly structured and complex, making the manipulation non-trivial. There are functors, that allow us to lift these objects into another category, where things become simpler. An example are locally ringed spaces, that can be lifted into the more general category of ringed spaces, where we can forget about the locality property.

**Definition 4.17** (Adjunction). Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  and  $G : \mathcal{D} \rightarrow \mathcal{C}$  be functors. An **adjunction between  $F$  and  $G$**  consists of two morphisms

$$\begin{aligned}\eta : \text{id}_{\mathcal{C}} &\rightarrow G \circ F \\ \varepsilon : F \circ G &\rightarrow \text{id}_{\mathcal{D}}\end{aligned}$$

such that the triangle identities

$$\begin{array}{ccc} F \circ \text{id}_{\mathcal{C}} & \xrightarrow{F \bullet \eta} & F \circ G \circ F \\ & \searrow & \downarrow \varepsilon \bullet F \\ & & \text{id}_{\mathcal{D}} \circ F \end{array} \quad \begin{array}{ccc} \text{id}_{\mathcal{C}} \circ G & \xrightarrow{\eta \bullet G} & G \circ F \circ G \\ & \searrow & \downarrow G \bullet \varepsilon \\ & & G \circ \text{id}_{\mathcal{D}} \end{array}$$

hold. An adjunction  $(F, G, \eta, \varepsilon) : \mathcal{C} \rightarrow \mathcal{D}$  between  $F$  and  $G$  is called **left adjoint** or we say that  $G$  **is right adjoint to  $F$** ; in signs,  $F \dashv G$ .

We call  $\eta$  **unit** and  $\varepsilon$  is called **co-unit** of the adjunction (see [3, Ch. 7, § 2, Def. 7.2.2]).

When we simplify a category, for example once we simplified a construction of locally ringed spaces to be a construction of the simpler ringed spaces, we are interested in the closest object in the original category. The reflector is a special functor, that allows us to "reverse" a simplification of a category.

**Definition 4.18** (Reflector). Let  $\mathcal{C}$  be a category. A full subcategory  $\mathcal{D}$  of  $\mathcal{C}$  is called **reflective** if there exists a left adjoint functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  for the inclusion functor  $G : \mathcal{D} \hookrightarrow \mathcal{C}$ . The functor  $F$  is called **reflector** (see [3, Ch. 7, § 6, Def. 7.6.1]).

**Remark 4.19** (Co-reflector). A full subcategory is called **coreflective** if the inclusion functor has a right adjoint functor. We call it **co-reflector** (see [3, Ch. 7, § 6, Def. 7.6.1]).

The category of sheaves is coreflexive, which motivates the sheafification of pre-sheaves (see [3, Ch. 7, § 6, Rem. 7.6.11]).

**Definition 4.20** (Presheaf). A **presheaf**  $F$  is a functor

$$F : \text{Open}(X)^{\text{op}} \rightarrow \text{Set}.$$

They form the category of presheaves, denoted by  $\text{PSh}(X)$ . Let  $U$  be an open subset in  $X$ . The elements  $s \in F(U)$  are called **sections over**  $U$ .

For any inclusion  $U \subseteq V$ , the corresponding morphism  $V \rightarrow U$  induces a **restriction**  $F(V) \rightarrow F(U)$ , denoted by

$$s \mapsto s|_U$$

(see [3, Ch. 3, § 5, Rem. 3.5.9]).

**Remark 4.21.** 1. Since presheaves are functors, they satisfy the following properties:

- a)  $F(U) \rightarrow F(U)$  is the identity  $\text{id}_{F(U)}$ .
- b) For any inclusion chain  $U \subseteq V \subseteq W$ , the diagram

$$\begin{array}{ccc} F(W) & \xrightarrow{\quad} & F(U) \\ & \searrow & \nearrow \\ & F(V) & \end{array}$$

commutes (see [3, Ch. 3, § 5, Rem. 3.5.9]).

- 2. The morphisms between presheaves are natural transformations. That is, any natural transformation between two presheaves  $F \rightarrow G$  consists of a family

of morphisms  $F(U) \rightarrow G(U)$ , such that for any inclusion  $U \subseteq V$ , the diagram

$$\begin{array}{ccc} F(V) & \longrightarrow & G(V) \\ \downarrow & & \downarrow \\ F(U) & \longrightarrow & G(U) \end{array}$$

commutes (see [3, Ch. 3, § 5, Rem. 3.5.9]).

3. We can replace the category  $\mathbf{Set}$  with an arbitrary algebraic category  $\mathcal{A}$ , like  $\mathbf{CRing}$  (see B). Then  $\mathbf{PSh}(X, \mathcal{A})$  is the category of  $\mathcal{A}$  **valued presheaves** (see [3, Ch. 3, § 5, Rem. 3.5.9]).
4. As we see in Chapter 6, the category  $\mathbf{PSh}$  has all limits and colimits. Thus, it is complete and cocomplete.

Before we can do any constructions with locally ringed spaces, we have to understand (co-)limits and the universal property. The Yoneda-construction in Chapter 5 builds on functors and the natural transformations. One of the functors we need is the so-called  $\mathbf{Hom}$ -functor.

**Definition 4.22** ( $\mathbf{Hom}$ -functor). Let  $\mathcal{C}$  be a locally small category and let  $A \in \mathcal{C}$  be an arbitrary but fixed object.

We assign to every object  $B \in \mathbf{Ob}(\mathcal{C})$  the *set* of morphisms

$$B \mapsto \mathbf{Hom}_{\mathcal{C}}(A, B)$$

and to every morphism  $f : B \rightarrow C$  in  $\mathcal{C}$ , we assign the morphism

$$\begin{aligned} f_* : \mathbf{Hom}_{\mathcal{C}}(A, B) &\rightarrow \mathbf{Hom}_{\mathcal{C}}(A, C) \\ g &\mapsto f \circ g \end{aligned} \tag{4.23}$$

for any  $g \in \mathbf{Hom}_{\mathcal{C}}(A, B)$ . The axioms (C1)-(C5) of a category (see Definition 3.1) ensure, that the properties of functors (F1)-(F4) (see Definition 4.1) hold:

1.  $(\mathrm{id}_B)_* = \mathrm{id}_{\mathbf{Hom}_{\mathcal{C}}(A, B)}$ , since  $\mathrm{id}_B \circ h = h$  for any  $h \in \mathbf{Hom}_{\mathcal{C}}(A, B)$ .

2.  $(f \circ g)_* = f_* \circ g_*$ , since

$$\begin{aligned} (f \circ g)_*(h) &= (f \circ g) \circ h \\ &= f \circ (g \circ h) \\ &= f_*(g \circ h) \\ &= f_*(g_*(h)) \\ &= (f_* \circ g_*)(h). \end{aligned}$$

Thus, we can define the **covariant** Hom **functor**

$$\mathrm{Hom}_{\mathcal{C}}(A, -) : \mathcal{C} \rightarrow \mathrm{Set}.$$

Applying the very same construction in the opposite category  $\mathcal{C}^{\mathrm{op}}$  yields the **contravariant** Hom-**functor**

$$\mathrm{Hom}(-, A) := \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}} : \mathcal{C}^{\mathrm{op}} \rightarrow \mathrm{Set}.$$

For any  $B \in \mathrm{Ob}(\mathcal{C})$  and an arbitrary morphism  $f : C \rightarrow B$ , we obtain morphisms such that

$$\begin{aligned} B &\mapsto \mathrm{Hom}_{\mathcal{C}}(B, A) \\ f^* &: \mathrm{Hom}_{\mathcal{C}}(C, A) \rightarrow \mathrm{Hom}_{\mathcal{C}}(B, A) \\ h &\mapsto h \circ f. \end{aligned}$$

For the contravariant Hom-functor, the following properties hold:

1.  $\mathrm{id}^* = \mathrm{id}$
2.  $(f \circ g)^* = g^* \circ f^*$

for all morphisms  $g \in \mathrm{Hom}_{\mathcal{C}}(A, B)$  (see [3, Ch. 5, § 2, Def. 5.2.1]).

*Remark 4.24.* We may even define the Hom-functor in "two variables":

$$\mathrm{Hom}_{\mathcal{C}}(-, -) : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \rightarrow \mathrm{Set}$$

(see [3, Ch. 5, § 2, Def. 5.2.1]).

We can ask ourselves if and how a  $\text{Hom}$ -functor is determined by the fixed element  $A$  and if we can reconstruct it. This leads to a very central lemma of category theory. In fact, this lemma is so powerful, that it deserves its own chapter (see [3, Ch. 5, § 2, Def. 5.2.1]).

## 5 Universal Properties and the Yoneda Lemma

Universal properties are one of the fundamental concepts in category theory. They characterise key constructions like cones and limits, as well as their dual constructions, cocones and colimits. The Yoneda Lemma explains why universal properties work, by defining a natural isomorphism between natural transformations and elements of a functor's codomain. This is the core idea behind the representability of functors. To fully understand universal properties, we study the Yoneda Construction and the Yoneda Lemma first. The term "Yoneda Construction" follows the source [3].

**Construction 5.1.** (Yoneda Construction) Let  $\mathcal{C}$  be a category. Given a functor  $F : \mathcal{C} \rightarrow \mathbf{Set}$  and  $A \in \mathbf{Ob}(\mathcal{C})$ , there is a correspondence between natural transformations

$$\alpha : \mathbf{Hom}(A, -) \rightarrow F$$

and elements  $u \in F(A)$ . In particular, any such natural transformation has a component at  $A$ ,

$$\alpha(A) : \mathbf{Hom}(A, A) \rightarrow F(A).$$

Moreover, any natural transformation  $\alpha$  gives rise to an element

$$u_\alpha := (\alpha(A))(\mathrm{id}_A) \in F(A). \tag{5.2}$$

On the other hand, any element  $u \in F(A)$  gives rise to a natural transformation

$$\begin{aligned} \alpha_u : \mathbf{Hom}(A, -) &\rightarrow F \\ (\alpha_u(B))(f) &:= (F(f))(u) \end{aligned} \tag{5.3}$$



with  $f \in \text{Hom}_{\mathcal{C}}(A, B)$  (see [3, Ch. 5, § 2, Def. 5.2.4]).

*Proof.* To show that  $\alpha_u$  is a natural transformation, it suffices to prove that the naturality diagram (see Definition 4.10, Diagram 4.11)

$$\begin{array}{ccc} \text{Hom}(A, B) & \xrightarrow{\alpha_u(B)} & F(B) \\ g_* \downarrow & & \downarrow F(g) \\ \text{Hom}(A, C) & \xrightarrow{\alpha_u(C)} & F(C) \end{array}$$

commutes for every morphism  $g : B \rightarrow C$ .

We consider a morphism  $g \in \text{Hom}_{\mathcal{C}}(B, C)$ . In Definition 4.22 we defined the covariant Hom-functor  $g_* : \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{C}}(A, C)$ . For every morphism  $f \in \text{Hom}_{\mathcal{C}}(A, B)$  it holds that

$$\begin{aligned} (F(g) \circ \alpha_u(B))(f) &\stackrel{5.3}{=} F(g \circ f)(u) \\ &\stackrel{5.3}{=} \alpha_u(C)(g \circ f) \\ &\stackrel{4.22}{=} (\alpha_u(C) \circ g_*)(f). \end{aligned}$$

Thus, the diagram commutes and  $\alpha_u$  is a natural transformation (see [3, Ch. 5, § 2, Def. 5.2.4]).  $\square$

The Yoneda Lemma shows how  $\alpha_u$  and  $u_\alpha$  in the Yoneda Construction are related. This is crucial when we define the universal property in Lemma 5.6.

**Lemma 5.4** (Yoneda Lemma). Let  $\mathcal{C}$  be a category,  $A \in \text{Ob}(\mathcal{C})$ , and let  $F : \mathcal{C} \rightarrow \text{Set}$  be a functor. Let  $\alpha_u$  be like in (5.3) and  $u_\alpha$  like in (5.2).

The correspondences  $u \mapsto \alpha_u$  and  $\alpha \mapsto u_\alpha$  define mutually inverse bijections, that is,

$$\text{Hom}(\text{Hom}(A, -), F) \cong F(A).$$

Here,  $\text{Hom}(\text{Hom}(A, -), F)$  denotes the set of natural transformations between the functors  $\text{Hom}(A, -)$  and  $F$  (see [3, Ch. 5, § 2, Lemma 5.2.5]).

*Proof.* Let  $u \in F(A)$ . By Definition (5.2)

$$u_{\alpha_u} = \alpha_u(A)(\text{id}_A) = F(\text{id}_A)(u) = \text{id}_{F(A)}(u) = u$$

holds. Let  $\alpha : \text{Hom}(A, -) \rightarrow F$  be a natural transformation. By definition of the naturality diagram 4.11,

$$\begin{array}{ccc} \text{Hom}(A, A) & \xrightarrow{\alpha(A)} & F(A) \\ \downarrow f_* & & \downarrow F(f) \\ \text{Hom}(A, B) & \xrightarrow{\alpha(B)} & F(B) \end{array}$$

commutes for all  $f \in \text{Hom}_{\mathcal{C}}(A, B)$ . Evaluating the diagram at  $\text{id}_A$  gives us

$$\begin{aligned} \alpha(B)(f) &= \alpha(B)(f_*(\text{id}_A)) \\ &\stackrel{\text{commutativity of diagram}}{=} F(f)(\alpha(A)(\text{id}_A)) \\ &= F(f)(u_\alpha) \\ &= \alpha_{u_\alpha}(B)(f). \end{aligned}$$

Thus,  $\alpha$  and  $\alpha_{u_\alpha}$  have the same components for every morphism  $f \in \text{Hom}_{\mathcal{C}}(A, B)$ , which makes them equal as natural transformations by definition. Hence  $\alpha = \alpha_{u_\alpha}$  (see [3, Ch. 5, § 2, Lemma 5.2.5]).  $\square$

Before we can understand universal elements and universal properties, we need one final definition, the representability of functors.

**Definition 5.5** (Representation). A functor  $F : \mathcal{C} \rightarrow \text{Set}$  is called **representable** if there exists an  $A \in \text{Ob}(\mathcal{C})$  and an isomorphism

$$\alpha : \text{Hom}(A, -) \cong F.$$

Then  $A$  is called **representing object** of  $F$  (and  $F$  is represented by  $A$ ). The pair  $(A, \alpha)$  is the **representation of  $F$**  (see [3, Ch. 5, § 2, Def 5.2.12]).

**Lemma and Definition 5.6** (Universal property). A functor  $F : \mathcal{C} \rightarrow \text{Set}$  is representable if and only if there exists an object  $A$  and an element  $u \in F(A)$  such that for every  $B \in \text{Ob}(\mathcal{C})$  and  $v \in F(B)$  there exists a **unique** morphism  $f : A \rightarrow B$  with  $v = F(f)(u)$ . We call such an element  $u$  the **universal element** of  $F$  and the property **universal property** (see [3, Ch. 5, § 2, Def. 5.2.14]).

*Proof.* " $\Rightarrow$ ": Let  $(A, \alpha_u)$  be a representation of  $F$ , such that

$$\alpha_u : \text{Hom}(A, -) \cong F$$

is a natural transformation, and let

$$u := \alpha_u(A)(\text{id}_A) \in F(A).$$

Then for any  $B \in \text{Ob}(\mathcal{C})$  and  $v \in F(B)$ , we define

$$f := \alpha_u(B)^{-1}(v).$$

Since  $\alpha_u$  is a natural isomorphism, it is invertible. In particular,  $f$  exists and is unique. By naturality,

$$v = \alpha_u(B)(f) = F(f)(\alpha_u(A)(\text{id}_A)) = F(f)(u).$$

Hence,  $u$  is a universal element.

" $\Leftarrow$ ": Assume  $u_\alpha \in F(A)$  has the universal property. Then we can define a natural transformation

$$\begin{aligned} \alpha_u : \text{Hom}(A, -) &\rightarrow F \\ \alpha_u(B)(f) &:= F(f)(u). \end{aligned}$$

By assumption, for every  $v \in F(B)$ , there exists a unique morphism  $f \in \text{Hom}_{\mathcal{C}}(A, B)$  such that  $F(f)(u) = v$ . Thus,  $\alpha_u(B)$  is bijective for all  $B$ . We saw the naturality in the proof of the Yoneda correspondence 5.1. Hence,  $F$  can be represented by  $(A, \alpha_u)$  (see [3, Ch. 5, § 2, Def. 5.2.14]).  $\square$

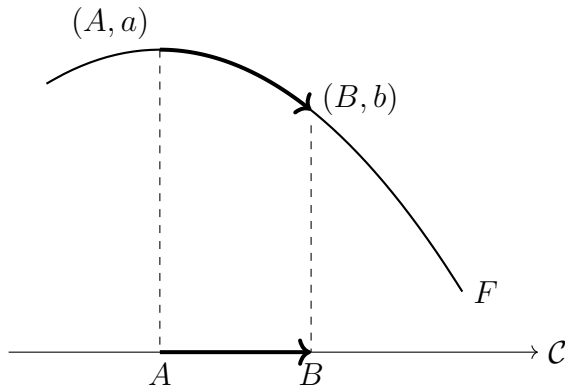
*Remark 5.7.* Let  $F : \mathcal{C} \rightarrow \text{Set}$  be a functor. We define the category of elements of  $F$ , denoted by  $\int F$ , as follows:

The objects  $\text{Ob}(\int F)$  are pairs  $(A, a)$ , with  $a \in F(A)$  and  $A \in \text{Ob}(\mathcal{C})$ . The morphisms  $(A, a) \rightarrow (B, b)$  are those morphisms  $f \in \text{Hom}(A, B)$  such that

$$F(f)(a) = b.$$

The universal element of  $F$  is the same as the initial object in  $\int F$ . We can visualise

this as follows:



The notation  $\int F$  reflects the idea of integrating over all elements of  $F$ , indexed by the objects of the category  $\mathcal{C}$  (see [3, Ch. 5, § 2, Def. 5.2.12]).

The universal element allows us to formulate a criterion for the representability of functors, which allows us to introduce limits as the representation of a functor in Chapter 6

**Corollary 5.8** (Criterion for representability). Let  $F : \mathcal{C} \rightarrow \mathbf{Set}$  be a functor.  $F$  is **representable** if and only if  $F$  has a universal element (see [3, Ch. 5, § 2, Kor. 5.2.15]).

*Proof.* This follows directly from the characterisation of representability in terms of universal elements (see Definition 5.5). □

We saw that representability of a functor is characterised entirely by the existence of a universal element. This allows us to introduce limits and colimits in the next chapter.

## 6 Introduction to Limits and Colimits

In this chapter, we introduce limits and colimits, which are an essential tool in category theory. We begin with the definition of limits and continue with exploring special cases of limits, namely products and equalisers. Equalisers play a crucial role, both in defining sheaves—and consequently in defining locally ringed spaces—and in the construction of limits. Furthermore, we discuss the dual concepts, as well as certain properties, like the commutativity of finite limits and directed colimits. The concrete construction of limits and colimits is illustrated with examples, some of which are used in the construction of colimits in the category of locally ringed spaces in Chapter 7.

### 6.1 Limits

Limits can be introduced as the representation of a functor we saw in Definition 5.5 in Chapter 5, or alternatively, as universal cone. Both definitions are mathematically equivalent; in this definition, we use the representability and see the definition using the universal cone in the remarks.

**Definition 6.1 (Cone).** Let  $\mathcal{C}$  be a category and let  $\Gamma = (V, E)$  be a directed graph. Let  $X \in \text{Diag}_\Gamma(\mathcal{C})$  (not necessarily commutative) and  $T \in \text{Ob}(\mathcal{C})$  arbitrary. A **cone with tip  $T$  to  $X$**  is a family of morphisms

$$(\text{pr}_v : T \rightarrow X(v))_{v \in V}$$

such that for all directed edges  $e : v \rightarrow w$  in  $X$ , the diagram

$$\begin{array}{ccc}
 & T & \\
 \text{pr}_v \swarrow & & \searrow \text{pr}_w \\
 X(v) & \xrightarrow{X(e)} & X(w)
 \end{array} \tag{6.2}$$

commutes. The set of cones with tips  $T$  to  $X$  is denoted by  $\text{Hom}(\Delta(T), X)$  (see [3, Ch. 6, § 2, Def. 6.2.1]).

**Remark 6.3.** 1. An alternative definition is provided via **constant diagrams**:

Let  $\Delta(T)$  be the constant diagram with  $\Delta(T)(v) = T$  for all  $v \in V$  and  $\Delta(T)(e) := \text{id}_T$  for every  $e \in E$ . Then  $\text{Hom}(\Delta(-), X)$  is the set of cones with tip  $T$  to  $X$ . The functor

$$\Delta^{\text{op}} : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$$

is the composite of the constant functor  $\Delta^{\text{op}} : \mathcal{C}^{\text{op}} \rightarrow \text{Diag}_{\Gamma}(\mathcal{C})^{\text{op}}$  and the Hom-functor  $\text{Hom}(-, X) : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$  (see [3, Ch. 6, § 2, Def. 6.2.1]).

2. Consider a cone with tip  $T'$  and a morphism  $T \rightarrow T'$ . Then there exists a cone with tip  $T$ , simply by pre-concatenating the morphism  $T \rightarrow T'$ . This motivates the functor of cones

$$\begin{aligned}
 \text{Hom}(\Delta(-), X) : \mathcal{C}^{\text{op}} &\rightarrow \text{Set} \\
 T &\mapsto \{\text{cone with tip } T \text{ to } X\}
 \end{aligned}$$

(see [3, Ch. 6, § 2, Def. 6.2.1]).

**Definition 6.4 (Limit).** The **limit** of an arbitrary diagram  $X \in \text{Diag}_{\Gamma}(\mathcal{C})$  is a representation of the functor of cones

$$\text{Hom}(\Delta(-), X) : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$$

(see [3, Ch. 6, § 2, Def. 6.2.2]).

**Remark 6.5.** 1. In literature, the limit is often referred to as **projective limit** or **inverse limit** (see [3, Ch. 6, § 2, Def. 6.2.2]).

2. Let  $X \in \text{Diag}_\Gamma(\mathcal{C})$ . The following objects are equivalent:

- a) The limit on  $X$
- b) The terminal object of  $\text{Hom}(\Delta(-), X)$
- c) The universal element of the functor  $\text{Hom}(\Delta(-), X)$

(see [10, Ch. 8, § 8.2, Rem. 8.2.6]).

3. By corollary 5.8, the limit of  $X$  consists of a cone  $\Delta(L) \rightarrow X$ , such that for every other cone  $\Delta(L') \rightarrow X$  there exists a unique morphism  $L' \rightarrow L$  such that diagram

$$\begin{array}{ccc} \Delta(L') & \xrightarrow{\quad} & \Delta(L) \\ & \searrow & \swarrow \\ & X & \end{array}$$

commutes. The tip  $L$  of a universal cone with base  $X$  is often referred to as limit of  $X$ , which is unique up to isomorphism (if it exists). In that case, we denote the tip by  $\lim(X)$  or  $\lim_{v \in V} X(v)$  (see [3, Ch. 6, § 2, Rem. 6.2.3]).

4. Limits are functorial in the following sense: Let  $\Gamma$  be a directed graph and let  $\mathcal{C}$  be a category. Let  $X$  and  $Y$  be diagrams of shape  $\Gamma$  in  $\mathcal{C}$  with limits  $\lim(X)$  and  $\lim(Y)$  respectively. Let  $X \rightarrow Y$  be a morphism between the diagrams. Then there exists a unique morphism  $\lim(X) \rightarrow \lim(Y)$  such that the diagram

$$\begin{array}{ccc} \Delta(\lim(X)) & \xrightarrow{\quad} & \Delta(\lim(Y)) \\ \downarrow & & \downarrow \\ X & \xrightarrow{\quad} & Y \end{array}$$

commutes. Note that the vertical morphisms are the universal cones. We can use the universal property of  $\lim(Y)$  on the cone  $\Delta(\lim(X)) \rightarrow X \rightarrow Y$ : Let  $f, g$  be two morphisms between diagrams. Due to the uniqueness, the following properties are satisfied:

- a)  $\text{id} = \lim(\text{id})$
- b)  $\lim(g \circ f) = \lim(g) \circ \lim(f)$ .

Thus, assuming that  $\mathcal{C}$  has limits of shape  $\Gamma$ , we get a functor

$$\lim : \text{Diag}_\Gamma(\mathcal{C}) \rightarrow \mathcal{C}.$$

The limits were chosen arbitrarily and their existence relies on the axiom of choice 2.1 (see [3, Ch. 6, § 2, Rem. 6.2.24]).

5. If every diagram in a category has a limit, the category is called **complete** (see [3, Ch. 6, § 4, Def. 6.4.3]).
6. If the underlying graph of a limit is finite, the limit on that graph is called **finite limit** (see [3, Ch. 6, § 2]).
7. The commutative diagram

$$\begin{array}{ccc}
 & T & \\
 f_v \swarrow & \downarrow \exists! f & \searrow f_w \\
 & L & \\
 \text{pr}_v \swarrow & & \searrow \text{pr}_w \\
 X(v) & \xrightarrow{X(e)} & X(w)
 \end{array} \tag{6.6}$$

visualises limits.

8. Although a limit object is often referred to as limit, strictly speaking, limits are pairs  $(L, (\text{pr}_i)_{i \in I})$  of a limit object  $L$  and a family of arrows  $(\text{pr}_i : L \rightarrow X_i)_{i \in I}$  (see [10, Ch. 8, § 2, Def. 8.2.3]).
9. The limit is a right-adjoint functor, that is,  $\Delta \dashv \lim$ .

### 6.1.1 Products

The definition of limits allows arbitrary diagrams. Thus, we may choose a diagram without arrows, which gives rise to the product.

**Definition 6.7** (Product). Let  $\Gamma = (V, \emptyset)$  be a directed graph (without edges). Let  $X \in \text{Diag}_\Gamma(\mathcal{C})$ . We call the limit of  $X$  the **product** of the family  $(X(v))_{v \in V}$ , consisting of:



1. An object  $P := \prod_{v \in V} X(v)$
2. A family of morphisms, the so-called **projections**

$$\text{pr}_w : P \rightarrow X(w)$$

for all  $w \in V$ .

The following universal property has to be satisfied:

**Universal property of products:**

For all  $T \in \text{Ob}(\mathcal{C})$  and every family of morphisms  $f_v : T \rightarrow X(v)$ , there exists a unique morphism

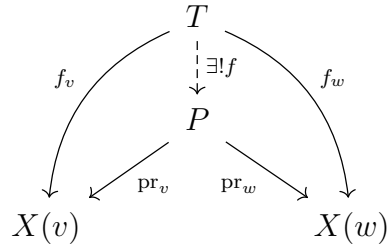
$$f : T \rightarrow P$$

such that

$$\text{pr}_v \circ f = f_v$$

for all  $v \in V$ .

We can visualise a binary product in the following diagram:



(see [3, Ch. 6, § 2, Def. 6.2.4]).

*Remark 6.8.* The product  $\prod_{i \in I} X_i$  of a family of objects  $(X_i)_{i \in I}$  is a universal object "over" the objects  $X_i$ . Since  $\text{Hom}(\Delta(-), X)$  identifies with the functor  $\prod_{i \in I} \text{Hom}(-, X_i)$ , we may characterise  $\prod_{i \in I} X_i$  via

$$\text{Hom}\left(-, \prod_{i \in I} X_i\right) \cong \prod_{i \in I} \text{Hom}(-, X_i)$$

(see [3, Ch. 6, § 2, def. 6.2.4]).

In the following, we see some important and illustrating examples of products:

**Example 6.9** (Products in  $\mathbf{Set}$ ). Let  $(X_i)_{i \in I}$  be a family of objects in  $\mathbf{Set}$ . The product in  $\mathbf{Set}$  is the Cartesian product (like in Definition 2.4). The projections are defined as

$$\begin{aligned} \mathrm{pr}_j : \prod_{i \in I} X_i &\rightarrow X_j \\ \mathrm{pr}_j((x_i)_{i \in I}) &\mapsto x_j. \end{aligned}$$

Let  $T \in \mathbf{Ob}(\mathbf{Set})$  be arbitrary and let  $(f_i : T \rightarrow X_i)_{i \in I}$  be a family of morphisms. Then the universal property is determined uniquely by

$$\begin{aligned} f : T &\rightarrow \prod_{i \in I} X_i, \\ f(t) &:= (f_i(t))_{i \in I}. \end{aligned}$$

By definition of  $f$ , it holds that  $\mathrm{pr}_i \circ f = f_i$  for all  $i \in I$  (see [3, Ch. 6, § 2, Ex. 6.2.5]).

**Example 6.10** (Products in  $\mathbf{Top}$ ). Let  $(X_i)_{i \in I}$  be a family of topological spaces. We define the product

$$\prod_{i \in I} X_i := \{(x_i)_{i \in I} \mid x_i \in X_i\}$$

as the coarsest topology, such that the projections

$$\begin{aligned} \mathrm{pr}_j : \prod_{i \in I} X_i &\rightarrow X_j \\ (x_i)_{i \in I} &\mapsto x_j \end{aligned}$$

are continuous. This is exactly the product topology we already saw in Example 2.18.

Let  $T \in \mathbf{Ob}(\mathbf{Top})$  be arbitrary with a family of morphisms  $(f_i : T \rightarrow X_i)_{i \in I}$ . Then there exists a unique morphism

$$\begin{aligned} f : T &\rightarrow \prod_{i \in I} X_i \\ f(t) &:= (f_i(t))_{i \in I}. \end{aligned}$$

satisfying  $\mathrm{pr}_i \circ f = f_i$  for all  $i \in I$  by definition. Since all components of  $f$  are

continuous,  $f$  is continuous by Theorem 2.20 (see [3, Ch. 6, § 4, Proof 6.4.12]).

**Example 6.11** (Products in CRing). Let  $(R_i)_{i \in I}$  be a family of commutative rings. We define the product  $\prod_{i \in I} R_i$  as the Cartesian product of rings (recall Remark 2.31, 2) with projections

$$\text{pr}_j : \prod_{i \in I} R_i \rightarrow R_j.$$

Let  $T \in \text{Ob}(\text{CRing})$  be arbitrary and let

$$f_i : T \rightarrow R_i$$

be a family of ring homomorphisms for all  $i \in I$ . We can define a unique morphism  $f : T \rightarrow \prod_{i \in I} R_i$  via

$$f := ((f_i)_{i \in I})$$

which is unique. By definition, it holds that  $\text{pr}_i \circ f = f_i$  (see [3, Ch. 6, § 2, Ex. 6.2.5], [7, pCh. 3, § 1]).

**Example 6.12** (Empty product). What happens if we construct the product over an empty indexing category? Let  $\mathcal{C}$  be a category and define

$$P := \prod_{i \in \emptyset} X_i$$

to be the product with a family of morphisms  $(\text{pr}_i : P \rightarrow X_i)_{i \in \emptyset}$ . By the universal property of products, for any  $T \in \text{Ob}(\mathcal{C})$  with a family of morphisms

$$f_i : T \rightarrow X_i,$$

there exists a unique morphism  $f : T \rightarrow P$  such that  $f_i = \text{pr}_i \circ f$ . Since the indexing set is empty, the universal property is vacuously satisfied, leaving us with the unique morphism  $f : T \rightarrow P$ . This is the definition of terminal objects (see Definition 3.20,2; [3, Ch. 6, § 9, Ex. 6.2]).

**Example 6.13** (Products in PSh). Let  $(F_i : \mathcal{C}^{\text{op}} \rightarrow \text{CRing})_{i \in I}$  be a family of presheaves.

We define the product presheaf

$$\left(\prod_{i \in I} F_i\right)(U) := \prod_{i \in I} F_i(U)$$

for all  $U \in \text{Ob}(\mathcal{C}^{\text{op}})$ . Let  $f \in \text{Hom}_{\mathcal{C}^{\text{op}}}(U, V)$  with restrictions

$$\begin{aligned} \left(\prod_{i \in I} F_i\right)(f) : \prod_{i \in I} F_i(U) &\rightarrow \prod_{i \in I} F_i(V) \\ (u_i)_{i \in I} &\mapsto (F_i(f)(u_i))_{i \in I} \end{aligned}$$

for  $u \in \prod_{i \in I} F_i(U)$ . The projections

$$\text{pr}_i : \prod_{i \in I} F_i \rightarrow F_i$$

are natural transformations, given by the componentwise projections in  $\mathbf{CRing}$ . For any presheaf  $H$  with natural transformations  $f_i : T \rightarrow F_i$  there exists a unique natural transformation

$$f : T \rightarrow \prod_{i \in I} F_i$$

with components

$$((f_i)_{i \in I}(h) = (f_i(h))_{i \in I})$$

for any  $h \in T(U)$  such that  $f_i : \text{pr}_i \circ f$  (see [16]).

Let  $(X_i)_{i \in I}$  be a family of objects with product  $(P, (\text{pr}_i)_{i \in I})$ . We saw that the arising diagram commutes for any  $(T, (f_i)_{i \in I})$ . This is not necessarily the case when we add morphisms  $g_{ij} : X_i \rightarrow X_j$ . In the next section, we study a construction, that solves that problem.

### 6.1.2 Equalisers

A special limit that plays a crucial role in the definition of sheaves and in the construction of limits is the equaliser. Intuitively, it "filters out" data where morphisms do not agree, producing a subobject where morphisms behave identically.

**Definition 6.14** (Equaliser). Let  $\mathcal{C}$  be a category and let  $f, g \in \text{Hom}_{\mathcal{C}}(A, B)$  be parallel morphisms. An **equaliser** of  $f$  and  $g$  is a pair  $(E, \iota)$ , where  $E$ -often denoted by  $\text{eq}(f, g)$ -is an object and  $\iota : E \rightarrow A$  is a monomorphism (recall Definition 3.25), satisfying the following properties:

1. **Equalising property**

$$f \circ \iota = g \circ \iota$$

2. **Universal property:** For any  $T \in \text{Ob}(\mathcal{C})$  with morphism  $h : T \rightarrow A$  such that  $f \circ h = g \circ h$ , there exists a **unique** morphism  $\eta : T \rightarrow E$  such that

$$\iota \circ \eta = h.$$

We denote the equaliser by  $\iota : E \hookrightarrow A$ .

Equivalently, we can define the equaliser as the limit on a diagram  $\Gamma$  of shape  $\bullet \rightrightarrows \bullet$ . The (by definition commutative) diagram

$$E \xrightarrow{\iota} A \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} B \quad (6.15)$$

visualises equalisers. Equalisers consist of an object  $E$ , together with a monomorphism (recall Definition 3.25)

$$\iota : E \hookrightarrow A.$$

The universal property of the equaliser is that for a cone (like in Diagram 6.15), the morphism  $h : T \rightarrow A$  factorises uniquely over  $\iota$ :

$$\begin{array}{ccc} T & \xrightarrow{h} & A \\ \downarrow \exists! \eta & \searrow & \downarrow \\ E & \xrightarrow{\iota} & A \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} B \end{array}$$

(see [3, Ch. 6, § 2, Def. 6.2.13]).

*Remark 6.16.* For an arbitrary family with more than two morphisms

$$(f_i : A \rightarrow B)_{i \in I},$$

we can define the multiple equaliser  $\text{eq}((f_i)_{i \in I})$ -if the equaliser exists (see [3, Ch. 6, § 2, Def. 6.2.13]).

In the following, we study some examples of equalisers:

**Example 6.17** (Equalisers in  $\text{Set}$ ). Let  $f, g \in \text{Hom}_{\text{Set}}(A, B)$ . The equaliser is

$$E = \text{eq}(f, g) := \{a \in A \mid f(a) = g(a)\}$$

$$\iota : E \hookrightarrow A.$$

Let  $T \in \text{Ob}(\text{Set})$  be arbitrary and let  $h : T \rightarrow A$  be a morphism such that

$$g \circ h = f \circ h$$

holds, then consequently,  $\text{im } h \subseteq E$ . Thus,  $h$  factorises uniquely over  $\iota$  (see [3, Ch. 6, § 2, Ex. 6.2.14]).

**Example 6.18** (Equalisers in  $\text{Top}$ ). Let  $f, g \in \text{Hom}_{\text{Top}}(X, Y)$  be parallel morphisms. The equaliser is

$$E = \text{eq}(f, g) := \{x \in X \mid f(x) = g(x)\}$$

$$\iota : E \hookrightarrow X.$$

Let  $T \in \text{Ob}(\text{Top})$  be arbitrary with a morphism  $h : T \rightarrow X$ , such that

$$f \circ h = g \circ h.$$

Since  $\text{im } h \subseteq E$ , consequently  $h$  factorises uniquely over  $\iota$ . The topological space  $E$  is equipped with the subspace topology (Example 2.25, see [3, Ch. 6, § 2, Proof 6.4.12]).

**Example 6.19** (Equalisers in  $\text{CRing}$ ). Let  $Q, R$  be commutative rings and let  $f, g \in \text{Hom}_{\text{CRing}}(Q, R)$  be parallel morphisms. We define the equaliser as

$$\text{eq}(f, g) := \{r \in Q \mid f(r) = g(r)\}$$

$$\iota : \text{eq}(f, g) \hookrightarrow Q.$$

The ring  $\text{eq}(f, g)$  is a subring of  $Q$ , since it is closed under multiplication and addition: Let  $q, r \in \text{eq}(f, g)$ , then  $f(r) = g(r)$  and  $f(q) = g(q)$  by definition. Since  $f$  and  $g$  are ring homomorphisms, it holds that

$$\begin{aligned} f(r + q) &= f(r) + f(q) = g(r) + g(q) = g(r + q) \\ f(r \cdot q) &= f(r) \cdot f(q) = g(r) \cdot g(q) = g(r \cdot q). \end{aligned}$$

Let  $T \in \text{Ob}(\text{CRing})$  be arbitrary with a morphism  $h : T \rightarrow Q$ , such that

$$g \circ h = f \circ h.$$

Then  $\text{im } h \subseteq E$ . Thus,  $h$  factorises uniquely over  $\iota$  (see [3, Ch. 6, § 2, Ex. 6.2.14]).

**Example 6.20** (Equalisers in  $\text{PSh}$ ). Let  $F, G : \mathcal{C}^{\text{op}} \rightarrow \text{CRing}$  be presheaves with parallel natural transformations  $f, g : F \rightarrow G$ . The equaliser is defined pointwise via

$$E(U) = \text{eq}(f, g)(U) := \{u \in F(U) \mid f_U(u) = g_U(u)\}$$

for each  $U \in \text{Ob}(\mathcal{C}^{\text{op}})$ .

Any restriction  $s \in \text{Hom}_{\mathcal{C}^{\text{op}}}(U, V)$  gives rise to

$$E(U) \rightarrow E(V),$$

which is a restriction of  $F$ . For any  $u \in E(U)$ ,  $f_U(u) = g_U(u)$ . By the naturality of  $f$  and  $g$ , their components satisfy

$$\begin{aligned} f_V(F(s)(u)) &= G(s)(f_U(u)) \\ &= G(s)(g_U(u)) \\ &= g_V(F(s)(u)). \end{aligned}$$

Thus,  $F(s)(u) \in E(V)$  and the restrictions are well-defined.

The canonical natural transformation

$$\iota : E \rightarrow F$$

is the componentwise inclusion. Let  $T$  be an arbitrary presheaf with a natural trans-

formation

$$h : T \rightarrow F$$

such that

$$f \circ h = g \circ h.$$

There exists a unique natural transformation  $\eta : T \rightarrow E$  such that  $h$  factorises over  $\eta$ :

For any  $U \in \text{Ob}(\mathcal{C}^{\text{op}})$ , we have components

$$h_U : T(U) \rightarrow F(U).$$

Since  $f \circ h = g \circ h$ , it follows that

$$f_U(h_U(t)) = g_U(h_U(t))$$

for all  $t \in T(U)$ . Thus,  $\text{im } h_U \subseteq E(U) \subseteq F(U)$ . This allows us to define

$$\begin{aligned} \eta_U : T(U) &\rightarrow E(U) \\ t &\mapsto h_U(t) \end{aligned}$$

(see [16]).

Now that we understand equalisers, we can carry out our Ansatz and construct limits.

### 6.1.3 Constructions with Limits

We saw how to construct equalisers and products, but how can we construct arbitrary limits? Can we construct limits in any category? The answer is no, not every category has limits. While not every category has products, equalisers and limits, any category that has both products and equalisers, also has all limits. In fact, we can construct the limit by first constructing the product and then constructing the equaliser on the product.

**Theorem 6.21** (Construction of Limits). Let  $\mathcal{C}$  be a category that has all equalisers and all products, and let  $X$  be a diagram in  $\mathcal{C}$  of shape  $\Gamma = (V, E)$ . Then  $\mathcal{C}$  has all



limits. Furthermore, it is constructed as equaliser of suitable morphisms

$$a, b : \prod_{v \in V} X(v) \rightrightarrows \prod_{e: v \rightarrow v' \in E} X(v')$$

with

$$\begin{aligned} \text{pr}_e \circ a &= X(e) \circ \text{pr}_v \\ \text{pr}_e \circ b &= \text{pr}_{v'} \end{aligned}$$

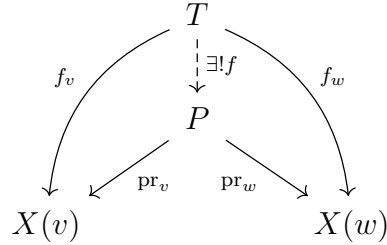
for any  $e : v \rightarrow v'$  (see [3, Ch. 6, § 4, Thm. 6.4.1]).

*Proof.* In order to make the proof more accessible, we prove the theorem first for the limit on two objects and give the general proof afterwards. We assume that the category has both products and equalisers. We construct the limit by first forming the product and then taking the equaliser of suitable morphisms.

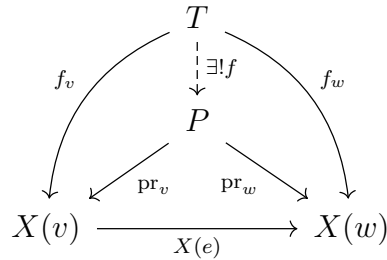
For two objects  $X(v)$  and  $X(w)$ , we construct the product  $P$  with projections

$$\begin{aligned} \text{pr}_v : P &\rightarrow X(v) \\ \text{pr}_w : P &\rightarrow X(w). \end{aligned}$$

By itself, the diagram



commutes. However if the diagram includes a morphism  $e : v \rightarrow w$ , the extended diagram



may fail to commute. This motivates taking the equaliser

$$E := \text{eq}(X(e) \circ \text{pr}_v, \text{pr}_w) \hookrightarrow P.$$

Since  $f : T \rightarrow P$  is a universal morphism by the definition of products (see Definition 6.7) and  $\iota : E \rightarrow P$  is a universal morphism by the definition of equalisers (see Definition 6.14), the morphism  $(g \circ \iota) : T \rightarrow P$  is a universal morphism as composite of universal morphisms, making the diagram

commute. Thus,  $E$  is the limit object, together with the morphisms

$$\begin{aligned} (\text{pr}_v \circ \iota) : E &\rightarrow X(v) \\ (\text{pr}_w \circ \iota) : E &\rightarrow X(w). \end{aligned}$$

Now we generalise the proof for an arbitrary family of objects  $(X(v))_{v \in V}$  in  $\mathcal{C}$ . We construct the product  $P$  with morphisms  $(\text{pr}_v : P \rightarrow X(v))_{v \in V}$ , which exist by assumption. To force commutativity, we take the equaliser of the following morphisms: By the universal property of the product, there exists a unique morphism

$$a : \prod_{v \in V} X(v) \rightarrow \prod_{e: v \rightarrow v'} X(v')$$

such that

$$\text{pr}_e \circ a = X(e) \circ \text{pr}_v$$

for all edges  $e : v \rightarrow v'$  with

$$\begin{aligned} \text{pr}_v &: \prod_{v \in V} X(v) \rightarrow X(v) \\ \text{pr}_e &: \prod_{e: v \rightarrow v'} X(v') \rightarrow X(v') \end{aligned}$$

and a parallel morphism

$$b : \prod_{v \in V} X(v) \rightarrow \prod_{e: v \rightarrow v'} X(v')$$

such that

$$\text{pr}_e \circ b = \text{pr}_{v'}$$

for all edges  $e : v \rightarrow v'$ , that is, the diagrams

$$\begin{array}{ccc} \prod_{v \in V} X(v) & \xrightarrow{a} & \prod_{e: v \rightarrow v'} X(v') \\ \downarrow \text{pr}_v & & \downarrow \text{pr}_e \\ X(v) & \xrightarrow{X(e)} & X(v') \end{array} \quad \begin{array}{ccc} \prod_{v \in V} X(v) & \xrightarrow{b} & \prod_{e: v \rightarrow v'} X(v') \\ & \searrow \text{pr}'_v & \downarrow \text{pr}_e \\ & & X(v') \end{array}$$

commute. A morphism

$$f : T \rightarrow \prod_{v \in V} X(v)$$

with  $a \circ f = b \circ f$ , corresponds to a family of morphisms

$$(f_v : T \rightarrow X(v))_{v \in V}$$

such that for every morphism  $e : v \rightarrow v'$  in  $E$

$$X(e) \circ f_v = f_{v'}.$$

This is exactly a cone on  $X$ , and an equaliser  $\text{eq}(a, b)$  is exactly a universal cone with base  $X$  and thus, a limit of  $X$  (see [3, Ch. 6, § 4, Proof 6.4.1]).  $\square$

*Remark 6.22.* It is not guaranteed that the product as well as the equaliser exist in

a given category:

In the category of fields  $\mathbf{Fld}$ , neither the product nor the equaliser exist. We may consider for example  $\mathbb{Q} \times \mathbb{F}_2$  which is a ring, but not a field. The operations  $+$  and  $\cdot$  on  $\mathbb{Q} \times \mathbb{F}_2$  are defined componentwise. Furthermore,  $(1, 0)$  is an element of  $\mathbb{Q} \times \mathbb{F}_2$ , but it has no inverse element, since

$$(1, 0) \cdot (a, b) = (a, 0) \neq (1, 1)$$

for all  $(a, b) \in \mathbb{Q} \times \mathbb{F}_2$ .

Categories that have limits are the categories  $\mathbf{Set}$ ,  $\mathbf{Top}$  and  $\mathbf{CRing}$ , as we see in the following example:

**Example 6.23** (Limits in  $\mathbf{Set}$ ). In Example 6.9, we saw that the product in  $\mathbf{Set}$  is the Cartesian product of a family of objects  $(X(v))_{v \in V}$ . Taking morphisms  $e_{v,w} : v \rightarrow w$  into consideration, they give rise to a family of morphisms  $X(e_{v,w}) : X(v) \rightarrow X(w)$ . We define the product over the morphisms

$$\prod_{e:v \rightarrow w \in E} X(w).$$

This gives rise to parallel morphisms

$$\begin{aligned} a_e &:= X(e) \circ \text{pr}_v \\ b_e &:= \text{pr}_w \end{aligned}$$

with the projections  $\text{pr}_v : \prod_{w \in V} X(w) \rightarrow X(v)$ . The limit is the equaliser we saw in Example 6.17, that is

$$\lim X \cong \text{eq}(a, b) \rightarrow \prod_{w \in V} X(w)$$

which is a subset of the product set (see [3, Ch. 6, § 4, Ex. 6.4.2]).

**Example 6.24** (Limits in  $\mathbf{Top}$ ). In Example 6.10, we constructed the product of topological spaces, which is exactly the product topology we saw in Example 2.18. In Example 6.18 we saw that the equaliser in topological spaces is the subspace topology we saw in Example 2.25. The limit of topological spaces is the topological

subspace of the product space, such that the diagram commutes (see [3, Ch. 6, § 4, Proof 6.4.12]).

**Example 6.25** (Limits in  $\mathbf{CRing}$ ). The product of commutative rings is the Cartesian product of rings, as we saw in Example 6.11 and in Remark 2.31,2. Since the equaliser is a subring as we saw in Example 6.19, the limit is a subring of the product ring (see [3, Ch. 6, § 4, Thm. 6.4.11]).

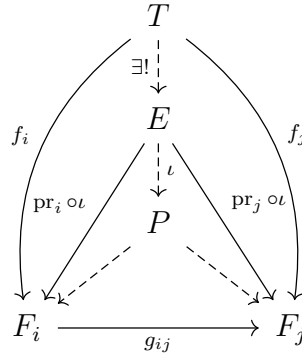
**Example 6.26** (Limits in  $\mathbf{PSh}$ ). Let  $(F_i)_{i \in I}$  be a family on presheaves and let  $P$  be the product sheaf with the universal natural transformation  $f : T \rightarrow P$  and a family of natural transformations  $f_i : T \rightarrow F_i$  for all  $i \in I$  and projections  $\text{pr}_i : P \rightarrow F_i$ . Let

$$g_{ij} : F_i \rightarrow F_j$$

be a collection of natural transformations. Then

$$E := \text{eq} \left( \prod_{i \in I} g_{ij} \circ \text{pr}_i, \prod_{j \in I} \text{pr}_j \right)$$

with a unique natural transformation  $f : T \rightarrow E$  such that  $f \circ \text{pr}_i \circ \iota = f_i$ , is the limit. By the definition of equalisers, any morphism  $T \rightarrow P$  factors uniquely through  $\iota : E \rightarrow P$ , making



commute (see [16]).

We saw that products, as well as equalisers are special limits. The construction of a limit is taking the equaliser of a product object. What about arbitrary nested limits?

**Theorem 6.27** (Nested Limits). Let  $\Gamma = (V, E)$  and  $\Lambda = (W, F)$  be directed graphs and let  $\Gamma \otimes \Lambda$  be their Cartesian product. Consider a legal diagram  $X$  of shape  $\Gamma \otimes \Lambda$

in an arbitrary category. Assume, that the limit of the diagram  $X(v, -)$  of shape  $\Lambda$  exists for every  $v \in V$ . We denote this limit by  $\lim_{w \in W} X(v, w)$ . By Remark 6.5, 4, there exists a natural morphism

$$\lim_{w \in W} X(v, w) \cong \lim_{w \in W} X(v', w)$$

for every edge  $v \rightarrow v'$  in  $E$ . Thus, we obtain a diagram  $\lim_{w \in W} X(-, w)$  of shape  $\Gamma$ .

The cones on that diagram  $\lim_{w \in W} X(-, w)$  correspond to the cones on  $X$ . Especially

$$\lim_{v \in V} \lim_{w \in W} X(v, w)$$

exists if and only if

$$\lim_{(v,w) \in V \times W} X(v, w)$$

exists. In that case, there exists a natural isomorphism

$$\lim_{v \in V} \lim_{w \in W} X(v, w) \cong \lim_{(v,w) \in V \times W} X(v, w)$$

(see [3, Ch. 6, § 6, Thm. 6.6.4]).

*Proof.* We want to show that

$$\phi : \left( \lim_{v \in V} \lim_{w \in W} X(v, w) \right) \xrightarrow{\cong} \left( \lim_{(v,w) \in V \times W} X(v, w) \right)$$

is an isomorphism.

For any fixed vertex  $v \in V$ , the limit  $\lim_{w \in W} X(v, w)$  gives rise to a diagram

$$\begin{aligned} \lim_{v \in V} \lim_{w \in W} X(v, w) : V &\rightarrow \mathcal{C} \\ v &\mapsto \lim_{w \in W} X(v, w). \end{aligned}$$

Since  $\lim_{v \in V} \lim_{w \in W} X(v, w)$  is a limit object, there are projections

$$\pi_v : \lim_{v \in V} \lim_{w \in W} X(v, w) \rightarrow \lim_{w \in W} X(v, w),$$

and by the limit  $\lim_{w \in W} X(v, w)$ , there are projections

$$\pi_{v,w} : \lim_{w \in W} X(v, w) \rightarrow X(v, w).$$

We can compose the projections and get

$$\pi_{v,w} \circ \pi_v : \lim_{v \in V} \lim_{w \in W} X(v, w) \rightarrow X(v, w).$$

By the universal property of the limit, this defines a unique morphism

$$\phi : \lim_{v \in V} \lim_{w \in W} X(v, w) \rightarrow \lim_{(v,w) \in V \times W} X(v, w).$$

On the other hand, the limit  $\lim_{(v,w) \in V \times W} X(v, w)$  comes with projections

$$\text{pr}_{v,w} : \lim_{(v,w) \in V \times W} X(v, w) \rightarrow X(v, w),$$

which induce projections

$$\text{pr}_v : \lim_{(v,w) \in V \times W} X(v, w) \rightarrow \lim_{w \in W} X(v, w)$$

to each  $\lim_{w \in W} X(v, w)$  by the universal property of limits. The projections  $\text{pr}_v$  assemble to a cone over the diagram  $\lim_{w \in W} X(v, w)$ , which yields a unique morphism

$$\psi : \lim_{(v,w) \in V \times W} X(v, w) \rightarrow \lim_{v \in V} \lim_{w \in W} X(v, w).$$

By the universality, it holds that  $\psi \circ \phi = \text{id}_{\lim_{(v,w) \in V \times W} X(v, w)}$  and  $\phi \circ \psi = \text{id}_{\lim_{v \in V} \lim_{w \in W} X(v, w)}$ . Therefore,  $\phi$  is an isomorphism (see [3, Ch. 6, § 6, Proof 6.6.4]).  $\square$

A consequence of this theorem is the commutativity of limits, which allows us to proof that certain presheaves satisfy the equaliser condition in Chapter 7.

**Theorem 6.28** (Commutativity of Limits). Let  $\Gamma := (V, E)$  and  $\Lambda := (W, F)$  be two directed graphs and let  $(X(v, w))_{v \in V, w \in W}$  be a legal diagram on the Cartesian product  $\Gamma \otimes \Lambda$ . If the limits exist, we have the following isomorphism:

$$\lim_{v \in V} \lim_{w \in W} X(v, w) \cong \lim_{w \in W} \lim_{v \in V} X(v, w)$$

(see [3, Ch. 6, § 6, Thm. 6.6.5]).

*Proof.* By Theorem 6.27, it holds that

$$\lim_{v \in V} \lim_{w \in W} X(v, w) \cong \lim_{(v, w) \in V \times W} X(v, w) \cong \lim_{w \in W} \lim_{v \in V} X(v, w)$$

(see [3, Ch. 6, § 6, Proof 6.6.5]). □

In the following section, we study the dual notation of limits, equalisers and products.

## 6.2 Colimits

Colimits are the categorical dual to limits. While limits describe the universal way of "matching" data across a diagram, colimits describe the universal way of "gluing" objects together along shared morphisms. Analogous to limits, we can define colimits using cone-like structures called cocones.

**Definition 6.29** (Cocone). Let  $\mathcal{C}$  be a category and let  $\Gamma = (V, E)$  be a directed graph. Let  $X \in \text{Diag}_\Gamma(\mathcal{C})$  be a diagram of shape  $\Gamma$  in  $\mathcal{C}$ . A **cocone** is a morphism between the diagrams

$$X \rightarrow \Delta(T)$$

for  $T \in \text{Ob}(\mathcal{C})$ . Concretely, this is a family of morphisms

$$(X(v) \rightarrow T)_{v \in V}$$

such that the diagram

$$\begin{array}{ccc} X(v) & \xrightarrow{X(e)} & X(w) \\ & \searrow & \swarrow \\ & T & \end{array}$$

commutes for all edges  $e : v \rightarrow w$ . We call  $T$  the **tip** of the cocone (see [3, Ch. 6, § 3, Def. 6.3.1]).



This allows us to define the colimits as the representation of a functor between cocones.

**Definition 6.30** (Colimit). The **colimit** of a diagram  $X$  in a category  $\mathcal{C}$  is the representation of the functor of cocones

$$\text{Hom}(X, \Delta(-)) : \mathcal{C} \rightarrow \text{Set}$$

(see [3, Ch. 6, § 3, Def. 6.3.2]).

*Remark 6.31.* 1. In literature, the colimit is often referred to as **inductive limit** or **direct limit** (see [3, Ch. 6, § 3, Def. 6.3.2]).

2. A colimit of a diagram  $X$  corresponds to a universal cocone by Corollary 5.8. That is, there exists a cocone

$$X \rightarrow \Delta(\text{colim}(X))$$

such that for every cocone  $X \rightarrow \Delta(T)$  there exists a unique morphism

$$\text{colim}(X) \rightarrow T$$

such that the following diagram commutes:

$$\begin{array}{ccc} & X & \\ \swarrow & & \searrow \\ \Delta(\text{colim}(X)) & \xrightarrow{\quad} & \Delta(T) \end{array}$$

(see [3, Ch. 6, § 3, Rem. 6.3.3]).

3. A category  $\mathcal{C}$  is called **cocomplete** if the colimit of every diagram exists in  $\mathcal{C}$ . This is the same as  $\mathcal{C}^{\text{op}}$  being complete (see [3, Ch. 6, § 5, Def. 6.5.2]).

4. Although  $L^{\text{co}}$  is often referred to as colimit, the actual colimit is a pair  $(L^{\text{co}}, (\iota_i)_{i \in I})$ , where  $L^{\text{co}}$  is the colimit object and  $(\iota_i : X_i \rightarrow L^{\text{co}})$  is a family of morphisms.

5. The colimit is left-adjoint, that is,  $\text{colim} \dashv \Delta$ .

Analogous to the limits, we have to understand two special cases of colimits, that is, coproducts and coequalisers, in order to construct colimits.

### 6.2.1 Coproducts

Coproducts are the categorical dual of products. They can be defined as the colimits over diagrams, indexed by a discrete category (recall Remark 8). Intuitively, while products "project" into objects, coproducts "include" objects into a larger object via canonical morphisms.

**Definition 6.32** (Coproduct). Let  $\Gamma := (V, \emptyset)$  be the underlying graph of a discrete category and let  $X \in \text{Diag}_\Gamma(\mathcal{C})$ . We call the colimit of  $X$  the **coproduct** of the family  $(X(v))_{v \in V}$ . That is, an object

$$\coprod_{v \in V} X(v)$$

together with a universal family of morphisms, the so-called **injections**

$$\iota_v : X(v) \rightarrow \coprod_{v \in V} X(v)$$

such that for every  $T \in \text{Ob}(\mathcal{C})$  and a family of morphisms  $(f_v : X(v) \rightarrow T)_{v \in V}$ , there exists a unique morphism

$$f : \coprod_{v \in V} X(v) \rightarrow T$$

such that  $f \circ \iota_v = f_v$  for every  $v \in V$ . The binary coproduct  $X(v) \sqcup X(w)$  can be illustrated with the following commutative diagram:

$$\begin{array}{ccc} X(v) & & X(w) \\ & \searrow \iota_v \quad \swarrow \iota_w & \\ & X(v) \sqcup X(w) & \\ & \downarrow \exists! f & \\ & T & \end{array}$$

$f_v$  (curved arrow from  $X(v)$  to  $T$ ) and  $f_w$  (curved arrow from  $X(w)$  to  $T$ )

(see [3, Ch. 6, § 3, Def. 6.3.5]).

**Remark 6.33.** 1. The coproduct  $\coprod_{v \in V} X(v)$  of a family of objects  $(X(v))_{v \in V}$  is the universal object that lies "under" the objects  $X(v)$ . We can characterise this

by the isomorphism

$$\mathrm{Hom}\left(\coprod_{v \in V} X(v), -\right) \cong \prod_{v \in V} \mathrm{Hom}(X(v), -)$$

(see [3, Ch. 6, § 3, Def. 6.3.5]).

2. Equivalently, the coproduct in a category  $\mathcal{C}$  is the product in  $\mathcal{C}^{\mathrm{op}}$  (see [3, Ch. 6, § 3, Def. 6.3.2]).

The following examples illustrate the construction of coproducts in different categories, that are used in the construction of colimits:

**Example 6.34** (Coproducts in  $\mathbf{Set}$ ). Let  $(X(v))_{v \in V}$  be a family of objects in  $\mathbf{Set}$ . Then the coproduct  $\coprod_{v \in V} X(v)$  is the disjoint union

$$\coprod_{v \in V} X(v) = \bigcup_{v \in V} (X(v) \times \{v\})$$

with injections

$$\begin{aligned} \iota_w : X(w) &\rightarrow \coprod_{v \in V} X(v) \\ x &\mapsto (x, w) \end{aligned}$$

for all  $x \in X(w)$ . The universal property is given by a family of morphisms

$$(f_v : X(v) \rightarrow T)_{v \in V}.$$

Any morphism  $f_v$  factors uniquely through  $f$  as  $\iota_v$  with

$$\begin{aligned} f : \coprod_{v \in V} X(v) &\rightarrow T \\ f((x, v)) &:= f_v(x) \end{aligned}$$

(see [3, Ch. 6, § 3, Ex. 6.3.6]).

**Example 6.35** (Coproducts in  $\mathbf{Top}$ ). Let  $(X(v))_{v \in V}$  be a family in  $\mathbf{Ob}(\mathbf{Top})$ . The

coproduct is the disjoint union of topological spaces like we saw in Example 2.23

$$\coprod_{v \in V} X(v) := \bigcup_{v \in V} (X(v) \times \{v\}).$$

The injections are defined as

$$\begin{aligned} \iota_w : X(w) &\rightarrow \coprod_{v \in V} (X(v), v) \\ x &\mapsto (x, w) \end{aligned}$$

for an element  $x \in X(w)$ . The topology on  $\coprod_{v \in V} X(v)$  is the finest topology, such that all inclusions are continuous.

Let  $T$  be an arbitrary object with a family of morphisms  $(f_v : X(v) \rightarrow T)_{v \in V}$ . We can define the unique isomorphism  $f : \coprod_{v \in V} X(v) \rightarrow T$  such that  $f_v = f \circ \iota_v$  for all  $v \in V$  as

$$f((x, v)) := f_v(x)$$

(see [3, Ch. 6, § 5, Proof. 6.5.6]).

**Example 6.36** (Coproducts in PSh). Let  $(F_i : \mathcal{C}^{\text{op}} \rightarrow \text{Set})_{i \in I}$  be a family of presheaves. Then

$$\left( \coprod_{i \in I} F_i \right)(U) = \coprod_{i \in I} F_i(U)$$

for all  $U \in \text{Ob}(\mathcal{C}^{\text{op}})$ , where  $\coprod_{i \in I} F_i(U)$  is the coproduct in Set. For any morphism  $s \in \text{Hom}_{\mathcal{C}^{\text{op}}}(U, V)$  the restriction is

$$\left( \coprod_{i \in I} F_i \right)(s) : \coprod_{i \in I} F_i(U) \rightarrow \coprod_{i \in I} F_i(V)$$

with

$$\left( \coprod_{i \in I} F_i \right)(s)(u) = F_i(s)(u)$$

if  $u \in F_i(U)$ . The injections  $\iota_i : F_i \rightarrow \coprod_{i \in I} F_i$  are determined by the injections of the components

$$\iota_i : F_i(U) \rightarrow \coprod_{i \in I} F_i(U).$$

Let  $T : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$  be an arbitrary presheaf with a family of natural transformations  $f_i : F_i \rightarrow T$ . Then there exists a unique morphism  $f : \coprod_{i \in I} F_i \rightarrow T$  such that  $\iota_i \circ f = f_i$ . We define  $\phi$  componentwise via a family of morphisms

$$f(u) := f_i(u)$$

such that  $u \in F_i(U)$  (see [17]).

Before we can construct colimits, we have to understand the categorical dual notion to equalisers.

## 6.2.2 Coequalisers

Intuitively, while equalisers "filter out" any data where two morphisms do not agree, coequalisers "force" the data to agree.

**Definition 6.37** (Coequaliser). Let  $\mathcal{C}$  be a category and let  $f, g \in \text{Hom}_{\mathcal{C}}(A, B)$  be parallel morphisms. A **coequaliser** of  $f$  and  $g$  is a pair  $(E^{\text{co}}, \pi)$  where  $E^{\text{co}}$  is an object—often denoted by  $\text{coeq}(f, g)$ —and  $\pi : B \rightarrow E^{\text{co}}$  is a morphism satisfying the following properties:

### 1. Coequalising property

$$\pi \circ f = \pi \circ g$$

### 2. Universal property: The following diagram commutes:

$$\begin{array}{ccccc} A & \xrightleftharpoons[g]{f} & B & \xrightarrow{\pi} & E^{\text{co}} \\ & & \searrow h & & \downarrow \exists! \varphi \\ & & & & T \end{array}$$

(see [3, Ch. 6, § 3, Def. 6.3.9]).

We discuss illustrating examples in the following, that are essential to construct the colimit in the categories we discussed in Examples 6.34–6.36:

**Example 6.38** (Coequalisers in  $\mathbf{Set}$ ). Let  $f, g \in \mathbf{Hom}_{\mathbf{Set}}(A, B)$  be two parallel morphisms. Then the coequaliser is

$$E^{\text{co}} = \text{coeq}(f, g) := B / \sim$$

where  $\sim$  is the smallest equivalence relation such that

$$f(a) \sim g(a)$$

holds for all  $a \in A$ . Let  $T$  be an arbitrary object with a morphism  $h : B \rightarrow T$  such that

$$h \circ g = h \circ f.$$

By the universal property there exists a unique morphism

$$\varphi : E^{\text{co}} \rightarrow T$$

such that

$$h = \varphi \circ \pi$$

where  $\pi : B \rightarrow E^{\text{co}}$  maps any element  $b \in B$  to its equivalence class  $[b]$  (see [3, Ch. 6, § 3, Ex. 6.3.10]).

**Example 6.39** (Coequalisers in  $\mathbf{Top}$ ). Let  $f, g \in \mathbf{Hom}_{\mathbf{Top}}(X, Y)$  be parallel morphisms. Then  $E^{\text{co}} = \text{coeq}(f, g) := Y / \sim$  is the coequaliser of sets, equipped with the quotient topology we studied in Example 2.22 and a morphism  $\pi : Y \rightarrow E^{\text{co}}$ . For an arbitrary  $T \in \mathbf{Ob}(\mathbf{Top})$  and a morphism  $h : Y \rightarrow T$  such that  $h \circ f = h \circ g$ , there exists a unique morphism  $\varphi : E^{\text{co}} \rightarrow T$  such that  $h = \varphi \circ \pi$  (see [3, Ch. 6, § 5, Proof 6.5.6]).

**Example 6.40** (Coequalisers in  $\mathbf{PSh}$ ). Let  $F, G : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$  be presheaves and let  $f, g : F \rightarrow G$  be parallel natural transformations. The coequaliser object  $E^{\text{co}}$  is defined objectwise for  $U \in \mathbf{Ob}(\mathcal{C}^{\text{op}})$  as

$$E^{\text{co}}(U) = G(U) / \sim_U$$

with an equivalence relation  $\sim_U$  generated by

$$f_U(u) \sim_U g_U(u)$$

for all  $u \in F(U)$ . The restrictions are induced by the restrictions of  $G$ , that is, for  $s \in \text{Hom}_{\mathcal{C}^{\text{op}}}(U, V)$ , it holds that

$$E^{\text{co}}(s)([u]) = [G(s)(u)]$$

for  $u \in G(U)$ . The presheaf  $E^{\text{co}}$  well-defined due to the naturality of  $f$  and  $g$ , ensuring that equivalence classes are preserved. The canonical natural transformation  $\pi : G \rightarrow E^{\text{co}}$  is given by the components

$$\pi_U : G(U) \rightarrow E^{\text{co}}(U)$$

satisfying  $\pi \circ f = \pi \circ g$ . Let  $T : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$  be an arbitrary presheaf with a morphism  $h : G \rightarrow T$  such that  $h \circ f = h \circ g$ . Then  $h$  factors uniquely over  $\pi$  by  $\varphi : E^{\text{co}} \rightarrow T$  with

$$\begin{aligned} \varphi(U) : G(U) &\rightarrow E^{\text{co}}(U) \\ \varphi_U([u]) &:= h_U(u) \end{aligned}$$

for  $u \in G(U)$  (see [17]).

In the following section, we discuss the construction of colimits.

### 6.2.3 Constructions with Colimits

In Chapter 6.1.3 we saw that limits can be constructed by first taking the product, followed by the equaliser of suitable morphisms. The construction of colimits follows an analogous construction, using coproducts and coequalisers.

**Theorem 6.41** (Construction of Colimits). Let  $\mathcal{C}$  be a category that has all coproducts and all coequalisers, and let  $X$  be a diagram of shape  $\Gamma = (V, E)$ . Then  $\mathcal{C}$  admits all colimits of shape  $\Gamma$ . Furthermore, the colimit is constructed as the

coequaliser of suitable morphisms

$$a, b : \coprod_{e:v \rightarrow w \in E} X(v) \rightrightarrows \coprod_{v \in V} X(v)$$

(see [3, Ch. 6, § 5, Thm. 6.5.1]).

*Proof.* This proof is dual to the proof of Theorem 6.21. To make the construction more accessible, we illustrate the construction of binary colimits. For two objects  $X(v)$  and  $X(w)$ , we first construct the coproduct  $P^{\text{co}}$  with injections  $\iota_v : X(v) \rightarrow P^{\text{co}}$  and  $\iota_w : X(w) \rightarrow P^{\text{co}}$ . By itself, the diagram

$$\begin{array}{ccc} X(v) & & X(w) \\ & \searrow \iota_v \quad \swarrow \iota_w & \\ & P^{\text{co}} & \\ & \downarrow \exists! f & \\ & T & \end{array}$$

$f_v$    $f_w$

commutes. However if the diagram includes a morphism  $e : v \rightarrow w$ , the expanded diagram

$$\begin{array}{ccc} X(v) & \xrightarrow{X(e)} & X(w) \\ & \searrow \iota_v \quad \swarrow \iota_w & \\ & P^{\text{co}} & \\ & \downarrow \exists! f & \\ & T & \end{array}$$

$f_v$    $f_w$

might fail to commute. This motivates the need to take the coequaliser

$$\pi : P^{\text{co}} \rightarrow E^{\text{co}} := \text{coeq}(\iota_v, X(e) \circ \iota_w).$$



In terms of diagrams,

$$\begin{array}{ccc}
 X(v) & \xrightarrow{X(e)} & X(w) \\
 \downarrow \iota_v & & \downarrow \iota_w \\
 & P^{\text{co}} & \\
 & \downarrow \pi & \\
 & E^{\text{co}} & \\
 \downarrow f_v & \text{---} \exists! \text{---} & \downarrow f_w \\
 & T &
 \end{array}
 \tag{6.42}$$

is the now considered diagram. Since the morphisms  $P^{\text{co}} \rightarrow T$  and  $P^{\text{co}} \rightarrow E^{\text{co}}$  are universal by the definitions of the coproduct and the coequaliser,  $P^{\text{co}} \rightarrow T$  factors uniquely through  $P^{\text{co}} \rightarrow E^{\text{co}}$ , making the diagram (6.42) commute and  $E^{\text{co}}$  the colimit object with morphisms  $\pi \circ \iota_v$  and  $\pi \circ \iota_w$ .  $\square$

Thus, we can construct the colimit of sets, topological spaces, and presheaves, using the constructions of Examples 6.34 - 6.36 and Examples 6.38-6.40. We make particular use of the colimit in topological spaces in the construction of colimits in locally ringed spaces.

**Example 6.43** (Colimits in Set). In Example 6.34, we saw that the coproduct in Set is the disjoint union of a family of objects  $(X(v))_{v \in V}$ . Taking morphisms  $e_{v,w} : v \rightarrow w$  into consideration, they give rise to a family of morphisms  $X(e_{v,w}) : X(v) \rightarrow X(w)$  for  $v, w \in V$ . We define the coproduct over the morphisms

$$\coprod_{e:v \rightarrow w \in E} X(v).$$

This gives rise to parallel morphisms

$$a_e := X(e) \circ \iota_w$$

$$b_e := \iota_v$$

with the inclusions  $\iota_v : X(v) \rightarrow \coprod_{w \in V} X(w)$ . The colimit is the coequaliser we saw in

Example 6.38, that is

$$\operatorname{colim} X \cong \coprod_{w \in V} X(w) \rightarrow \operatorname{coeq}(a, b)$$

which is a set of equivalence classes (see [3, Ch. 6, § 5, Proof 6.5.1]).

**Example 6.44** (Colimits in  $\mathbf{Top}$ ). In Example 6.35, we constructed the coproduct of topological spaces, which is exactly the disjoint union topology we saw in Example 2.23, and we saw in Example 6.39 that the coequaliser of topological spaces is the quotient topology we saw in Example 2.22. The colimit of topological spaces is the quotient space of the disjoint union topology space, such that the diagram commutes (see [3, Ch. 6, § 5, Proof 6.5.6]).

**Example 6.45** (Colimits in  $\mathbf{PSh}$ ). Let  $(F_i)_{i \in I} : \mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$  be a family of presheaves and let  $P^{\mathrm{co}} := \coprod_{i \in I} F_i$  with injections  $\iota_i : F_i \rightarrow P^{\mathrm{co}}$  be the coproduct we constructed in Example 6.36. Let  $T : \mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$  be an arbitrary presheaf with a family of natural transformations  $f_i : F_i \rightarrow T$ . Then there exists a unique natural transformation  $f : P^{\mathrm{co}} \rightarrow T$ . Let  $g_{ij} : F_i \rightarrow F_j$  be a collection of natural transformations. We define a presheaf

$$E^{\mathrm{co}} := \operatorname{coeq} \left( \prod_{i \in I} \iota_i, \prod_{i \in I} \iota_j \circ g_{ij} \right)$$

with  $\pi : P^{\mathrm{co}} \rightarrow E^{\mathrm{co}}$ . Then there exists a unique natural transformation  $\gamma : E^{\mathrm{co}} \rightarrow T$  such that  $f_i$  factorises over  $\gamma$ , and  $(E^{\mathrm{co}}, (\pi \circ \iota_i)_{i \in I})$  is the colimit of presheaves (see [17]).

In the next section, we see under which circumstances we are allowed to interchange limits and colimits. This is especially useful when we construct colimits on the category of locally ringed spaces.

## 6.3 The Commutativity of Directed Colimits and Finite Limits

In general, limits and colimits do not commute. However, in some categories such as  $\mathbf{Set}$ , finite limits commute with directed colimits. We have not seen the dual notion

of directed limits, since it is rarely used and does not share the same properties as directed colimits. Interestingly enough, directed colimits are a common concept.

**Definition 6.46** (Directed colimit). A colimit on a directed diagram (see Definition 3.8) is called **directed colimit** (see [3, Ch. 6, § 5, Def. 6.5.15]).

Filtered colimits are a special kind of colimit that arise naturally when studying stalks in locally ringed spaces. Since stalks can be defined as colimits over directed systems of neighbourhoods, understanding filtered colimits is essential for analysing local behaviour.

**Definition 6.47** (Filtered colimit). A **filtered colimit** is a colimit taken over a filtered category (see [9, Ch. 9, § 1]).

Notice, that although filtered colimits and directed colimits are closely related, but not the same. While every directed colimit is a filtered colimit, not any filtered colimit is a directed colimit. The following lemma allows us to restrict filtered colimits to cofinal indexing sets, which simplifies the handling of stalks.

**Lemma 6.48.** If  $\mathcal{I}$  is a filtered category and  $\mathcal{J}$  is a full subcategory such that every object in  $\mathcal{I}$  maps to some object in  $\mathcal{J}$ , then  $\mathcal{J}$  is filtered and cofinal in  $\mathcal{I}$ .

As a result, for any diagram  $X : \mathcal{I} \rightarrow \mathcal{C}$  in a cocomplete category,

$$\operatorname{colim}_{\mathcal{I}} X \cong \operatorname{colim}_{\mathcal{J}} X|_{\mathcal{J}}$$

(see [18]).

*Proof. Claim:*  $\mathcal{J}$  is filtered.

Since  $\mathcal{I}$  is filtered, it is non-empty. By assumption, there exists a morphism  $i \rightarrow j$  with  $i \in \operatorname{Ob}(\mathcal{I})$  and  $j \in \operatorname{Ob}(\mathcal{J})$ . Thus,  $\mathcal{J}$  is non-empty.

Let  $j_1, j_2 \in \operatorname{Ob}(\mathcal{J})$ . Since  $\mathcal{I}$  is filtered, there exists  $i \in \operatorname{Ob}(\mathcal{I})$  and morphisms  $j_1 \rightarrow i$  and  $j_2 \rightarrow i$ . By assumption, there exists another morphism  $i \rightarrow j_3$ , giving rise to

$$\begin{array}{c} j_1 \rightarrow i \rightarrow j_3 \\ j_2 \rightarrow i \rightarrow j_3 \end{array}$$

and thus, there exist morphisms  $j_1 \rightarrow j_3$  and  $j_2 \rightarrow j_3$  in  $\mathcal{J}$ .

Let  $f, g : \text{Hom}_{\mathcal{J}}(j_1 \rightarrow j_2)$ . Since  $\mathcal{I}$  is filtered, there exists a morphism  $\pi : j_2 \rightarrow i$  such that  $\pi \circ f = \pi \circ g$ . By assumption, there exists a morphism  $i \rightarrow j_3$  with  $j_3 \in \mathcal{J}$  and  $j_2 \rightarrow i \rightarrow j_3$  equalising  $f$  and  $g$  in  $\mathcal{J}$ . Hence,  $\mathcal{J}$  is filtered.

**Claim:**  $\mathcal{J} \hookrightarrow \mathcal{I}$  is cofinal.

By assumption of the lemma, for every  $i \in \text{Ob}(\mathcal{I})$  there exists a  $j \in \text{Ob}(\mathcal{J})$  and a morphism  $i \rightarrow j$ , and thus,  $\mathcal{J} \hookrightarrow \mathcal{I}$  is cofinal.

**Claim:**  $\text{colim}_{\mathcal{I}} X \cong \text{colim}_{\mathcal{J}} X \mid_{\mathcal{J}}$ .

By the definition of colimits, there exists a universal cone

$$(\phi_i : X(i) \rightarrow \text{colim}_{\mathcal{I}} X)_{i \in \mathcal{I}}$$

that is natural in  $i$ , such that any other cocone  $(\psi_i : X(i) \rightarrow Y)_{i \in \mathcal{I}}$  for some object  $Y$ , factors uniquely through  $(\phi_i)$ . Analogous, there exists a universal cone

$$(\lambda_j : X(j) \rightarrow \text{colim}_{\mathcal{J}} X \mid_{\mathcal{J}})_{j \in \mathcal{J}}$$

that is natural in  $j$ , such that any other cocone  $(\gamma_i : X(j) \rightarrow Z)_{j \in \mathcal{J}}$  for some object  $Z$ , factors uniquely through  $(\lambda_i)$ .

Our strategy is to construct a cocone  $(\alpha_i : X(i) \rightarrow \text{colim}_{\mathcal{J}} X \mid_{\mathcal{J}})_{i \in \mathcal{I}}$  that satisfies the universal property over  $\mathcal{I}$ .

Let  $i \in \mathcal{I}$ . Since  $\mathcal{J}$  is cofinal, there exists a morphism  $f_i : i \rightarrow j$  with  $j \in \text{Ob}(\mathcal{J})$ . Let

$$\alpha_i := \lambda_j \circ X(f_i) : X(i) \rightarrow X(j) \rightarrow \text{colim}_{\mathcal{J}} X \mid_{\mathcal{J}},$$

which is independent of our choice for  $j$ , since for any other morphism  $f_k : i \rightarrow k$ , the filteredness of  $\mathcal{I}$  gives a common target  $l \in \text{Ob}(\mathcal{J})$  with morphisms  $j \rightarrow l$  and  $k \rightarrow l$ .

The cocone  $\lambda$  ensures that both paths to  $\text{colim}_{\mathcal{J}} X \mid_{\mathcal{J}}$  agree. Thus, the cocone  $\alpha$  with tip  $\text{colim}_{\mathcal{J}} X \mid_{\mathcal{J}}$  is well-defined up to a unique morphism that is independent of the choice of  $j$ .

Assume  $\{\psi_i : X(i) \rightarrow Y\}_{i \in \mathcal{I}}$  is a cocone. Then its restriction to  $\mathcal{J}$  is a cocone

over  $X|_{\mathcal{J}}$ . Since  $\operatorname{colim}_{\mathcal{J}} X|_{\mathcal{J}}$  is the colimit of  $X|_{\mathcal{J}}$ , there exists a unique morphism  $f : \operatorname{colim}_{\mathcal{J}} X|_{\mathcal{J}} \rightarrow Y$  such that

$$\psi_j = f \circ \lambda_j$$

for all  $j \in \operatorname{Ob}(\mathcal{J})$ . For any  $i \in \operatorname{Ob}(\mathcal{I})$ , we defined  $\alpha_i := \lambda_j \circ X(f_i)$  and thus

$$f \circ \alpha_i = f \circ (\lambda_j \circ X(f_i)) = \psi_j \circ X(f_i) = \psi_i$$

by the naturality of the cone  $\psi$ . This shows that  $f \circ \alpha_i = \psi_i$  for all  $i$ , so the cone  $\alpha$  is universal. Hence,  $\operatorname{colim}_{\mathcal{I}} X \rightarrow \operatorname{colim}_{\mathcal{J}} X|_{\mathcal{J}}$  (see [9, Ch. 9, § 3]).  $\square$

A standard example of a directed colimit is the directed colimit in  $\mathbf{Set}$ . Understanding directed colimits in  $\mathbf{Set}$  makes the notion of directed colimits much more understandable.

**Example 6.49** (Directed colimit in  $\mathbf{Set}$ ). Let  $X = (X(v))_{v \in V}$  be a directed diagram of sets. We define an equivalence relation  $\sim$  on  $\coprod_{v \in V} X(v)$  by declaring  $x \sim y$  (for  $x \in X(v)$  and  $y \in X(w)$ ) if there exists  $k \in V$  with  $v \leq k$  and  $w \leq k$  such that  $X(v \rightarrow k)(x) = X(w \rightarrow k)(y)$ . The quotient set  $\coprod_{v \in V} X(v) / \sim$  is the colimit of  $X$ .

*Proof. Claim:*  $\sim$  is an equivalence relation:

It is reflexive, since every object  $X(v)$  is equipped with the identity, and thus,  $x \sim x$  holds for every element  $x \in X(v)$  for all  $v \in V$ .

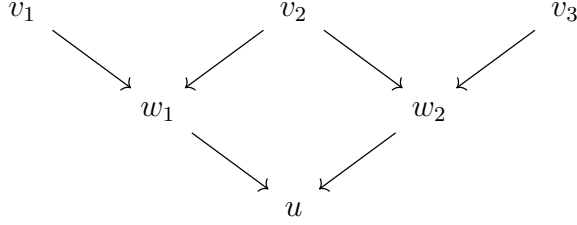
Assume that  $x \sim y$  holds (for  $x \in X(v)$  and  $y \in X(w)$ ). By definition of  $\sim$  and since  $X$  is directed, there exists  $k \in V$  such that  $v \leq k$  and  $w \leq k$  such that  $X(v \rightarrow k)(x) = X(w \rightarrow k)(y)$ . This is exactly the same as  $y \sim x$ .

Let  $v_1, v_2, v_3 \in V$  be vertices and let  $x \in X(v_1)$ ,  $y \in X(v_2)$  and  $z \in X(v_3)$  be elements such that  $x \sim y$  and  $y \sim z$ . By definition of  $\sim$  and since  $V$  is directed, there exist elements  $w_1, w_2 \in V$  such that  $v_1, v_2 \leq w_1$  and  $v_2, v_3 \leq w_2$  such that

$$\begin{aligned} X(v_1 \rightarrow w_1)(x) &= X(v_2 \rightarrow w_1)(y) =: x' \\ X(v_2 \rightarrow w_2)(y) &= X(v_3 \rightarrow w_2)(z) =: y'. \end{aligned}$$

We can apply the same argument once more: There exists  $u \in V$  such that  $w_1, w_2 \leq u$  with  $X(w_1 \rightarrow u)(x') = X(w_2 \rightarrow u)(y')$ . Thus,  $\sim$  is transitive and

hence an equivalence relation. The diagram



visualises our strategy. Assume  $v \leq w$ , then for every  $x \in X(v)$ , it holds that

$$x \sim X(v \rightarrow w)(x)$$

with  $X(v \rightarrow w)(x) \in X(w)$ . Thus, we get a cocone with tip  $\coprod_{v \in V} X(v) / \sim$ .

**Claim:** The cocone is universal.

Consider a family of morphisms

$$(f_v : X(v) \rightarrow T)_{v \in V}$$

to some object  $T$ , such that for every  $v \leq w$  the diagram

$$\begin{array}{ccc} X(v) & \xrightarrow{X(v \rightarrow w)} & X(w) \\ & \searrow f_v \quad \swarrow f_w & \\ & T & \end{array}$$

commutes. Then for every  $x \in X(v)$ , any element  $f_v(x) \in T$  depends only on the equivalence classes of  $x$  regarding  $\sim$ . Thus, we can continue the morphisms  $f_v$  uniquely on  $\coprod_{v \in V} X(v) / \sim$  (see [3, Ch. 6, § 5, Ex. 6.5.16]).  $\square$

While the following theorem is not used in this thesis, it deserves to be stated here for the sake of completeness.

**Theorem 6.50** (Finite limits commute with directed colimits). Let  $\Gamma = (V, E)$  be a finite graph and let  $(J, \leq)$  be a directed poset. Then the limit of shape  $\Gamma$  commutes with the directed colimit of shape  $(J, \leq)$  (see [3, Ch. 6, § 6, Thm. 6.6.16]).

*Remark 6.51.* Commutativity in the second variable means that

$$\begin{array}{ccc}
 X(v, i) & \xrightarrow{X(e, \text{id}_i)} & X(w, i) \\
 X(\text{id}_v, g) \downarrow & & \downarrow X(\text{id}_w, g) \\
 X(v, j) & \xrightarrow{X(e, \text{id}_j)} & X(w, j)
 \end{array} \tag{6.52}$$

commutes. where  $e : v \rightarrow w \in E$  and  $g : i \rightarrow j$  is a morphism in  $J$  (see [3, Ch. 6, § 6, Proof 6.6.16]).

A proof can be consulted in [3, Ch. 6, § 6, Proof 6.6.16] or in Chapter C. When we work with locally ringed spaces, we come across exact sequences.

**Definition 6.53** (Exact sequence). Let  $\mathcal{C}$  be an arbitrary category with a zero object  $0$ . The **kernel** of a morphism  $f : A \rightarrow B$  is an equaliser of  $f$  and the unique morphism  $0_{A,B} : A \rightarrow B$  which factorises over  $0$ . The **cokernel** of  $f$  is the coequaliser of  $f$  and  $0_{A,B}$ .

A sequence of morphisms  $0 \rightarrow A \rightarrow B \rightarrow C$  is **exact** if  $A \rightarrow B$  is a kernel of  $B \rightarrow C$ . Dually, a sequence  $A \rightarrow B \rightarrow C \rightarrow 0$  is **exact** if  $B \rightarrow C$  is a cokernel of  $A \rightarrow B$ . Finally, a sequence  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  is **exact** if both  $0 \rightarrow A \rightarrow B \rightarrow C$  and  $A \rightarrow B \rightarrow C \rightarrow 0$  are exact (see [3, Ch. 6, § 3, Def. 6.3.14]).

# 7 Construction of Colimits in the Category of Locally Ringed Spaces

In this chapter, we study the category of locally ringed spaces. We discuss how the objects and the morphisms are defined in Section 7.1 and proceed to construct coproducts in Section 7.2 and coequalisers in Section 7.3. In Section 7.4, we use our gained knowledge to construct colimits in the category of locally ringed spaces. For the constructions of coproducts, coequalisers, and colimits, we follow [12, Ch.1, §1, Thm. 1.6] and present its proof in full detail.

## 7.1 Introduction to Locally Ringed Spaces

Locally ringed spaces consist of the category  $\text{Open}(X)^{\text{op}}$ , the dual category to the one we saw in Example 3.14, and a structure sheaf. Sheaves are presheaves that satisfy the equaliser property, which makes sheaves agree on sections of overlapping sets.

**Definition 7.1 (Sheaf).** Let  $F \in \text{Ob}(\text{PSh}(X))$  and let  $U = \bigcup_{i \in I} U_i$  be a covering. If  $F$  satisfies the **equaliser condition**

$$F(U) \rightarrow \prod_{i \in I} F(U_i) \rightrightarrows \prod_{i, j \in I} F(U_i \cap U_j) \quad (7.2)$$



we call  $F$  a **sheaf**. The parallel morphisms are the **restrictions**

$$\begin{aligned} s_i &\mapsto s_i|_{U_i \cap U_j} \\ s_j &\mapsto s_j|_{U_i \cap U_j} . \end{aligned}$$

That is if  $s_i \in F(U_i)$  are restrictions for  $i \in I$  such that

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

in  $F(U_i \cap U_j)$  for all  $i, j \in I$ , then there exists a restriction  $s \in F(U)$  such that

$$s|_{U_i} = s_i$$

(see [3, Ch. 7, § 5, Ex. 7.5.5]).

- Remark 7.3.*
1. As we saw in Chapter 6, presheaves have all limits and colimits, making  $\mathbf{PSh}$  complete and cocomplete. Furthermore, limits in  $\mathbf{PSh}$  can be interchanged.
  2. Sheaves form a category. We denote the category of sheaves on  $X$  by  $\mathbf{Sh}(X)$ . They are a full subcategory of presheaves ([3, Ch. 7, § 6, Rem. 7.6.11]).
  3. Since limits commute with limits as we saw in Theorem 6.28,  $\mathbf{Sh}$  is closed under limits and thus, a complete category ([3, Ch. 7, § 6, Rem. 7.6.11]).
  4. Furthermore,  $\mathbf{Sh} \subseteq \mathbf{PSh}$  is reflective. The reflector maps a presheaf  $F$  to its associated sheaf  $F^\#$ . This makes  $\mathbf{Sh}(X)$  cocomplete ([3, Ch. 7, § 6, Rem. 7.6.11]).

Sheaves are defined on an open subset  $U$  of a topological space  $X$ . If we want to know how the sheaf behaves at one point  $x \in U$ , we can study the behaviour of the colimit object, indexed by a family of open subsets  $U_i$  such that  $x \in U_i$  for all  $i \in I$ . This motivates the definition of stalks.

**Definition 7.4** (Stalk and germ). Let  $X$  be a topological space and let  $F$  be a sheaf on  $X$ . The directed colimit

$$F_x := \operatorname{colim}_{x \in U} F(U)$$

over the directed poset  $(U(x), \supseteq)$  of open neighbourhoods of  $x$  is the **stalk**. **Germ**s are the equivalence classes  $[s]_x$  of elements  $s \in F(U(x))$ , such that

$$[s]_x = [s|_V]_x$$

for  $V \subseteq U(x)$  (see [3, Ch. 6, § 6, Ex. 6.6.17]).

*Remark 7.5.* According to Theorem C.1, the stalk functor

$$\begin{aligned} \text{PSh}(X) &\rightarrow \text{Set} \\ F &\mapsto F_x \end{aligned}$$

preserves finite limits (see [3, Ch. 6, § 6, Ex. 6.6.17]).

With these foundational concepts in place, we define locally ringed spaces.

**Definition 7.6** (Locally ringed space). A **locally ringed space** is a pair  $(X, \mathcal{O}_X)$ , where  $X$  is a topological space and  $\mathcal{O}_X$  is a sheaf

$$\mathcal{O}_X : \text{Open}(X)^{\text{op}} \rightarrow \text{CRing}$$

such that all stalks  $\mathcal{O}_{X,x}$  are local rings (see [3, Ch. 7, § 5, Ex. 7.5.5], [12, Ch. 1, § 1, Def. 1.1]).

Locally ringed spaces form the category,  $\text{LRS}$ , where the objects are locally ringed spaces. How can we define morphisms between locally ringed spaces? To understand these morphisms, we need to understand two special kinds of sheaves:

**Definition 7.7** (Direct image sheaf). Let  $\mathcal{C}$  be a cocomplete category. Let  $X$  and  $Y$  be topological spaces and let  $f : X \rightarrow Y$  be a continuous function. Let  $\mathcal{O}_X$  be a  $\mathcal{C}$ -valued sheaf on  $X$  and let  $\mathcal{O}_Y$  be a  $\mathcal{C}$ -valued sheaf on  $Y$ . The **direct image sheaf** (also called **push forward sheaf**) on  $Y$  is defined as

$$(f_*\mathcal{O}_X)(V) := \mathcal{O}_X(f^{-1}(V))$$

for every open subset  $V \subseteq Y$  (see [3, Ch. 9, § 5, Ex. 9.5.12]).

The inverse image sheaf is more subtle than the direct image sheaf and relies on the *sheafification* of presheaves.

**Definition 7.8** (Inverse image sheaf). The **inverse image presheaf** is defined as

$$(f^{\text{pre}}\mathcal{O}_Y)(U) := \operatorname{colim}_{V \subseteq Y \text{ open with } U \subseteq f^{-1}(V)} \mathcal{O}_Y(V)$$

for every open subset  $U \subseteq X$ . Sheafification of the inverse image presheaf gives rise to the **inverse image sheaf** (also called **pullback functor**)

$$(f^{-1}\mathcal{O}_Y) := (f^{\text{pre}}\mathcal{O}_Y)^{\#}$$

(see [3, Ch. 9, § 5, Ex. 9.5.12]).

**Remark 7.9.** 1.  $f_*\mathcal{O}_X$  is a sheaf on  $Y$ .

2. The functor  $f_* : \operatorname{Sh}(X) \rightarrow \operatorname{Sh}(Y)$  is left exact and preserves limits.
3.  $(f^{-1}\mathcal{O}_Y)$  is a sheaf on  $X$ .
4.  $f^{-1}$  is left adjoint to  $f_*$  at the level of sheaves of sets.
5. Let  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  be continuous functions between topological spaces. Then the following properties are satisfied:

- a)  $(g \circ f)_* = g_* \circ f_*$
- b)  $(g \circ f)^{-1} \cong f^{-1} \circ g^{-1}$
- c)  $(g \circ f)^* \cong f^* \circ g^*$ .

(see [3, Ch. 9, § 5, Ex. 9.5.12]).

**Definition 7.10** (Category of locally ringed spaces LRS). Let  $(X, \mathcal{O}_X)$  and  $(Y, \mathcal{O}_Y)$  be locally ringed spaces and let

$$f : X \rightarrow Y$$

be a morphism between the underlying topological spaces (like we defined it in Example 3.13). A morphism between two locally ringed spaces  $(X, \mathcal{O}_X)$  and  $(Y, \mathcal{O}_Y)$

is a pair of morphisms

$$(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y),$$

where  $f^\#$  is a morphism between sheaves of CRing

$$f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X. \quad (7.11)$$

Furthermore, for any  $x \in X$ , we require that the unique ring morphism

$$f_x^\# : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x} \quad (7.12)$$

such that for all open subsets  $V \subseteq Y$  with  $f(x) \in V$ , the diagram

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \longrightarrow & f_* \mathcal{O}_X(V) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y, f(x)} & \xrightarrow{\exists! f_x^\#} & \mathcal{O}_{X, x} \end{array} \quad (7.13)$$

commutes to be local (see [3, Ch. 7, § 5, Ex. 7.5.5]). In essence, for any point  $x \in X$  and open subsets  $V \subseteq Y$  such that  $f(x) \in V$ , the compositions

$$\mathcal{O}_Y(V) \xrightarrow{f^\#(V)} \mathcal{O}_X(f^{-1}(V)) \longrightarrow \mathcal{O}_{X, x}$$

are compatible with restriction morphisms of  $\mathcal{O}_Y$ . Hence, they induce a ring homomorphism

$$f_x^\# : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$$

as claimed (see [2, Ch. 6 § 6, Rem. 7], [12, Ch. 1, § 1, Def. 1.4]).

**Remark 7.14.** We can compose morphisms between locally ringed spaces

$f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  and  $g : (Y, \mathcal{O}_Y) \rightarrow (Z, \mathcal{O}_Z)$  as follows:

The composite  $g \circ f : X \rightarrow Z$  is the composite of continuous functions in  $\mathbf{Top}$ . On the level of sheaves, the composite of

$$\begin{aligned} f^\# : \mathcal{O}_Y &\rightarrow f_* \mathcal{O}_X \\ g^\# : \mathcal{O}_Z &\rightarrow g_* \mathcal{O}_Y. \end{aligned}$$

is defined as

$$(g \circ f)^\# = g^\# \circ g_* f^\# : \mathcal{O}_Z \rightarrow (g \circ f)_* \mathcal{O}_X.$$

This gives rise to the following diagram:

$$\mathcal{O}_Z \xrightarrow{g^\#} g_* \mathcal{O}_Y \xrightarrow{g_* f^\#} g_* \circ f_* \mathcal{O}_X.$$

**Claim:** On the level of the local rings, the composite of local ring homomorphisms

$$(g \circ f)_x^\# = g_{f(x)}^\# \circ f_x^\# : \mathcal{O}_{Z, (g \circ f)(x)} \rightarrow \mathcal{O}_{X, x}$$

is a local ring homomorphism.

The stalks  $\mathcal{O}_{X, x}$  are filtered colimits of the sections of  $\mathcal{O}_X$  over all open neighbourhoods of  $x$ ,  $\mathcal{U}(x)$ . For each  $U \in \mathcal{U}(x)$ , there exists a canonical ring homomorphism

$$\begin{aligned} \lambda_U : \mathcal{O}_X(U) &\rightarrow \mathcal{O}_{X, x} \\ s &\mapsto [s]_x, \end{aligned}$$

where  $s \in \mathcal{O}_X(U)$  is a section. The universal property of colimits states, that given an arbitrary commutative ring  $T$  and a compatible family of ring homomorphisms

$$(f_U : \mathcal{O}_X(U) \rightarrow T)_{U \in \mathcal{U}(x)}$$

with

$$f_V = f_U|_V$$

for  $U \subseteq V$ , there exists a unique ring homomorphism

$$f : \mathcal{O}_{X, x} \rightarrow T$$

such that

$$f \circ \lambda_U = f_U.$$

Let  $x \in X$  and  $y = f(x) \in Y$  and  $z = g(y) = (g \circ f)(x) \in Z$ . For an open neighbourhood  $W$  of  $z$ , we consider the canonical morphism

$$\lambda_W : \mathcal{O}_Z(W) \rightarrow \mathcal{O}_{Z, z}.$$

Due to the universal property of the colimit, it is sufficient to show that

$$(g \circ f)_x^\# \circ \lambda_W = (g_y^\# \circ f_x^\#) \circ \lambda_W \quad (7.15)$$

holds for any open neighbourhood  $W$  of  $z$ . The diagram

$$\begin{array}{ccccc}
 & & (g^\# \circ f^\#)(W) = f^\#(g^{-1}(W)) \circ g^\#(W) & & \\
 & \nearrow & & \searrow & \\
 \mathcal{O}_Z(W) & \xrightarrow{g^\#(W)} & \mathcal{O}_Y(g^{-1}(W)) & \xrightarrow{(g_*(f^\#))(W)} & \mathcal{O}_X(f^{-1}(g^{-1}(W))) \\
 \downarrow \lambda_W & & \downarrow \lambda_{g^{-1}(W)} & & \downarrow \lambda_{(g \circ f)^{-1}(W)} \\
 \mathcal{O}_{Z,z} & \xrightarrow{g_{f(x)}^\#} & \mathcal{O}_{Y,f(x)} & \xrightarrow{f_x^\#} & \mathcal{O}_{X,x} \\
 & \searrow & & \nearrow & \\
 & & (g^\# \circ f^\#)_x & & 
 \end{array}$$

visualises that. Notice, that the left half of the diagram

$$\begin{array}{ccc}
 \mathcal{O}_Z(W) & \xrightarrow{g^\#(W)} & \mathcal{O}_Y(g^{-1}(W)) \\
 \downarrow \lambda_W & & \downarrow \lambda_{g^{-1}(W)} \\
 \mathcal{O}_{Z,z} & \xrightarrow{g_{f(x)}^\#} & \mathcal{O}_{Y,f(x)}
 \end{array}$$

as well as the right half of the diagram

$$\begin{array}{ccc}
 \mathcal{O}_Y(g^{-1}(W)) & \xrightarrow{(g_*(f^\#))(W)} & \mathcal{O}_X(f^{-1}(g^{-1}(W))) \\
 \downarrow \lambda_{g^{-1}(W)} & & \downarrow \lambda_{(g \circ f)^{-1}(W)} \\
 \mathcal{O}_{Y,f(x)} & \xrightarrow{f_x^\#} & \mathcal{O}_{X,x}
 \end{array}$$

commute by (7.13). Thus, for an open neighbourhood  $W$  of  $z = (g \circ f)(x)$ , (7.15) holds. By the universal property of the colimit  $(\mathcal{O}_{Z,z})$ , it holds that

$$(g^\# \circ f^\#)_x = f_x^\# \circ g_{f(x)}^\#.$$

**Corollary 7.16.** Let  $(X, \mathcal{O}_X)$  be a locally ringed space. Then  $\mathcal{O}_X(\emptyset) = \{0\}$  by Example 3.21.

*Proof.*  $\emptyset$  is the initial object in  $\text{Open}(X)$ , hence a terminal object in  $\text{Open}(X)^{\text{op}}$ . As we saw in Example 3.21, the zero ring is a terminal object in  $\text{CRing}$ . The equaliser condition of sheaves satisfies the exact sequence

$$\mathcal{O}_X(\emptyset) \rightarrow \prod_{i \in \emptyset} \mathcal{O}_X(U_i) \rightrightarrows \prod_{i,j \in \emptyset} \mathcal{O}_X(U_i \cap U_j) \quad (7.17)$$

for any open cover  $\emptyset = \bigcup_{i \in I} U_i$ . As we saw in Example 6.12, the product over an empty set is the terminal object in the target category, which is  $\{0\}$  in  $\text{CRing}$ . Therefore, (7.17) becomes

$$\mathcal{O}_X(\emptyset) \rightarrow \{0\} \rightrightarrows \{0\},$$

which forces  $\mathcal{O}_X(\emptyset) = \{0\}$  (see [3, Ch. 2, § 5, Ex. 2.5.4, 6]).  $\square$

We can consider the set of points, where global sections are invertible. These sets are called *distinguished principal open subsets* and connect localisations in Algebra and open sets in geometry.

**Lemma 7.18.** Let  $(X, \mathcal{O}_X)$  be a locally ringed space and let  $f \in \mathcal{O}_X(X)$  be a global function on  $X$  and let  $\mathfrak{m}_x$  be the maximal ideal of  $\mathcal{O}_{X,x}$  and  $f_x$  the germ. The distinguished principal open subset

$$D(f) = \{x \in X \mid \text{im } f_x \notin \mathfrak{m}_x\} \quad (7.19)$$

is open and  $f|_{D(f)}$  has an inverse and therefore  $f$  is locally a unit in  $D(f)$  (see [19]).

*Proof.* Let  $U, V \subseteq X$  be open subsets such that  $f|_U$  has an inverse  $g$  and  $f|_V$  has an inverse  $h$ . By definition, it holds that

$$g|_{U \cap V} = h|_{U \cap V}.$$

Therefore, it is sufficient to show that  $f$  is invertible on an open neighbourhood of any  $x \in D(f)$ . Since  $f_x \notin \mathfrak{m}_x$ ,  $f_x$  is a unit, there exists  $g_x$  such that  $f_x \cdot g_x = 1$ . Thus,  $g_x$  is the inverse of  $f_x$ . Moreover, there exists an open neighbourhood  $x \in U \subseteq X$  so that  $g \in \mathcal{O}_X(U)$  and  $g \cdot f|_U = 1$  (see [19]).  $\square$

**Remark 7.20.** For an open subset  $U \subseteq X$  be open and let  $(U, \mathcal{O}_{X|U})$  be a restriction of the locally ringed space  $(X, \mathcal{O}_X)$ . For  $f \in \mathcal{O}_X(U)$ , we define

$$D_U(f) := \{x \in U \mid \text{im } f_x \notin \mathfrak{m}_x\}.$$

Furthermore, distinguished principal open subsets allow us to characterise the domain.

**Lemma 7.21.** Let  $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  be a morphism of locally ringed spaces. Let  $V \subseteq Y$  be open, and for  $s \in \mathcal{O}_Y(V)$  let  $D_V(s)$  be the principal open subset of  $V$ . Let  $U = f^{-1}(V) \subseteq X$  and  $t = f^\#(V)(s) \in \mathcal{O}_X(U)$ .

Then  $f^{-1}(D_V(s)) = D_U(t)$  is a principal open subset of  $U$ .

*Proof.* First, we show that  $f^{-1}(D_V(s)) \subseteq D_U(t)$ . Let  $x \in f^{-1}(D_V(s))$ . By the definition of the preimage,  $x \in X$  and  $y := f(x) \in D_V(s)$ . The condition that  $y \in D_V(s)$  means that

1.  $y \in V$
2. The section  $s$  is invertible in the stalk  $\mathcal{O}_{Y,y}$ . This is equivalent to saying that the germ of  $s$  at  $y$ , denoted by  $[s]_y \in \mathcal{O}_{Y,y}$  is a unit in the local ring  $\mathcal{O}_{Y,y}$ .

Since  $y = f(x) \in V$ , it follows that  $x \in f^{-1}(V) = U$ . Thus, the first condition for  $x$  to be in  $D_U(t)$  is satisfied.

Furthermore, it is to show that  $t$  is invertible in the stalk  $\mathcal{O}_{X,x}$ . Let  $[t]_x \in \mathcal{O}_{X,x}$  be the germ of  $t$  at  $x$ . By the definition of the induced map on stalks,

$$[t]_x = f_x^\#([s]_y),$$

that is, the germ  $[t]_x$  is the image of the germ  $[s]_y$  under the map  $f_x^\#$ . Since  $(f, f^\#)$  is a morphism of locally ringed spaces, the map

$$f_x^\# : \mathcal{O}_{Y,y} \rightarrow \mathcal{O}_{X,x}$$

is a local ring homomorphism. Thus if an element is a unit in the domain, its image is a unit in the codomain. Since we know that  $[s]_y$  is a unit in  $\mathcal{O}_{Y,y}$ , it follows that its image  $[t]_x = f_x^\#([s]_y)$  is a unit in  $\mathcal{O}_{X,x}$ . Thus, the section  $t$  is invertible at the



point  $x$ . Thus,  $x \in D_U(t)$ . Since our choice of  $x$  was arbitrary, we have shown that  $f^{-1}(D_V(s)) \subseteq D_U(t)$ .

We want to prove that  $D_U(t) \subseteq f^{-1}(D_V(s))$ .

Let  $x \in D_U(t)$ . We want to prove that  $x \in f^{-1}(D_V(s))$ . Since  $x \in D_U(t)$ , it holds that

1.  $x \in U$ .
2. The section  $t$  is invertible in the stalk  $\mathcal{O}_{X,x}$ , that is, its germ  $[t]_x \in \mathcal{O}_{X,x}$  is a unit in the local ring  $\mathcal{O}_{X,x}$ .

Since  $x \in U = f^{-1}(V)$ , it follows that  $y := f(x) \in V$ . Thus, the first condition for  $y$  to be in  $D_V(s)$  is satisfied.

We want to show that  $s$  is invertible in the stalk  $\mathcal{O}_{Y,y}$ . Let  $[s]_y \in \mathcal{O}_{Y,y}$  be the germ of  $s$  at  $y$ . By the relation

$$[t]_x = f_x^\#([s]_y)$$

via the local ring homomorphism  $f_x^\# : \mathcal{O}_{Y,y} \rightarrow \mathcal{O}_{X,x}$ . We know that  $[t]_x$  is a unit in  $\mathcal{O}_{X,x}$ . If the image of an element is a unit, the element itself must have been a unit. Since  $f_x^\#([s]_y) = [t]_x$  is a unit in  $\mathcal{O}_{X,x}$ , the element  $[s]_y$  must be a unit in  $\mathcal{O}_{Y,y}$ . Thus,  $s$  is a unit and therefore invertible at the point  $y$ . Hence,  $y = f(x) \in D_V(s)$ , which implies that  $x \in f^{-1}(D_V(s))$ .

We may conclude that  $D_U(t) = f^{-1}(D_V(s))$ . □

Now that we understood locally ringed spaces, we can construct the coproduct in LRS.

## 7.2 Coproducts in LRS

Knowing what locally ringed spaces and their morphisms are, we can construct the coproduct of locally ringed spaces. Therefore, let  $(X_i, \mathcal{O}_{X_i})_{i \in I}$  be a family of locally ringed spaces. In Chapter 6, we saw the coproduct of topological spaces. We use that construction on the underlying topological spaces and let

$$P^{\text{co}} := \coprod_{i \in I} X_i$$

be the disjoint union of the underlying sets, equipped with the disjoint union topology (see Example 2.23) and canonical injections

$$\iota_i : X_i \rightarrow P^{\text{co}}.$$

Furthermore, there exists a unique morphism

$$f : P^{\text{co}} \rightarrow T$$

for an arbitrary topological space  $T$ , such that for any family of morphisms

$$(f_i : X_i \rightarrow T)_{i \in I},$$

it holds that  $f_i = f \circ \iota_i$ .

We want to expand that construction on locally ringed spaces. Therefore, we construct an object  $\mathcal{O}_{P^{\text{co}}}$  and show that it is a structure sheaf, that  $(P^{\text{co}}, \mathcal{O}_{P^{\text{co}}})$  is a locally ringed space and that  $((P^{\text{co}}, \mathcal{O}_{P^{\text{co}}}), (\iota_i, \iota_i^\#)_{i \in I})$  is the coproduct (see [12, Ch. 1, § 1, Thm. 1.6]). To construct  $\mathcal{O}_{P^{\text{co}}}$ , we use the direct image sheaf (see Definition 7.7), since morphisms induce pullbacks on structure sheaves in LRS. Furthermore, since the direct image functor is contravariant on spaces but covariant on sheaves, we use the product of sheaves to make the morphisms between the structure sheaves compatible with the universal property. For any  $Q \subseteq P^{\text{co}}$ , we define

$$\mathcal{O}_{P^{\text{co}}} := \prod_{i \in I} (\iota_i)_* \mathcal{O}_{X_i}.$$

For any  $Q \in P^{\text{co}}$ , it holds that

$$\begin{aligned} \mathcal{O}_{P^{\text{co}}}(Q) &= \prod_{i \in I} (\iota_i)_* \mathcal{O}_{X_i}(Q) \\ &= \prod_{i \in I} \mathcal{O}_{X_i}(\iota_i^{-1}(Q)) \\ &= \prod_{i \in I} \mathcal{O}_{X_i}(X_i \cap Q). \end{aligned}$$

**Claim:**  $\mathcal{O}_{P^{\text{co}}}$  is a structure sheaf.

By the construction of the product of presheaves in Example 6.13,  $\mathcal{O}_{P^{\text{co}}}$  is a presheaf

on  $P^{\text{co}}$ . Thus it is left to show that the equaliser condition is satisfied. Let  $Q = \bigcup_{j \in J} Q_j$  be an arbitrary covering. Since  $\mathcal{O}_{X_i}$  is a sheaf for all  $i \in I$ , any  $\mathcal{O}_{X_i}$  satisfies the equaliser condition

$$\mathcal{O}_{X_i}(Q) = \text{eq} \left( \prod_{j \in J} \mathcal{O}_{X_i}(X_i \cap Q_j) \rightrightarrows \prod_{j,k \in J} \mathcal{O}_{X_i}(X_i \cap Q_j \cap Q_k) \right).$$

By Theorem 6.28, equalisers commute with products. Therefore

$$\begin{aligned} \mathcal{O}_{P^{\text{co}}}(Q) &= \prod_{i \in I} (\iota_i)_* \mathcal{O}_{X_i}(Q) \\ &= \prod_{i \in I} \mathcal{O}_{X_i}(\iota_i^{-1}(Q)) \\ &= \prod_{i \in I} \mathcal{O}_{X_i}(X_i \cap Q) \\ &= \prod_{i \in I} \text{eq} \left( \prod_{j \in J} \mathcal{O}_{X_i}(X_i \cap Q_j) \rightrightarrows \prod_{j,k \in J} \mathcal{O}_{X_i}(X_i \cap Q_j \cap Q_k) \right) \\ &\cong \text{eq} \left( \prod_{i \in I} \prod_{j \in J} \mathcal{O}_{X_i}(X_i \cap Q_j) \rightrightarrows \prod_{i \in I} \prod_{j,k \in J} \mathcal{O}_{X_i}(X_i \cap Q_j \cap Q_k) \right) \\ &\cong \text{eq} \left( \prod_{j \in J} \prod_{i \in I} \mathcal{O}_{X_i}(X_i \cap Q_j) \rightrightarrows \prod_{j,k \in J} \prod_{i \in I} \mathcal{O}_{X_i}(X_i \cap Q_j \cap Q_k) \right) \\ &= \text{eq} \left( \prod_{j \in J} \mathcal{O}_{P^{\text{co}}}(P^{\text{co}} \cap Q_j) \rightrightarrows \prod_{j,k \in J} \mathcal{O}_{P^{\text{co}}}(P^{\text{co}} \cap Q_j \cap Q_k) \right). \end{aligned} \tag{7.22}$$

Hence, the equaliser condition holds and  $\mathcal{O}_{P^{\text{co}}}$  is a structure sheaf on  $P^{\text{co}}$ .

**Claim:** The object  $(P^{\text{co}}, \mathcal{O}_{P^{\text{co}}})$  is a locally ringed space, and the morphisms  $(\iota_i^{\#})_x$  are local ring isomorphisms.

To show that  $(P^{\text{co}}, \mathcal{O}_{P^{\text{co}}})$  is a locally ringed space, we want to show that the stalks

$$\mathcal{O}_{P^{\text{co}},p} = \text{colim}_{x \in Q} \mathcal{O}_{P^{\text{co}}} = \text{colim}_{x \in Q} \prod_{i \in I} \mathcal{O}_{X_i}(X_i \cap Q)$$

are local rings for any  $p \in P^{\text{co}}$ . Note that we take the colimit over the set of neigh-

bourhoods of  $p$ ,

$$\mathcal{U}_{P^{\text{co}}}(p) := \{Q \subseteq P^{\text{co}} \mid x \in Q, Q \text{ open in } P^{\text{co}}\}.$$

Since  $p$  lies in exactly one of the underlying topological spaces  $X_j$  by the definition of the coproduct of topological spaces, Lemma 6.48 allows us to restrict the colimit on the cofinal subset

$$\mathcal{U}_{X_j}(p) := \{Q \cap X_j \mid Q \in \mathcal{U}_{P^{\text{co}}}(p)\}.$$

Thus,  $\mathcal{O}_{P^{\text{co}}, \iota_j(p)} = \text{colim}_{U \in \mathcal{U}_{X_j}(p)} \mathcal{O}_{P^{\text{co}}}(U)$ . Since

$$\begin{aligned} \prod_{i \in I} \mathcal{O}_{X_i}(X_i \cap U) &= \prod_{i \in I \setminus \{j\}} \underbrace{\mathcal{O}_{X_i}(X_i \cap U)}_{=\emptyset} \times \mathcal{O}_{X_j}(X_j \cap U) \\ &= \prod_{i \in I \setminus \{j\}} \underbrace{\mathcal{O}_{X_i}(\emptyset)}_{=\{0\}} \times \mathcal{O}_{X_j}(X_j \cap U) \\ &\cong \{0\} \times \mathcal{O}_{X_j}(X_j \cap U) \\ &\cong \mathcal{O}_{X_j}(X_j \cap U), \end{aligned}$$

it holds that

$$\prod_{i \in I} \mathcal{O}_{X_i}(X_i \cap U) \cong \mathcal{O}_{X_j}(U).$$

Therefore, for  $U \in \mathcal{U}_{X_j}(p)$ ,

$$\begin{array}{ccc} \underbrace{\prod_{i \in I} \mathcal{O}_{X_i}(X_i \cap U)}_{=\mathcal{O}_{P^{\text{co}}}(U)} & \xrightarrow{\text{pr}_j} & \mathcal{O}_{X_j}(X_j \cap U) \\ \parallel & & \parallel_{U \subseteq X_j} \\ \mathcal{O}_{X_j}(U) \times \prod_{i \in I \setminus \{j\}} \mathcal{O}_{X_i}(X_i \cap U) & \xrightarrow{\cong} & \mathcal{O}_{X_j}(U) \end{array}$$

is commutative, and thus, the canonical projection  $\text{pr}_j$  is an isomorphism. Thus, the

diagram

$$\begin{array}{ccc}
 \mathcal{O}_{P^{\text{co}}}(U) & \xrightarrow{\text{pr}_j = \iota_j^\#(U)} & (\iota_j)_* \mathcal{O}_{X_j}(U) = \mathcal{O}_{X_j}(U) \\
 \downarrow & & \downarrow \\
 \mathcal{O}_{P^{\text{co}}, \iota_j(p)} & \xrightarrow{(\iota_j^\#)_p} & \mathcal{O}_{X_j, p}
 \end{array} \tag{7.23}$$

commutes. Since  $(\iota_j^\#)_p$  is a colimit of  $(\mathcal{O}_{P^{\text{co}}} \rightarrow \mathcal{O}_{X_j}(U))_{U \in \mathcal{U}_{X_j}(p)}$ , it follows that  $(\iota_j^\#)_p$  is an isomorphism (see 6.5). Since  $(X_j, \mathcal{O}_{X_j})$  is a locally ringed space,  $\mathcal{O}_{X_j, p}$  is a local ring and thus,  $(P^{\text{co}}, \mathcal{O}_{P^{\text{co}}})$  is a locally ringed space.

**Claim:** The ring morphisms  $(\iota_i^\#)_x : \mathcal{O}_{P^{\text{co}}, \iota_i(x)} \rightarrow \mathcal{O}_{X_i, x}$  are local.

Since the diagram (7.23) in the previous claim commutes, the ring morphisms  $(\iota_i^\#)_x$  are local.

**Claim:** The object  $(P^{\text{co}}, \mathcal{O}_{P^{\text{co}}})$ , together with a family of morphisms  $(\iota_i, \iota_i^\#)_{i \in I}$ , is a coproduct in LRS.

Let  $(T, \mathcal{O}_T)$  be an arbitrary locally ringed space with a family of morphisms

$$((f_i, f_i^\#) : (X_i, \mathcal{O}_{X_i}) \rightarrow (P^{\text{co}}, \mathcal{O}_{P^{\text{co}}}))_{i \in I}.$$

On the level of topological spaces, we defined the unique morphism

$$\begin{aligned}
 f : P^{\text{co}} &\rightarrow T \\
 f(Q) &= (f_i(Q))_{i \in I}
 \end{aligned}$$

such that  $f_i = f \circ \iota_i$  holds. On the level of structure sheaves, the definition of such a morphism is much more subtle. We apply  $f_*$  to

$$\iota_j^\# : \mathcal{O}_{P^{\text{co}}} \rightarrow (\iota_j)_* \mathcal{O}_{X_j}$$

and get

$$f_*(\iota_j^\#) : f_* \mathcal{O}_{P^{\text{co}}} \rightarrow f_* \circ (\iota_j)_* \mathcal{O}_{X_j}.$$

Since

$$\begin{aligned}
 f_* \circ (\iota_j)_* \mathcal{O}_{X_j} &= \underbrace{(f \circ \iota_j)}_{=f_j} \mathcal{O}_{X_j} \\
 &= (f_j)_* \mathcal{O}_{X_j},
 \end{aligned}$$

it follows that

$$f_*(\iota_j^\#) : f_*\mathcal{O}_{P^{\text{co}}} \rightarrow (f_j)_*\mathcal{O}_{X_j}.$$

Moreover, since  $f_*$  is right adjoint, limits commute with  $f_*$  and therefore,  $f_*\mathcal{O}_{P^{\text{co}}}$  is the product of

$$f_* \circ (\iota_j)_*\mathcal{O}_{X_j} = (f_j)_*\mathcal{O}_{X_j}.$$

Hence, the family of morphisms

$$(f_j^\# : \mathcal{O}_T \rightarrow (f_j)_*\mathcal{O}_{X_j})_{j \in I}$$

motivate the existence of a unique morphism

$$f^\# : \mathcal{O}_T \rightarrow \prod_{j \in I} (f_j)_*\mathcal{O}_{X_j} = f_*\mathcal{O}_{P^{\text{co}}}$$

such that

$$\begin{array}{ccc} \mathcal{O}_T & \xrightarrow{f^\#} & f_*\mathcal{O}_{P^{\text{co}}} \\ & \searrow f_j^\# & \downarrow f_*(\iota_j^\#) \\ & & (f_j)_*\mathcal{O}_{X_j} \end{array}$$

commutes. It follows that  $(f_i, f_i^\#) = (f, f^\#) \circ (\iota_i, \iota_i^\#)$ . Furthermore,

$$f_p^\# : \mathcal{O}_{T, f(p)} \rightarrow \mathcal{O}_{P^{\text{co}}, p}$$

is a local ring homomorphism, since

$$f_p^\# = (\iota_i^\#)_x^{-1} \circ (f_i^\#)_x$$

and  $(f_i^\#)_x$  is local and  $(\iota_i^\#)_x$  is an isomorphism. Therefore,  $((P^{\text{co}}, \mathcal{O}_{P^{\text{co}}}), (\iota_i, \iota_i^\#)_{i \in I})$  is the coproduct.

In order to construct colimits in LRS, we study coequalisers of locally ringed spaces.

### 7.3 Coequalisers in LRS

We want to construct coequalisers on locally ringed spaces. Therefore, let  $(X, \mathcal{O}_X)$  and  $(Y, \mathcal{O}_Y)$  be locally ringed spaces and

$$(f, f^\#), (g, g^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$$

parallel morphisms. In Example 6.39 we defined the coequaliser on topological spaces as the quotient space

$$E^{\text{co}} := \text{coeq}(f, g) = Y / \sim,$$

where  $\sim$  is the smallest equivalence relation generated by  $f(x) \sim g(x)$  for all  $x \in X$ , with morphism

$$\pi : Y \rightarrow E^{\text{co}}$$

such that  $\pi \circ f = \pi \circ g$ . We want to lift that construction to locally ringed spaces (see [12, Ch. 1, § 1, Thm. 1.6]).

*Remark 7.24.* In literature, the sheaf on  $E^{\text{co}}$  is often defined as

$$\mathcal{O}_{E^{\text{co}}}(W) := \text{eq} \left( \mathcal{O}_Y(V) \rightrightarrows \mathcal{O}_X(U) \right)$$

for  $W \subseteq E^{\text{co}}$  with

$$\begin{aligned} V &:= \pi^{-1}(W) \subseteq Y \\ U &:= f^{-1}(V) = g^{-1}(V) \end{aligned}$$

and parallel morphisms

$$\begin{aligned} f_V^\# &: \mathcal{O}_Y(V) \rightarrow f_* \mathcal{O}_X(U) \\ g_V^\# &: \mathcal{O}_Y(V) \rightarrow g_* \mathcal{O}_X(U). \end{aligned}$$

This is the same Ansatz as ours, but more compact.

On the level of topological spaces, we have projections

$$\varpi : X \rightarrow E^{\text{co}}$$

$$\pi : Y \rightarrow E^{\text{co}}$$

such that

$$\begin{array}{ccc} X & \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} & Y \\ & \searrow \varpi \quad \swarrow \pi & \\ & E^{\text{co}} & \end{array}$$

commutes. We define

$$\mathcal{O}_{E^{\text{co}}} = \text{eq} \left( \pi_* \mathcal{O}_Y \rightrightarrows \varpi_* \mathcal{O}_X \right) \quad (7.25)$$

where the parallel morphisms are

$$\pi_*(f^\#) : \pi_* \mathcal{O}_Y \rightarrow \varpi_* \mathcal{O}_X$$

$$\pi_*(g^\#) : \pi_* \mathcal{O}_Y \rightarrow \varpi_* \mathcal{O}_X.$$

As in the construction of coproducts, we use the dual notion to coequalisers, since the direct image sheaf functor is contravariant on the topological spaces but covariant on sheaves.

**Claim:**  $\mathcal{O}_{E^{\text{co}}}$  is a structure sheaf on  $E^{\text{co}}$ .

Since  $\mathcal{O}_X$  and  $\mathcal{O}_Y$  are sheaves, it follows that  $\pi_* \mathcal{O}_Y$  and  $\varpi_* \mathcal{O}_X$  are sheaves. Furthermore, since limits commute with limits (recall Theorem 6.28 and that products and equalisers are limits),  $\mathcal{O}_{E^{\text{co}}}$  is a sheaf. In detail, the computation is analogous to (7.22).

For  $(E^{\text{co}}, \mathcal{O}_{E^{\text{co}}})$  to be a locally ringed space, it is not sufficient for  $\mathcal{O}_{E^{\text{co}}}$  to be a structure sheaf. We have to prove that all stalks are local rings.

*Remark 7.26.* For any  $W \subseteq E^{\text{co}}$ , it holds that

$$\mathcal{O}_{E^{\text{co}}}(W) = \text{eq} \left( \mathcal{O}_Y(\underbrace{\pi^{-1}(W)}_{:=V}) \rightrightarrows \mathcal{O}_X(\underbrace{\varpi^{-1}(W)}_{=f^{-1}(V)=g^{-1}(V)}) \right).$$

**Claim:** Any stalk  $\mathcal{O}_{E^{\text{co}},e}$  is a local ring, and the morphisms  $\pi_y^\# : \mathcal{O}_{E^{\text{co}},e} \rightarrow \mathcal{O}_{Y,y}$  are



local ring homomorphisms.

For an arbitrary  $W \subseteq E^{\text{co}}$ , we fix a section  $s \in \mathcal{O}_{E^{\text{co}}}(W)$  and a point  $y \in V$ . We assume, that  $\pi^\#(W)(s)$  is invertible in  $\mathcal{O}_{Y,y}$ , then

$$y \in D_V(\pi^\#(W)(s)) := \{y \in V \mid \pi(W)(s) \in \mathcal{O}_{Y,y}^\times\},$$

and  $D_V(\pi^\#(W)(s))$  is open by Lemma 7.18. By the definition of  $\mathcal{O}_{E^{\text{co}}}$ ,

$$f^\# \pi^\#(W)(s) = g^\# \pi^\#(W)(s)$$

holds. We call this section  $t \in \mathcal{O}_X(U)$ . Since  $f$  and  $g$  induce local morphisms, it follows that

$$f^{-1}(D_V(\pi^\#(s))) \stackrel{7.21}{=} D_V(t) \stackrel{7.21}{=} g^{-1}(D_V(\pi^\#(W)(s))).$$

Since

$$t = f^\#(\pi^\#(W)(s)) = g^\#(\pi^\#(W)(s)),$$

and by the definition of  $\pi^\#$ , it holds that

$$\mathcal{O}_{E^{\text{co}}} \xrightarrow{\pi^\#} \pi_* \mathcal{O}_X \xrightarrow[\pi_*(g^\#)]{\pi_*(f^\#)} \varpi_* \mathcal{O}_Y$$

and  $D(\pi^\#(W)(s))$  is the inverse image of some open subset  $W' \subseteq E^{\text{co}}$  such that  $\pi(y) \in W' \subseteq W$ . Therefore,  $s|_{W'}$  is invertible and  $s$  is locally invertible at  $\pi(y)$ .

Hence,  $(\pi_y^\#)^{-1}(\mathcal{O}_{Y,y}^\times) = \mathcal{O}_{E^{\text{co}},\pi(y)}^\times$ .

We want to show that  $\mathcal{O}_{Y,y}$  is a local ring using Lemma 2.35. Thus, we fix a point  $w \in W$ . Let  $s, t \in \mathcal{O}_{E^{\text{co}}}(W)$  be two sections that are not invertible at  $e$ . By the previous part, it follows that  $\pi^\#(W)(s)$  and  $\pi^\#(W)(t)$  are not invertible at any point  $y$  in  $\pi^{-1}(e)$ .

Since the stalks of  $Y$  are local, we may conclude that  $\pi^\#(W)(s+t)$  is not invertible at any point  $\pi^{-1}(e)$ . By

$$(\pi_y^\#)^{-1}(\mathcal{O}_{Y,y}^\times) = \mathcal{O}_{E^{\text{co}},\pi(y)}^\times,$$

it follows that  $s+t$  is not invertible at  $e$ . Thus,  $\pi$  defines local morphisms at stalks. Thus,  $(E^{\text{co}}, \mathcal{O}_{E^{\text{co}}})$  is a locally ringed space.

We can verify, that  $((E^{\text{co}}, \mathcal{O}_{E^{\text{co}}}), (\pi, \pi^\#))$  is the coequaliser. Let  $(T, \mathcal{O}_T)$  be an ar-

bitrary locally ringed space with a morphism  $(h, h^\#) : (Y, \mathcal{O}_Y) \rightarrow (T, \mathcal{O}_T)$  such that

$$(h, h^\#) \circ (f, f^\#) = (h, h^\#) \circ (g, g^\#).$$

The diagram

$$\begin{array}{ccccc} (X, \mathcal{O}_X) & \xrightarrow[(g, g^\#)]{(f, f^\#)} & (Y, \mathcal{O}_Y) & \xrightarrow{(\pi, \pi^\#)} & (E^{\text{co}}, \mathcal{O}_{E^{\text{co}}}) \\ & & & \searrow (h, h^\#) & \downarrow \exists! (\varphi, \varphi^\#) \\ & & & & (T, \mathcal{O}_T) \end{array}$$

visualises this. This leaves to show that a morphism  $(\varphi, \varphi^\#) : (E^{\text{co}}, \mathcal{O}_{E^{\text{co}}}) \rightarrow (T, \mathcal{O}_T)$  such that

$$(h, h^\#) = (\varphi, \varphi^\#) \circ (\pi, \pi^\#)$$

exists and that it is unique. We saw this on the level of topological spaces. On the level of structure sheaves, we apply  $\varphi_*$  to

$$\mathcal{O}_{E^{\text{co}}} \xrightarrow{\pi^\#} (\pi)_* \mathcal{O}_Y \xrightarrow[\pi_*(g^\#)]{\pi_*(f^\#)} \varpi_* \mathcal{O}_X$$

and get

$$\begin{array}{ccccc} \varphi_* \mathcal{O}_{E^{\text{co}}} & \xrightarrow{\varphi_*(\pi^\#)} & \varphi_* \circ \pi_* \mathcal{O}_Y & \xrightarrow[\varphi_* \circ \pi_*(g^\#)]{\varphi_* \circ \pi_*(f^\#)} & \varphi_* \circ \varpi_* \mathcal{O}_X \\ & & \parallel & & \parallel \\ & & h_* \mathcal{O}_Y & \xrightarrow[h_*(g^\#)]{h_*(f^\#)} & (h \circ f)_* \mathcal{O}_X \end{array}$$

Since  $\varphi_*$  commutes with limits by Remark 7.9, it follows that  $\varphi_* \mathcal{O}_{E^{\text{co}}}$  is the equaliser object. Furthermore, since

$$\begin{array}{ccccc} & & h_* \mathcal{O}_Y & & \\ & \swarrow h_*(f^\#) & & \searrow h_*(g^\#) & \\ h_* \circ f_* \mathcal{O}_X & & & & h_* g_* \mathcal{O}_X \\ \parallel & & & & \parallel \\ (h \circ f)_* \mathcal{O}_X & \xlongequal{\quad} & & & (h \circ g)_* \mathcal{O}_X \end{array}$$

is commutative, it follows that

$$\begin{array}{ccc}
 \mathcal{O}_T & \xrightarrow{h^\#} & h_*\mathcal{O}_Y \\
 \downarrow \exists! \varphi^\# & & \parallel \\
 \varphi_*\mathcal{O}_{E^{\text{co}}} & \xrightarrow{\varphi_* \circ \pi_* \mathcal{O}_Y} & \varphi_* \circ \pi_* \mathcal{O}_Y \rightrightarrows (h \circ f)_*\mathcal{O}_X
 \end{array} \tag{7.27}$$

commutes with

$$(\varphi^\# \circ \pi^\#)(W) = (\varphi \circ \pi)^\#(W) = \underbrace{(\varphi \circ \pi)_*}_{=h} \mathcal{O}_T(W) = h^\#(W)$$

for any  $W \subseteq T$ .

The morphism  $\varphi_x^\#$  is local, since the diagram 7.27 is local, as well as the morphism

$$\varphi_* \pi_* \mathcal{O}_Y : \varphi_* \mathcal{O}_{E^{\text{co}}} \rightarrow \varphi_* \circ \pi_* \mathcal{O}_Y.$$

The coequaliser allows us to define the colimit of locally ringed spaces via the coproduct.

## 7.4 Colimits in LRS

The constructed coproducts and coequalisers allow us to construct arbitrary colimits on locally ringed spaces. Let  $(X_i, \mathcal{O}_{X_i})_{i \in I}$  be a family of locally ringed spaces and let

$$(P^{\text{co}}, \mathcal{O}_{P^{\text{co}}}) := \coprod_{i \in I} (X_i, \mathcal{O}_{X_i})$$

be the coproduct with inclusion morphisms

$$(\iota_i, \iota_i^\#) : (X_i, \mathcal{O}_{X_i}) \rightarrow (P^{\text{co}}, \mathcal{O}_{P^{\text{co}}})$$

such that for any locally ringed space  $(T, \mathcal{O}_T)$  with a family of morphisms

$$(f_i, f_i^\#) : (X_i, \mathcal{O}_{X_i}) \rightarrow (T, \mathcal{O}_T)$$

there exists a unique morphism

$$(f, f^\#) : (P^{\text{co}}, \mathcal{O}_{P^{\text{co}}}) \rightarrow (T, \mathcal{O}_T)$$

and for all  $i \in I$ , such that  $(f_i, f_i^\#)$  factorises uniquely through  $(f, f^\#)$ . This is exactly the coproduct we constructed in Chapter 7.2. Let

$$(g_{ij}, g_{ij}^\#) : (X_i, \mathcal{O}_{X_i}) \rightarrow (X_j, \mathcal{O}_{X_j})$$

be an arbitrary family of morphisms. Then the induced diagram is no longer commutative. Let

$$(E^{\text{co}}, \mathcal{O}_{E^{\text{co}}}) := \text{coeq} \left( \coprod_{i \in I} \iota_i, \coprod_{i \in I} \iota_j \circ g_{ij} \right)$$

be the coequaliser we constructed in Chapter 7.3. We saw that  $(E^{\text{co}}, \mathcal{O}_{E^{\text{co}}})$  is a locally ringed space and the diagram

$$\begin{array}{ccccc}
 (X_i, \mathcal{O}_{X_i}) & \xrightarrow{(g_{ij}, g_{ij}^\#)} & & & (X_j, \mathcal{O}_{X_j}) \\
 & \searrow (\iota_i, \iota_i^\#) & & \swarrow (\iota_j, \iota_j^\#) & \\
 & & (P^{\text{co}}, \mathcal{O}_{P^{\text{co}}}) & & \\
 & & \downarrow (\pi, \pi^\#) & & \\
 & & (E^{\text{co}}, \mathcal{O}_{E^{\text{co}}}) & & \\
 & \searrow (f_i, f_i^\#) & \downarrow \exists! (\varphi, \varphi^\#) & \swarrow (f_j, f_j^\#) & \\
 & & (T, \mathcal{O}_T) & & 
 \end{array}$$

commutes.

**Claim:** Any ring homomorphism induced by  $\varphi^\# : \mathcal{O}_{E^{\text{co}}} \rightarrow \mathcal{O}_T$  is a local ring homomorphism.

In the construction of coequalisers on LRS, we saw that  $\varphi^\#$  and  $\pi^\#$  induce local ring homomorphisms. The composite of local ring homomorphisms is a local ring homomorphism, since maximal ideals are respected. Furthermore,  $(\varphi, \varphi^\#)$  is unique, as we saw in the construction of the coequaliser in LRS. In particular,  $(\varphi, \varphi^\#) \circ (\pi, \pi^\#)$  is a universal morphism as composite of universal morphisms. Hence, we have indeed constructed the colimit of locally ringed spaces, using the coproduct and the

coequaliser.

# A Elaborations on Preliminaries

To understand the proof of Theorem 2.20, we need to understand the concepts of a topological base and subbase. Additionally, we see the definition of the box topology and why the theorem is explicitly for the product topology and not for the box topology.

**Definition A.1** (Base). Let  $X$  be a set. A **base** for a topology on  $X$  is a collection  $\mathcal{B}$  of subsets  $(B_i)_{i \in I}$  of  $X$ , called **base elements**, such that

1. For each  $x \in X$  there exists a base element with  $x \in B_i$ .
2. If  $x \in B_i$  and  $x \in B_j$ , then there exists a base element  $B_k \subseteq B_i \cap B_j$  such that  $x \in B_k$ .

The **topology**  $\tau$  **generated by**  $\mathcal{B}$  as follows: A subset  $U \subseteq X$  is open if for each  $x \in U$  there exists a base element  $B \in \mathcal{B}$  such that  $x \in B$  and  $B \subseteq U$ . Every base element itself is element of  $\tau$ . [13, Ch. 2, § 13, p. 78]

**Example A.2.** A base for  $\mathbb{R}$  are all open intervals in  $\mathbb{R}$  (see [13, Ch. 2, § 19, Ex. 1, p. 116]).

In order to define the product topology, we need the notion of the subbase.

**Definition A.3** (Subbase). A **subbase**  $\mathcal{S}$  for a topology  $\tau$  on  $X$  is a collection of subsets  $S_i$  of  $X$  such that

$$\bigcup_{i \in I} S_i = X.$$

The **topology generated by the subbase**  $\mathcal{S}$  is defined to be the collection  $\mathcal{T}$  of all finite intersections and all unions of elements of  $\mathcal{S}$ . [13, Ch. 2, § 13, p. 82]

**Definition A.4** (Product topology). Let  $S_i$  denote the collection

$$S_i = \{\text{pr}_i^{-1}(U_i) \mid U_i \text{ open in } X_i\}$$

and let  $S$  denote the union of these collections

$$s = \bigcup_{i \in I} S_i.$$

The topology generated by the subbase  $S$  is called **product topology**. In this topology,

$$\prod_{i \in I} X_i$$

is called **product space** (see [13, Ch. 2, § 19, p. 114]).

**Definition A.5** (Box topology). Let  $(X_i)_{i \in I}$  be a family of topological spaces. We define the base for the product of topological spaces as

$$\prod_{i \in I} U_i$$

where  $U_i$  is open in  $X_i$  for each  $i \in I$ . The topology generated by this base is called **box topology** (see [13, Ch. 2, § 19, p. 114]).

*Remark A.6.* In general, the box topology is finer than the product topology.

Theorem 2.20 stated, that a function  $f : Y \rightarrow \prod_{j \in J} X_j$  into the product topology with  $f(y) = (f_j(y))_{j \in J}$  is continuous if and only if all functions  $f_j : Y \rightarrow X_j$  are continuous.

*Proof.* Let  $(X_j)_{j \in J}$  be a family of topological spaces and let  $\pi_j$  be the projection on the  $j$ -th factor. Since  $\pi_j$  is continuous,  $\pi_j^{-1}(U_j)$  is a subbase element for the product topology  $\prod_{j \in J} X_j$  for every open set  $U_j \subseteq X_j$ .

Assume that  $f : Y \rightarrow \prod_{j \in J} X_j$  is continuous. By definition, it holds that  $\pi_j \circ f = f_j$ . Since the composite of two continuous functions is continuous,  $f_j$  is continuous.

Conversely, assume that  $(f_i)_{i \in J}$  is continuous for all  $j \in J$ . If  $f$  is continuous, then the preimage under  $f$  of every subbase element is open in  $Y$ . A typical subbase

element of the product topology is  $\pi_j^{-1}(U_j)$  for  $U_j \subseteq X_j$  open. Since  $f_j = \pi_j \circ f$ ,

$$f^{-1}(\pi_j^{-1}(U_j)) = f_j^{-1}(U_j).$$

By assumption,  $f_j$  is continuous and thus,  $f_j^{-1}(U_j)$  is open in  $Y$ . [13, Ch. 2, § 19, p. 117] □

This illustrates why we use the product topology and not the box topology. While the box topology and the product topology agree on finitely many topological spaces, we can not conclude that  $f$  is continuous on the box topology on infinitely many topological spaces, since we cannot make any claims on the continuity of

$$\bigcap_{j \in J} f_j^{-1}(U_j)$$

in that case.

**Example A.7.** Consider  $\mathbb{R}^n$ . Since a base of  $\mathbb{R}$  consists of all open intervals in  $\mathbb{R}$ , a base for the product topology consists of all products of the form

$$(a_1, b_1) \times (a_2, b_2) \times \cdots \times (a_n, b_n).$$

Since  $n$  is finite, so is the product and the product topology agrees with the box topology.

**Example A.8.** Let  $q : \mathbb{R} \rightarrow \{a, b, c\}$  with

$$q(x) = \begin{cases} a & \text{if } x > 0 \\ b & \text{if } x < 0 \\ c & \text{if } x = 0. \end{cases}$$

The quotient topology on  $\{a, b, c\}$  induced by  $q$  is  $\{\{a\}, \{b\}, \{a, b\}, \{a, b, c\}, \emptyset\}$  (see [13, Ch. 2, § 22, p. 138]).

**Example A.9.** Consider  $\mathbb{R}$  and  $\mathbb{R}$ . Although  $\mathbb{R}$  and  $\mathbb{R}$  are not disjoint, we can make them disjoint via

$$\mathbb{R} \sqcup \mathbb{R} := \mathbb{R} \times \{1\} \cup \mathbb{R} \times \{2\}.$$



Although  $-3 \in \mathbb{R} \cap \mathbb{R}$ ,  $(-3, 1) \in (\mathbb{R}, 1)$  but  $(-3, 1) \notin (\mathbb{R}, 2)$ .

**Example A.10.** Consider  $\mathbb{R}$  and an interval  $(-3, 4) \subset \mathbb{R}$ . Then any set  $(-3, 4) \cap (a, b)$  with  $a < b$  in  $\mathbb{R}$  is open in  $(-3, 4)$ .

**Definition A.11** (Group). A **group** is a set  $G$ , equipped with a binary operation  $\star$ , such that the following properties are satisfied:

1. For  $a, b, c \in G$ ,  $\star$  is associative, that is

$$(a \star b) \star c = a \star (b \star c) = a \star b \star c.$$

2. There exists a unique element  $e \in G$ , called the identity, such that

$$e \star a = a \star e = a$$

for all  $a \in G$ .

3. For each  $a \in G$ , there exists a unique element  $a^{-1}$  such that

$$a \star a^{-1} = a^{-1} \star a = e.$$

A group is called **abelian** if it is commutative (see [7, p. 7]).

## B Algebraic Categories

When we defined presheaves, it was mentioned that we can define  $\mathcal{A}$ -valued presheaves on algebraic categories  $\mathcal{A}$ . While it was stated that  $\mathbf{CRing}$  is an algebraic category, it was not explained what an algebraic category is. To understand them in their depth, it is necessary to understand structures and categories of structures first.

**Definition B.1** (Signatures, types). A **signature** is a set  $\Omega$  together with a map  $\Omega \rightarrow \mathbb{N}$ . The elements  $F \in \Omega$  are called **logical operators** and their image  $n \in \mathbb{N}$  is called **arity**. Thus, we may denote a logical operator of arity  $n$  by  $F^{[n]}$ .

An **expression** in the variables  $X_1, \dots, X_m$  with regards to the signature  $\Omega$  is a symbolical expression, arising from the following rules:

1.  $X_i$  is an expression in the variables  $X_1, \dots, X_m$  for  $1 \leq i \leq m$ .
2. If  $T_1, \dots, T_n$  are expressions in the variables  $X_1, \dots, X_m$  and if  $F^{[n]} \in \Omega$ , then  $F(T_1 \dots, T_n)$  is an expression in the variables  $X_1, \dots, X_m$ .
3. An expression can be generated only by these rules.

An **equation** is an ordered pair of two expressions in finitely many variables.

A **type** of an algebraic structure is a tuple  $\tau = (\Omega, \mathcal{E})$ , where  $\Omega$  is a signature and  $\mathcal{E}$  is a set of equations.

Knowing what a type is allows us to define structures of a type  $\tau$ .

**Definition B.2** (Structures of a type). Let  $\tau = (\Omega, \mathcal{E})$  be a type. A **structure A of type  $\Omega$**  consists out of a set  $X = U(A)$ , the so-called **underlying set** and an association that maps any symbol  $F^{[n]} \in \Omega$  to a  $n$ -ary operation

$$| F |_A: X^n \rightarrow X.$$

For any structure  $A$  of type  $\Omega$  and  $x \in X^n$ , we can assign an element  $|T|_A(x) \in X$  to any expression  $T$  in the variables  $X_1, \dots, X_m$ , by defining recursively

$$\begin{aligned} |X_i|_A(x) &= x_i \\ |F(T_1, \dots, T_n)|_A(x) &:= |F|_A(|T_1|_A(x), \dots, |T_n|_A(x)). \end{aligned}$$

A structure  $A$  of type  $\Omega$  **satisfies the equation (S,T)** in the variables  $X_1, \dots, X_m$  if

$$|S|_A(x) = |T|_A(x)$$

for all  $x \in X^m$ .

A **structure of type  $\tau$**  is a structure  $A$  of type  $\Omega$ , that satisfies all equations in  $\mathcal{E}$  (see [3, p. 84]).

To fully understand the category of structures, we have to understand not only the objects, but also the morphisms.

**Definition B.3** (Homomorphisms of structures). Let  $A, B$  be structures of type  $\tau = (\Omega, \mathcal{E})$  with underlying sets  $U(A)$  and  $U(B)$ . A **homomorphism**  $f : A \rightarrow B$  is a mapping  $f : U(A) \rightarrow U(B)$  that is compatible with all operations, that is, for all  $F^{[n]} \in \Omega$  the diagram

$$\begin{array}{ccc} U(A)^n & \xrightarrow{f^n} & U(B)^n \\ \downarrow |F|_A & & \downarrow |F|_B \\ U(A) & \xrightarrow{f} & U(B) \end{array}$$

commutes. Thus, for all  $(x_1, \dots, x_n) \in X^n$

$$f(|F|_A(x_1, \dots, x_n)) = |F|_B(f(x_1), \dots, f(x_n))$$

and  $U(f) : U(A) \rightarrow U(B)$  is the underlying mapping between sets of the homomorphism  $f$  (see [3, p. 84]).

This allows us to define categories of structures and especially algebraic structures:

**Definition B.4** (Category of structures). In the category of structures, denoted by  $S(\tau)$ , we consider the objects to be structures of type  $\tau$  and the morphisms are

homomorphisms. An **algebraic category** of finite arity is a category  $\mathcal{C}$  together with a functor  $\mathcal{C} \rightarrow \mathbf{Set}$ , such that the diagram of functors

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\quad} & \mathbf{S}(\tau) \\ & \searrow & \swarrow \\ & \mathbf{Set} & \end{array}$$

commutes (see [3, p. 85]).

## C Elaborations on Limits and Colimits

We want to prove the following Theorem:

**Theorem C.1** (Finite limits commute with directed colimits). Let  $\Gamma = (V, E)$  be a finite graph and let  $(J, \leq)$  be a directed poset. Then the limit of shape  $\Gamma$  commutes with the directed colimit of shape  $(J, \leq)$  (see [3, Ch. 6, § 6, Thm. 6.6.16]).

*Proof.* Let  $X$  be a legal diagram of shape  $(V, E) \otimes (J, \leq)$  which is commutative in the second variable. We show that the canonical morphism

$$\alpha : \operatorname{colim}_{j \in J} \lim_{v \in V} X(v, j) \rightarrow \lim_{v \in V} \operatorname{colim}_{j \in J} X(v, j) \quad (\text{C.2})$$

is an isomorphism:

**Claim:** The morphism  $\alpha$  is injective.

We consider two objects  $[x], [y] \in \operatorname{colim}_{j \in J} \lim_{v \in V} X(v, j)$  such that

$$\alpha([x]) = \alpha([y]).$$

Recall, that the objects  $[x], [y]$  are equivalence classes, and that both objects consists of a compatible family  $(x_v)_{v \in V}$  at some indexing element  $i \in J$ , and a compatible family  $(y_v)_{v \in V}$  at some indexing element  $j \in J$  respectively.

Since by assumption,  $\alpha([x]) = \alpha([y])$ , it holds for all  $v \in V$  that

$$[x_v] = [y_v]$$

in  $\operatorname{colim}_{j \in J} X(v, j)$ . Since  $J$  is directed, there exists an indexing element  $k_v \in J$  for all  $v \in V$ , and morphisms

$$\phi_v : i \rightarrow k_v \tag{C.3}$$

$$\psi_v : j \rightarrow k_v \tag{C.4}$$

such that

$$X(\operatorname{id}_v, \phi_v)(x_v) = X(\operatorname{id}_v, \psi_v)(y_v) \tag{C.5}$$

in  $X(v, k_v)$ .

Since  $V$  is finite, the set

$$K := \{k_v \mid v \in V\}$$

is finite, and since  $J$  is filtered,  $K$  has an upper bound  $k$  in  $J$  with morphisms

$$\lambda_v : k_v \rightarrow k. \tag{C.6}$$

This gives rise to morphisms

$$\beta_v := \lambda_v \circ \phi_v : i \rightarrow k \tag{C.7}$$

$$\gamma_v := \lambda_v \circ \psi_v : j \rightarrow k. \tag{C.8}$$

It follows that

$$X(\operatorname{id}_v, \phi_v)(x_v) \stackrel{\text{(C.3)}}{\stackrel{\text{(C.4)}}{=}} X(\operatorname{id}_v, \psi_v)(y_v) \tag{C.9}$$

$$X(\operatorname{id}_v, \beta_v)(x_v) \stackrel{\text{(C.7)}}{=} X(\operatorname{id}_v, \lambda_v)(X(\operatorname{id}_v, \phi_v)(x_v)) \tag{C.10}$$

$$X(\operatorname{id}_v, \gamma_v)(y_v) \stackrel{\text{(C.8)}}{=} X(\operatorname{id}_v, \lambda_v)(X(\operatorname{id}_v, \psi_v)(y_v)) \tag{C.11}$$

by the definitions of the respective morphisms. Combined, (C.9)-(C.11) give rise to

$$X(\operatorname{id}_v, \beta_v)(x_v) = X(\operatorname{id}_v, \gamma_v)(y_v). \tag{C.12}$$

We can define two cones in  $X(-, k)$  with

$$x' := (X(\operatorname{id}_v, \beta_v)(x_v))_{v \in V}$$

and

$$y' := (X(\operatorname{id}_v, \gamma_v)(y_v))_{v \in V}.$$

Since  $x_v$  and  $y_v$  become equal in  $X(v, k)$  after suitable morphisms, and since this holds for all  $v \in V$ , the resulting families  $(X(\text{id}_v, \beta_v)(x_v))$  and  $(X(\text{id}_v, \gamma_v)(y_v))$  are equal elements of  $\lim_{v \in V} X(v, k)$ . Thus, their equivalence classes in the colimit over  $j \in J$  coincide and  $[x] = [y]$ . This shows that  $\alpha$  is injective.

**Claim:** The morphism  $\alpha$  is surjective.

The central argument in showing surjectivity of  $\alpha$  is that due to the filteredness of  $V$ , there are only finitely many indices to "amalgamate", and due to the directedness of  $J$ , there exists a single index  $j'$ , "dominating" any other index, which makes all equalities become strict equalities in the component  $j'$ . We consider an element  $([x_v])_{v \in V} \in \lim_{v \in V} \text{colim}_{j \in J} X(v, j)$ . We can choose for each equivalence class  $[x_v] \in \text{colim}_{j \in J} X(v, j)$  a representative

$$x'_v \in X(v, j_v)$$

for some  $j_v \in J$ . By the compatibility in the limit, for any morphism  $f : u \rightarrow v$  in  $V$ , the images of  $[x_u]$  and  $[x_v]$  in  $\text{colim}_{j \in J} X(v, j)$  are equal. Let  $X(w \rightarrow k)$  denote the structure morphism  $X(w, j_w) \rightarrow X(w, k)$ , which is induced by the morphism  $j_w \rightarrow k$  in  $J$ . By the definition of the colimit in  $\text{Set}$ , for each morphism  $f : u \rightarrow v$  in  $V$ , there exists an index  $k_f \in J$  and morphisms

$$j_u \rightarrow k_f$$

$$j_v \rightarrow k_f$$

in  $J$ , such that

$$X(f, k_f)(X(u \rightarrow k_f)(x_u)) = X(v \rightarrow k_f)(x_v)$$

is satisfied by the images of  $x_u$  and  $x_v$  in  $X(u, k_f)$  and  $X(v, k_f)$ . Since  $V$  is finite, the set

$$S := \{j_v \mid v \in \text{Ob}(V)\} \cup \{k_f \mid f \in \text{Mor}(V)\}$$

is finite. There exists an index  $j' \in J$  and morphisms  $\tau_s : s \rightarrow j'$  for every  $s \in S$ , because  $J$  is filtered. We may define

$$y_v := X(v, \tau_{j_v})(x_v) \in X(v, j')$$

for each  $v \in V$ . It holds that

$$X(f, k_f)(X(u \rightarrow k_f)(x_u)) = X(v \rightarrow k_f)(x_v)$$

for any morphism  $f : u \rightarrow v$  and index  $k_f \in S$ , and the functoriality of the structure maps, that is the composition with morphisms  $k_f \rightarrow j'$ , gives

$$X(f, j')(X(u \rightarrow j')(x_u)) = X(v \rightarrow j')(x_v),$$

which is in essence

$$X(f, j')(y_u) = y_v.$$

Thus the family  $(y_v)_{v \in V}$  is a cone in  $\lim_{v \in V} X(v, j')$ , and its image under the canonical map

$$\lim_{v \in V} X(v, j') \rightarrow \lim_{v \in V} \operatorname{colim}_{j \in J} X(v, j)$$

is precisely the original element  $([x_v])_v$ .

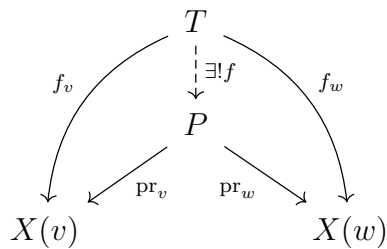
This shows that the preimage of any element in  $\lim_{v \in V} \operatorname{colim}_{j \in J} X(v, j)$  exists, and  $\alpha$  is surjective.

Hence,  $\alpha$  is injective and surjective and therefore defines an isomorphism (see [3, Ch. 6, § 6, Proof 6.6.16]). □

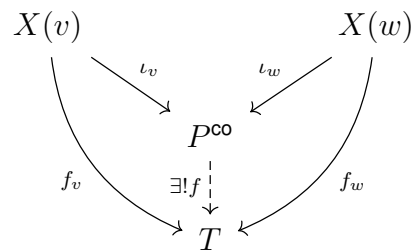


# D Overview over Universal Property Diagrams

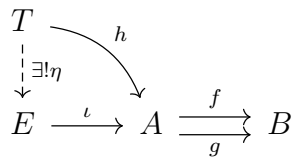
Binary product



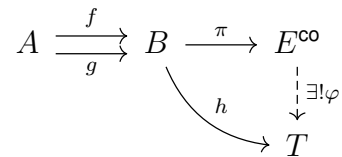
Binary coproduct



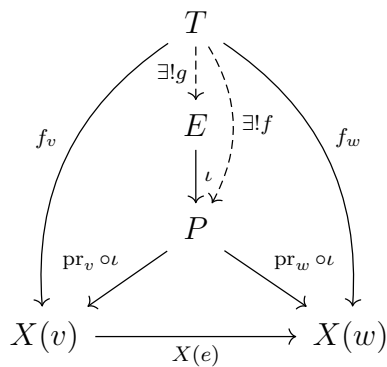
Equaliser



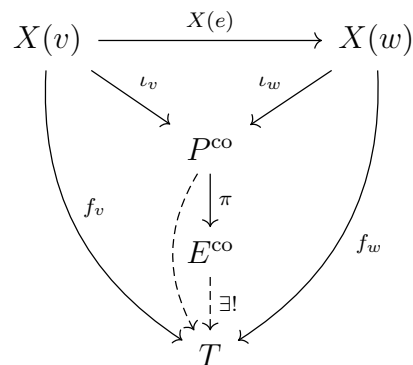
Coequaliser



Limit



Colimit



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